ASSOCIATION OF ZOOS & AQUARIUMS

CAPYBARA (*Hydrochoerus hydrochaeris*) CARE MANUAL

CREATED BY THE
AZA Capybara Species Survival Plan® Program
IN ASSOCIATION WITH THE
AZA Rodent, Insectivore and Lagomorph TAG

Capybara (*Hydrochoerus hydrochaeris*) Care Manual

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Disclaimer: This manual presents a compilation of knowledge provided by recognized animal experts based on the current science, practice, and technology of animal management. The manual assembles basic requirements, best practices, and animal care recommendations to maximize capacity for excellence in animal care and welfare. The manual should be considered a work in progress, since practices continue to evolve through advances in scientific knowledge. The use of information within this manual should be in accordance with all local, state, and federal laws and regulations concerning the care of animals. While some government laws and regulations may be referenced in this manual, these are not all-inclusive nor is this manual intended to serve as an evaluation tool for those agencies. The recommendations included are not meant to be exclusive management approaches, diets, medical treatments, or procedures, and may require adaptation to meet the specific needs of individual animals and particular circumstances in each institution. Commercial entities and media identified are not necessarily endorsed by AZA. The statements presented throughout the body of the manual do not represent AZA standards of care unless specifically identified as such in clearly marked sidebar boxes.

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Introduction

Preamble

AZA accreditation standards, relevant to the topics discussed in this manual, are highlighted in boxes such as this throughout the document (Appendix A).

AZA accreditation standards are continuously being raised or added. Staff from AZA-accredited institutions are required to know and comply with all AZA accreditation standards, including those most recently listed on the AZA website (http://www.aza.org), which might not be included in this manual.

Taxonomic Classification

Table 1. Taxonomic classification for capybara

Classification	Taxonomy
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Rodentia
Suborder	Hystricomorpha
Family	Caviidae
Genus	*Hydrochaeris*

Genus, Species, and Status

Table 2. Genus, species, and status information for capybara

Genus	Species	Common Name	USA Status	IUCN Status	AZA Status
Hydrochoerus	*hydrochaeris*	Capybara	n/a	Least Concern	SSP
Hydrochoerus	*isthmius*	Lesser Capybara	n/a	Data Deficient	Included in SSP

General Information

The information contained within this Animal Care Manual (ACM) provides a compilation of animal care and management knowledge that has been gained from recognized species experts, including AZA Taxon Advisory Groups (TAGs), Species Survival Plan® Programs (SSPs), biologists, veterinarians, nutritionists, reproduction physiologists, behaviorists, and researchers (visit the AZA Animal Program page to contact these individuals). It is based on the most current science, practices, and technologies used in animal care and management and is a valuable resource that enhances animal welfare by providing information about the basic requirements needed and best practices known for caring for *ex situ* capybara populations. This ACM is considered a living document that is updated as new information becomes available and at a minimum of every five years.

Information presented is intended solely for the education and training of zoo and aquarium personnel at AZA-accredited institutions. Recommendations included in the ACM are not exclusive management approaches, diets, medical treatments, or procedures, and may require adaptation to meet the specific needs of individual animals and particular circumstances in each institution. Statements presented throughout the body of the manuals do not represent specific AZA accreditation standards of care unless specifically identified as such in clearly marked sidebar boxes. AZA-accredited institutions which care for capybaras must comply with all relevant local, state, and federal wildlife laws and/or regulations; AZA accreditation standards that are more stringent than these laws and/or regulations must be met (AZA Accreditation Standard 1.1.1).

AZA Accreditation Standard

(1.1.1) The institution must comply with all relevant local, state/provincial, and federal wildlife laws and/or regulations. It is understood that, in some cases, AZA accreditation standards are more stringent than existing laws and/or regulations. In these cases the AZA standard must be met.

The ultimate goal of this ACM is to facilitate excellent capybara management and care, which will ensure superior capybara welfare at AZA-accredited institutions. Ultimately, success in our capybara management and care will allow AZA-accredited institutions to contribute to capybara conservation, and ensure that capybaras are in our future for generations to come.

Capybaras are the largest living rodent. The family Caviidae includes the species *Hydrochoerus hydrochaeris* (common capybara) and *Hydrochoerus isthmius* (lesser capybara). According to the Guarani Indians, the name capybara (pronounced "kapibar'u") means the "master of the grasses" (Montiani-Ferreira, 2008). Capybaras have a combined head and body length of 100–130 cm (39–51 in.), with a shoulder height as tall as 50 cm (20 in.) and a body weight that can range from 27–79 kg (60–174 lb.). The average weight of managed capybaras is 50 kg (110 lb.) for adult males and 61 kg (135 lb.) for adult females. The lesser capybara is smaller than the common capybara in all external and cranial measurements, weighing an average of 28 kg (62 lb.). The lesser capybara exhibits thicker and wider frontal bones, shorter and thicker pterygoids, and a longer diastema. The diploid number of *H. isthmius* is 2n=64 (FN=104) which differs from *H. hydrochaeris* whose diploid number is 2n=66 (FN=102). Sexual maturity occurs at 12–18 months of age in both *H. hydrochaeris* and *H. isthmius*, with the length of gestation varying between 150 days (*H. hydrochaeris*) and 108 days (*H. isthmius*) (Moreira et al., 2013). The average lifespan in their natural environment is 4 years; however, they may live up to 10 years (Macdonald, 2006).

Capybaras' long, coarse pelage is sparse, leaving the skin visible. Their coloration is generally reddish brown to grayish on the upper parts and yellowish brown on the under parts. Occasionally they have some black on the face, the outer surface of the limbs, and the rump. Their ears are short and rounded and their eyes are placed dorsally and relatively far back on the head. The forefeet of the capybara have four digits and the hindfeet have three. These digits, arranged in a radial pattern, are partially webbed and armed with short, strong claws, making capybaras very strong swimmers. They are able to stay underwater for up to 5 minutes at a time. Capybaras have a muzzle that is heavy and truncated, with an enlarged upper lip. In mature males, a bare, raised area on the top of the snout contains a greatly enlarged sebaceous gland, which plays a role in signaling dominance status. The proportions of the various chemicals present in the secretions of capybaras differ by individual, providing potential for individual recognition via personal "olfactory fingerprints" (Macdonald, Krantz, & Aplin, 1984).

The native range of *H. hydrochaeris* occurs to the east of the Canal Zone in Panama, and on the east side of the Andes in South America from Colombia and the Guianas to Uruguay and northeastern Argentina (Mones & Ojasti, 1986). The native range of *H. isthmius* occurs in eastern Panama, northwestern Colombia, and northwestern Venezuela (Lord, 2009). *H. isthmius* current population numbers are unknown. Both *H. hydrochaeris* and *H. isthmius* live in a rainforest environment that contains estuaries, marshes, rivers and/or streams. Their habitat typically experiences approximately 2.3 m (7.5 ft.) of rain annually, with the dry season occurring from May to November tempered by a flood season running from December to April. *H. hydrochaeris* usually lives in groups averaging ten in number. Individual herds may temporarily create larger aggregations, which may contain up to one hundred individuals. The herd size varies according to the season, with larger groups found during the dry season (Lord, 2009). Likewise, *H. isthmius* lives in social groups during the dry season; however, *H. isthmius* lives individually during the rest of the year.

Capybaras have at least seven distinct vocalizations (Nogueira et al., 2012). Infants and the young constantly emit a guttural purr, potentially to maintain contact with their mothers or other group members. Subordinate males also make this sound during altercations, perhaps to appease their adversaries. Another vocalization, the alarm bark, is generally used when a predator is detected. This coughing sound is often repeated several times, and the reaction of nearby animals may be to stand alert or to rush into the water. Capybaras also use clicks, squeaks, whistles, and grunts to communicate with others in the herd.

Capybaras are naturally crepuscular; however, they have become nocturnal in areas populated by humans in order to avoid contact. The main threats to the species are those presented by humans, both in terms of hunting and deforestation.

Chapter 1. Ambient Environment

1.1 Temperature and Humidity

The animals must be protected or provided accommodation from weather, and any adverse conditions. (AZA Accreditation Standard 1.5.7). Animals not normally exposed to cold weather/water temperatures should be provided heated enclosures/pool water. Likewise, protection from excessive cold weather/water temperatures should be provided to those animals normally living in warmer climates/water temperatures.

> **AZA Accreditation Standard**
>
> **(1.5.7)** The animals must be protected or provided accommodation from weather or other conditions clearly known to be detrimental to their health or welfare.

Capybaras can withstand great temperature extremes and may be kept outdoors in areas with lows of 4 °C (40 °F) as long as a heated shelter with bedding such as straw, hay, or mulch is provided. Capybaras can tolerate temperatures as high as 32–38 °C (90–100 °F) as long as they have access to water and shade. Water can be provided in the form of a pool, mister, or mud wallows. Shade may be provided by trees, shelters, shade cloth, or access to an indoor area. In an effort to mimic the capybara's natural environment, humidity levels for indoor enclosures should range from 30–70% relative humidity (RH), but capybaras can tolerate lower levels of humidity (15–20% RH) in drier climates without apparent difficulty (Joslin et al., 1998).

AZA institutions with exhibits which rely on climate control must have critical life-support systems for the animal collection and emergency backup systems available. Warning mechanisms and backup systems must be tested periodically (AZA Accreditation Standard 10.2.1).

> **AZA Accreditation Standard**
>
> **(10.2.1)** Critical life-support systems for the animals, including but not limited to plumbing, heating, cooling, aeration, and filtration, must be equipped with a warning mechanism, and emergency backup systems must be available. Warning mechanisms and emergency backup systems must be tested periodically.

Indoor enclosures require heating, ventilation, and air conditioning (HVAC) systems to maintain their climates, and these systems must have a secondary power source in the event of an emergency or a power outage. Secondary power sources include backup generators for the building housing the exhibit, and these systems should be maintained by certified HVAC workers (T. Schoffner, personal communication, 2014).

1.2 Light

Careful consideration should be given to the spectral, intensity, and duration of light needs for all animals in the care of AZA-accredited zoos and aquariums. Typically, capybaras are housed outdoors during at least part of the year, but when housed indoors, capybaras respond well to a 12-hour light and 12-hour dark cycle (Morgan & Tromborg, 2007). Natural fluorescent or incandescent lighting is acceptable to provide the light part of the cycle. When animals are housed indoors, it is recommended that a UV light also be used for a minimum of 7 hours per day as a source of visual stimulation. For purposes of visibility, facilities maintaining capybaras in a nocturnal house may find it useful to incorporate a reverse light cycle, using either red or blue light during the "night" part of the cycle and a bright white light for the "day" part (Morgan & Tromborg, 2007). It has yet to be documented if a variation in the light cycle plays an important role in affecting capybara behavior.

1.3 Water and Air Quality

AZA-accredited institutions must have a regular program of monitoring water quality for collections of aquatic animals and a written record must document long-term water quality results and chemical additions (AZA Accreditation Standard 1.5.9). Monitoring selected water quality parameters provides confirmation of the correct operation of filtration and disinfection of the water supply available for the collection. Additionally, high quality water enhances animal health programs instituted for aquatic collections.

> **AZA Accreditation Standard**
>
> **(1.5.9)** The institution must have a regular program of monitoring water quality for fish, marine mammals, and other aquatic animals. A written record must be maintained to document long-term water quality results and chemical additions.

Water elements should be included in the enclosure in the form of pools, streams, or natural water features. Pools may be on a filtered system, but most in capybara enclosures are the type that requires

draining, cleaning, and daily refilling. Pools should be disinfected a minimum of once a week, using bleach or other approved cleaning agents; however, it is not necessary to chemically treat the water supply (via chlorine, for example). When disinfecting enclosures using bleach, a 5% dilution should be used and all items that come into contact with the dilution should be rinsed thoroughly before animals are exposed to them (Heuschele, 1995). Water quality tests may be performed at the facility's discretion for coliform and *E. coli* if there is a possibility of contamination.

Capybaras housed outdoors do not require any type of ventilation. For capybaras housed in a closed-system exhibit, there should be procedures in place for filtering recirculated air. For example, one AZA accredited facility with an indoor capybara enclosure maintains 2831.685 cubic meters (100,000 cubic feet) per hour of fresh air coming in from outside; this volume provides a 100% turnover of air approximately 5–11 times per hour. Other methods for filtering recirculated air include using live plants and air filters at the point of air intake.

1.4 Sound and Vibration

Consideration should be given to controlling sounds and vibrations that can be heard by animals in the care of AZA-accredited zoos and aquariums. At this time, it is unknown what the tolerances are for vibration; however, as with any wildlife, these disturbances should be kept to a minimum.

In their natural environment, capybaras are typically exposed to sounds that are approximately 27 decibels (Morgan & Tromborg, 2007). In a zoo/aquarium (or laboratory) setting, research has shown that animals can be exposed to sounds that are between 40–80 decibels with animals beginning to show signs of stress at approximately 85 decibels (Anthony, Ackerman, & Lloyd, 1959). Sources of sound that capybaras are typically exposed to while housed in managed care include construction within or surrounding the facility, noise from highways/roadways, trains, airports, sirens, guests, other animals, and the cleaning/daily maintenance process. The intensity of these sounds can be measured in decibels using a decibel meter; sounds measuring above 80 decibels must be dampened. Methods to dampen sounds may include moving animals to an indoor enclosure and/or adding soundproofing materials to the enclosure.

Chapter 2. Habitat Design and Containment

2.1 Space and Complexity

Careful consideration should be given to exhibit design so that all areas meet the physical, social, behavioral, and psychological needs of the species. Animals must be well cared for and presented in a manner reflecting modern zoological practices in exhibit design (AZA Accreditation Standard 1.5.1). All animals must be housed in enclosures that meet their physical and psychological needs, as well as their social needs. (AZA Accreditation Standard 1.5.2, 1.5.2.1, 1.5.2.2).

AZA Accreditation Standard

(1.5.1) All animals must be well cared for and presented in a manner reflecting modern zoological practices in exhibit design, balancing animals' welfare requirements with aesthetic and educational considerations.

AZA Accreditation Standard

(1.5.2) All animals must be housed in enclosures which are safe for the animals and meet their physical and psychological needs.

AZA Accreditation Standard

(1.5.2.1) All animals must be kept in appropriate groupings which meet their social and welfare needs.

AZA Accreditation Standard

(1.5.2.2) All animals should be provided the opportunity to choose among a variety of conditions within their environment.

Social needs: Capybaras are social animals that should be kept in groups ranging from neonate to geriatric individuals (Moreira et al., 2013). In smaller social groups, large age gaps require more monitoring, as some older animals can be intolerant of younger individuals. In nature, social groups contain more females than males, because when males reach sexual maturity the dominant males expel them from the group. Males should not be housed together, especially if females are nearby, as this will increase the probability of fighting. Introductions should be carried out very carefully, as bonded adults may attack or kill strange adults (see Chapter 5.3 for more on introductions).

Enclosures for isolated/separated individuals require standards similar to those for primary enclosures because individuals may become easily stressed if housed in small areas. Use caution when deciding to separate an individual, as it is very difficult to reintroduce capybaras. Separated individuals may be housed outdoors providing there is adequate shelter from extreme weather. It is not necessary to separate parturient females; provide them instead with sufficient space within the enclosure, such as access to a secluded area away from other animals, in order to protect infants from other members of the group during birthing (Alvarez & Kravetz, 2006).

Exhibit space: Exhibit size varies based on the size of the herd and other species housed in the enclosure. The minimum exhibit space recorded in an unpublished 2014 survey of AZA-accredited institutions that house capybaras was 95 m^2 (1023 ft^2) which sufficiently held one nonbreeding pair of capybaras, while the maximum was 15,933 m^2 (171,501 ft^2) for a mixed species exhibit containing 8 other animals including 4 mammals and 4 birds. The median exhibit size was 523 m^2 (5,630 ft^2).

Based on the 2014 survey, capybaras are able to stay on exhibit 24 hours a day as they are not considered dangerous animals. Facilities that are located in areas with extreme temperature changes by season may choose to seasonally leave their capybaras on exhibit or bring them into a holding space overnight. Additionally, because capybaras are not capable of climbing, exhibits do not need to be fully enclosed. When capybaras are housed in fully enclosed exhibits, these exhibits usually have a mixed species composition with fully flighted birds or species that have the ability to climb.

Figure 1. Example of a capybara exhibit
Photo courtesy of T. Schoffner

Exhibit substrate: Enclosures should contain natural substrates including mulch, sod, hay, leaf litter, river rock, straw, gravel, sand, and flagstone. Substrates and nesting/bedding material such as hay, pine straw, soil, grass, or other materials that can be easily removed and replaced but not ingested are appropriate and should be provided daily. The entire exhibit does not need to have thick substrate, just localized spots containing softer substrate for resting and sleeping. These spots are

important for joint health and footpad quality, especially if there are geriatric animals within the enclosure (Morgan & Tromborg, 2007).

Exhibit cleaning: Capybaras normally scent mark on trees and rocks around their living spaces, and these may need periodic cleaning or replacement. It is typically more efficient to replace logs and plants than to clean them. Enclosures should be cleaned, with wet or soiled bedding replaced, on a daily basis. Capybaras defecate in water, so artificial pools without recirculating, filtered water should be cleaned and disinfected at least once a week. Natural pools need to have filtration for the solid waste, especially if the water source is shared with other habitats.

Enclosure complexity: Capybaras do not typically show signs of stress due to the presence of visitors; therefore, it is not necessary to house capybaras behind glass or a great distance away from visitors. However, capybaras are flight animals and should be provided with a hide for cover; this may necessitate that more than one area is available within the enclosure. Examples of hides include, but are not limited to, wooden shelters, caves, access to a holding facility, vegetation, rocks, logs, bamboo, and waterfalls. The addition of rocks, fallen logs, trees, bushes, and leaves within the exhibit encourages increased activity levels for all group members. Exhibits should be well planted to give individuals visual barriers from each other.

Capybaras rest through the hot portion of the day and nap on and off throughout the night. Rest areas should be sheltered and well bedded. Plants commonly used include palms, tall grasses, and trees and shrubs native to the area. If artificial vining is necessary, capybaras should be monitored for ingestion.

Injuries caused by enclosure materials are rare, but superficial wounds and teeth damage may be associated with chain link fences.

Figure 2. Capybara in water feature
Photo courtesy of C. Atherton

Figure 3. Capybaras near water feature
Photo courtesy of C. Atherton

Water sources: Capybaras should have frequent access to bodies of water since a significant amount of their behavior, including feeding, mating, escape, and hiding revolves around water. Pools should be provided in both indoor and outdoor enclosures. The median pool size in AZA capybara exhibits is 88 m^2 (950 ft^2), with a minimum size of 14 m^2 (150 ft^2) and a maximum size of 1,214 m^2 (13,068 ft^2). The water in the exhibits can be provided as a pool, a pond, a river, or a lagoon. Pools may be as shallow as 1.1 m (3.5 ft.), however, the norm is at least 1.8 m (6 ft.), including a gradual incline. The AZA Rodent, Insectivore and Lagomorph TAG has not yet set minimum guidelines for space—these dimensions were provided from the results of an unpublished 2014 AZA survey.

AZA Accreditation Standard

(10.3.3) All animal enclosures (exhibits, holding areas, hospital, and quarantine/isolation) must be of a size and complexity sufficient to provide for the animal's physical, social, and psychological well-being. AZA housing guidelines outlined in the Animal Care Manuals should be followed.

AZA Accreditation Standard

(10.3.4) When sunlight is likely to cause overheating of or discomfort to the animals, sufficient shade (in addition to shelter structures) must be provided by natural or artificial means to allow all animals kept outdoors to protect themselves from direct sunlight.

The same careful consideration regarding exhibit size and complexity and its relationship to the capybaras' overall well-being must be given to the design and size of all enclosures, including those used in exhibits, holding areas, hospital, and quarantine/isolation (AZA Accreditation Standard 10.3.3). Sufficient shade must be provided by natural or artificial means when sunlight is likely to cause overheating or discomfort to the animals (AZA Accreditation Standard 10.3.4).

2.2 Safety and Containment

Animals housed in free-ranging environments should be carefully selected, monitored and treated humanely so that the safety of these animals and persons viewing them is ensured (AZA Accreditation Standard 11.3.3).

Although capybaras are usually docile creatures, they are large and relatively quick (35 km/hr. (22 mph)) and have 7.6 cm (3 in.) long incisors. If scared, they can seriously injure keepers or veterinary staff. Capybaras can be managed with a light touch; with reinforcement, many learn routine calls and commands. However, caution should be maintained when pups are present or animal introductions are taking place. Caretakers should pay close attention to the warning calls and barks given by the animals and be particularly aware of the location of pups. When restraint of individuals is necessary, it is preferable to separate the individuals from the herd to prevent counter attacks from group members, particularly if pups are being removed. Individual capybaras have different degrees of safe distance; it is up to the keepers to determine what is comfortable for their animals.

Animal exhibits and holding areas in all AZA-accredited institutions must be secured to prevent unintentional animal egress (AZA Accreditation Standard 11.3.1). All animal exhibit and holding area air and water inflows and outflows must also be securely protected to prevent animal injury or egress (AZA Accreditation standard 1.5.15). Pest control methods must be administered so there is no threat to the animals, staff, public, and wildlife (AZA Accreditation Standard 2.8.1). Exhibit design must be considered carefully to ensure that all areas are secure and particular attention must be given to shift doors, gates, keeper access doors, locking mechanisms, and exhibit barrier dimensions and construction.

Exhibits in which the visiting public is not intended to have contact with animals must have a barrier of sufficient strength and/or design to deter such contact (AZA Accreditation Standard 11.3.6).

There are currently no known risks associated with housing capybaras in un-topped exhibits. Capybaras are not capable of climbing or jumping high/long distances so a barrier of appropriate height, approximately 1.8 m (6 ft.), should suffice. Barriers should be composed of materials that allow a clear path of vision without posing a health risk to the animal. The barrier should keep native fauna out of the exhibit; however, birds are not a threat to capybaras on exhibit (Moreira et al., 2013). Glass barriers should be between 14-20ml and wire mesh should be at least 8 gauge.

All emergency safety procedures must be clearly written, provided to appropriate paid and unpaid staff, and readily available for reference in the event of an actual emergency (AZA Accreditation Standard 11.2.4).

AZA Accreditation Standard

(11.3.3) Special attention must be given to free-ranging animals so that no undue threat is posed to either the institution's animals, free-ranging animals, or the visiting public. Animals maintained where they will be in contact with the visiting public must be carefully monitored, and treated humanely at all times.

AZA Accreditation Standard

(11.3.1) All animal exhibits and holding areas must be secured to prevent unintentional animal egress.

AZA Accreditation Standard

(1.5.15) All animal exhibit and holding area air and water inflows and outflows must be securely protected to prevent animal injury or egress.

AZA Accreditation Standard

(2.8.1) Pest control management programs must be administered in such a manner that the animals, paid and unpaid staff, the public, and wildlife are not threatened by the pests, contamination from pests, or the control methods used.

AZA Accreditation Standard

(11.3.6) There must be barriers in place (for example, guardrails, fences, walls, etc.) of sufficient strength and/or design to deter public entry into animal exhibits or holding areas, and to deter public contact with animals in all areas where such contact is not intended.

AZA Accreditation Standard

(11.2.4) All emergency procedures must be written and provided to appropriate paid and unpaid staff. Appropriate emergency procedures must be readily available for reference in the event of an actual emergency.

Depending on the nature of the incident, different emergency plans are recommended. Institutions should be prepared for, and have plans for, any natural or human-caused emergencies. For example, in case of fire, caretakers should either remove the capybaras from the area or lock them out on exhibit where they can remove themselves from the situation. It is the responsibility of each institution to be prepared for relevant emergencies. Staff training for emergencies must be undertaken and records of such training maintained. Security personnel must be trained to handle all emergencies in full accordance

with the policies and procedures of the institution and in some cases, may be in charge of the respective emergency (AZA Accreditation Standard 11.6.2).

Capybaras are not designated as dangerous animals, so specific guidelines for emergencies are not necessary beyond the facility's general policies, including those for visitor intrusion and animal escape and health emergencies. In the event of a medical emergency, the proper procedure should include notifying curators and veterinary staff (e.g., initiating a phone tree if after hours).

Emergency drills must be conducted annually for each basic type of emergency to ensure all staff is aware of emergency procedures and to identify potential problematic areas that may require adjustment. These drills must be recorded and results evaluated for compliance with emergency procedures, efficacy of paid/unpaid staff training, aspects of the emergency response that are deemed adequate are reinforced, and those requiring improvement are identified and modified (AZA Accreditation Standard 11.2.5). AZA-accredited institutions must have a communication system that can be quickly accessed in case of an emergency (AZA Accreditation Standard 11.2.6). A paid staff member or a committee must be designated as responsible for ensuring that all required emergency drills are conducted, recorded, and evaluated in accordance with AZA accreditation standards (AZA Accreditation Standard 11.2.0).

AZA-accredited institutions must also ensure that written protocols define how and when local police or other emergency agencies are contacted and specify response times to emergencies (AZA Accreditation Standard 11.2.7). Animal attack emergency response procedures must be defined and personnel must be trained.

Capybaras are not considered dangerous, so police are not necessary in case of an on grounds escape. Instead, staff should be trained to corral the animals. If an animal should get off grounds, every effort for recapture should be made. Institutions should adhere to all local ordinances and work with local law enforcement as necessary.

AZA Accreditation Standard

(11.6.2) Security personnel, whether employed by the institution, or a provided and/or contracted service, must be trained to handle all emergencies in full accordance with the policies and procedures of the institution. In some cases, it is recognized that Security personnel may be in charge of the respective emergency (i.e. shooting teams).

AZA Accreditation Standard

(11.2.5) Live-action emergency drills (functional exercises) must be conducted at least once annually for each of the four basic types of emergency (fire; weather or other environmental emergency appropriate to the region; injury to visitor or paid/unpaid staff; and animal escape). Four separate drills are required. These drills must be recorded and results evaluated for compliance with emergency procedures, efficacy of paid/unpaid staff training, aspects of the emergency response that are deemed adequate are reinforced, and those requiring improvement are identified and modified. (See 11.5.2 and 11.7.4 for other required drills).

AZA Accreditation Standard

(11.2.6) The institution must have a communication system that can be quickly accessed in case of an emergency.

AZA Accreditation Standard

(11.2.0) A paid staff member or a committee must be designated as responsible for ensuring that all required emergency drills are conducted, recorded, and evaluated in accordance with AZA accreditation standards (see 11.2.5, 11.5.2, and 11.7.4).

AZA Accreditation Standard

(11.2.7) A written protocol should be developed involving local police or other emergency agencies and include response times to emergencies.

Chapter 3. Records

3.1 Definitions

In the zoo and aquarium world, animal records are defined as "data, regardless of physical form or medium, providing information about individual animals, samples or parts thereof, or groups of animals". Capybara data should be recorded for each individual animal.

3.2 Types of Records

There are many types of records kept for the animals in our care, including but not limited to, veterinary, husbandry, behavior, enrichment (see section 9.2), nutrition, and collection management. These types of records may be kept as separate records, as logs in separate locations, or as part of the collection records and some may be required by regulatory agencies (e.g., primate enrichment records) or per AZA Accreditation Standards (e.g., emergency drill records).

Recordkeeping is an important element of animal care and ensures that information about individual animals or groups of animals is always available. The institution must show evidence of having a zoological records management program for managing animal records, veterinary records, and other relevant information (AZA Accreditation Standard 1.4.0). These records contain important information about an individual animal or group of animals, including but not limited to taxonomic name, transaction history, parentage, identifiers, gender, weights, enclosure locations and moves, and reproductive status (see Appendix C for Guidelines for Creating and Sharing Animal and Collection Records). It is helpful to have as much reproductive information as possible, specifically if an animal receives any contraceptive measures. Dietary information is also helpful to share. Including diet offered and preferred items animal consumes. This is helpful for behavioral husbandry reinforcement troubleshooting.

Individual husbandry records must be maintained, even though capybaras are generally housed in herds. Records should be kept electronically on databases such as ZIMS and TRACKS.

A designated paid staff member must be responsible for maintaining the animal record-keeping system and for conveying relevant laws and regulations to the animal care staff (AZA Accreditation Standard 1.4.6). Animal and veterinary records must be accurate and current (AZA Accreditation Standard 1.4.7). Complete and up-to-date animal and veterinary records must be duplicated and stored at a separate location (AZA Accreditation Standard 1.4.4) and at least one set of historical records safely stored and protected (AZA Accreditation Standard 1.4.5).

AZA member institutions must inventory their capybara population at least annually and document all capybara acquisitions, transfers, euthanasias, releases, and reintroductions (AZA Accreditation Standard 1.4.1). All capybaras owned by an AZA institution must be listed on the inventory, including those animals on loan to and from the institution (AZA Accreditation Standard 1.4.2). All AZA-accredited institutions must abide by the AZA Policy on Responsible Population Management (Appendix D) and the long-term welfare of animals should be considered in all acquisition, transfer, and transition decisions. For an example

AZA Accreditation Standard

(1.4.0) The institution must show evidence of having a zoological records management program for managing animal records, veterinary records, and other relevant information.

AZA Accreditation Standard

(1.4.6) A paid staff member must be designated as being responsible for the institution's animal record-keeping system. That person must be charged with establishing and maintaining the institution's animal records, as well as with keeping all paid and unpaid animal care staff members apprised of relevant laws and regulations regarding the institution's animals.

AZA Accreditation Standard

(1.4.7) Animal and veterinary records must be kept current.

AZA Accreditation Standard

(1.4.4) Animal records and veterinary records, whether in electronic or paper form, must be duplicated and stored in a separate location. Animal records are defined as data, regardless of physical form or medium, providing information about individual animals, or samples or parts thereof, or groups of animals.

AZA Accreditation Standard

(1.4.5) At least one set of the institution's historical animal and veterinary records must be stored and protected. Those records should include permits, titles, declaration forms, and other pertinent information.

AZA Accreditation Standard

(1.4.1) An animal inventory must be compiled at least once a year and include data regarding acquisition, transfer, euthanasia, release, and reintroduction.

AZA Accreditation Standard

(1.4.2) All species owned by the institution must be listed on the inventory, including those animals on loan to and from the institution.

of an animal transfer data sheet, see Appendix I.

3.3 Permit Considerations

Capybara are not currently regulated by federal and/or state governments. They are currently listed as IUCN: Least Concern, CITES: Not Listed, and USFWS: Not Listed. It is important to always check the current requirements of the state sending and receiving capybara before transporting.

3.4 Government Ownership

Capybara are not government owned and do not require additional documentation.

3.5 Identification

Ensuring that capybaras are identifiable through various means increases the ability to care for individuals more effectively. All animals held at AZA facilities must be individually identifiable whenever practical, and have corresponding identification (ID) numbers. For example, capybaras may receive tattoos, ear tags, or transponders to ease identification. For animals maintained in colonies or groups, or other animals not considered readily identifiable, institutions must have a procedure for identification of and recording information about these groups or colonies. (AZA Accreditation Standard 1.4.3). These IDs should be included in specimen reports, collection and/or transaction records and veterinary records. Types of identifiers include:

AZA Accreditation Standard

(1.4.3) Animals must be identifiable, whenever practical, and have corresponding ID numbers. For animals maintained in colonies/groups or other animals not considered readily identifiable, the institution must provide a statement explaining how record keeping is maintained.

Physical identifier: These include, but are not limited to, ear notching, tattoos, microchips/transponder and RFID devices and obvious visual differences (gender, size, etc.). Permanent physical identifiers are often used to distinguish separate animals in studbooks. Some common capybara identifiers are plastic or metal ear clips, ear notching, tags, and transponders. Please note tags or clips placed in ears are often lost as capybara ears are small and thin and tear easily. Transponders and/or tattoos are more reliable for long term identification. These are usually assigned during the first exam of the pups. Due to the thickness of capybara skin, transponders are generally placed surgically between the shoulder blades.

Intangible identifiers (called ‘logical identifiers’ in the Zoological Information Management System [ZIMS]): These include, but are not limited to, institutional accession number, house name, public name, studbook number, and ZIMS Global Accession Number. All of the mentioned methods are useful for identifying individual animals. Due to large litters, it is best to assign these identifiers soon after birth. Temporary markers such as coloring the nails or fur are best until animals can be permanently identified.

Chapter 4. Transport

4.1 Preparations

Animal transportation must be conducted in a manner that adheres to all laws, is safe, and minimizes risk to the animal(s), employees, and general public (AZA Accreditation Standard 1.5.11). All temporary, seasonal, and traveling live animal exhibits must meet the same accreditation standards as the institution's permanent resident animals, with foremost attention to animal welfare considerations (AZA Accreditation Standard 1.5.10). Safe animal transport requires the use of appropriate conveyance and equipment that is in good working order. Include copies of appropriate permits and authorizations in transport documentation. If the animal is not owned by the shipping institution, permission is to be obtained from the owner well in advance of the move.

AZA Accreditation Standard

(1.5.11) Animal transportation must be conducted in a manner that is safe, well-planned and coordinated, and minimizes risk to the animal(s), employees, and general public. All applicable laws and/or regulations must be adhered to.

AZA Accreditation Standard

(1.5.10) Temporary, seasonal and traveling live animal exhibits, programs, or presentations (regardless of ownership or contractual arrangements) must be maintained at the same level of care as the institution's permanent resident animals, with foremost attention to animal welfare considerations, both onsite and at the location where the animals are permanently housed.

Before the transfer of a capybara from one facility to another, an Animal Data Transfer Form (ADTF) should be completed and sent to the receiving institution. A Certificate of Veterinary Inspection should accompany the animal stating that it is healthy and ready to be put into quarantine at the new facility. The ADTF should have detailed information about the age, sex, and behavioral husbandry profile of the animal. Pedigree information is available in the studbook and a full medical history would have been received and reviewed before the animal is moved.

When transporting the capybara domestically each state may require different transit permits, including entry permits, as a result individual states should be contacted before transport. If the animal is crossing international borders, International shipment also requires inspection by US Fish and Wildlife Service and a Declaration (Form 3-177), use of a designated port, Customs declarations, international veterinary certification, origin documentation, etc. If the animal is being flown, the International Air Transport Association (IATA) Live Animal Regulations should be consulted for crate requirements in addition to all necessary permits.

The equipment must provide for the adequate containment, life support, comfort, temperature control, food/water, and safety of the animal(s). Equipment necessary for the transport of capybaras includes a crate with a locking mechanism, food/water, and barriers/push boards to be used in the event that the capybara does not shift into the crate (CITES, 2013). Additionally, the vehicle itself should be able to protect the capybara from weather. All equipment should be inspected for rough or sharp edges, holes in the wires, and anything wrong with the crate.

Capybaras need to be crated for the duration of transport. The crate should be maneuvered in a manner to avoid physical trauma to the animal. The transit crate should be secured in the vehicle with a non-slip surface on the floor. Wooden crates can be used, and should be lined with welded mesh, sheet metal, rigid plastic, or wire mesh. Crates should also have the joints bolted or screwed together. The entire interior should be lined to deter the animal from gnawing. Ventilation openings lined with wire mesh should be 2.5 cm (1 in.) in diameter and present on the sides and roof of the crate. The diameter of the mesh covering the ventilation openings should not permit the snout or feet of the animal to protrude. The floor should be solid metal and leak-proof. A sliding door should be provided at the front and rear of the crate to allow access. The front door should have a narrow gap of 2.5 cm (1 in.) at the bottom of the welded mesh so that food and water may be passed into the crate. The welded mesh should be braced on the inside of the bottom of the crate. All doors should be provided with secure fasteners to prevent accidental opening. In order to avoid injury during transport, the crate should be large enough that the animal can freely stand up, lie down, and be able to turn around completely. Crate size will likely depend on maturity (males and females being approximately the same size), so juveniles may require a smaller crate. Please check current state regulations and IATA guidelines before transporting a capybara as these are subject to change.

Round-edged food and water containers should be provided. The crate's floor should be covered with a non-slip material and a thick layer of absorbent material, such as wood shavings for domestic moves and shredded paper for international transport, for bedding and absorption of urine. No additional tray is needed for transport to remove feces and urine.

Documents needed for transit should be attached to the outside of the crate and easily accessible. The crate should have clear “this side up” signage on the outside, on all four sides, to indicate the correct orientation. The crate should also have “Live Animal” stickers, especially for international shipments.

Safe transport also requires the assignment of an adequate number (two to four) of appropriately trained personnel (from the institution or by contract) who are equipped and prepared to handle contingencies and/or emergencies that may occur in the course of transport if a long flight or drive is anticipated. Domestic transport does not usually require this. Planning and coordination for animal transport requires good communication among all affected parties, plans for a variety of emergencies and contingencies that may arise, and timely execution of the transport. At no time should the animal(s) or people be subjected to unnecessary risk or danger (AZA Accreditation Standard 1.5.11).

Transport should not occur if the animal: is more than 135 days through the gestation period, gave birth the previous week, is a newborn whose navel healing is not yet complete, or is too sick or injured to move on its own. Veterinarians can waive these restrictions if deemed appropriate (CITES, 2013). Animals should not be sedated unless absolutely necessary and sedation should be supervised by a veterinarian.

The animal should be accompanied by an individual deemed competent in the following:

- Knowledge of transport regulations, as applicable
- Knowledge of animal health and welfare regulations, and document requirements applicable to the countries of origin, transit, and destination
- Knowledge of the handling and care of animals before, during, and after loading/unloading and transport
- Ability to recognize an animal which is ill or becomes unfit for transport
- Ability to recognize signs of stress and their causes, and how to reduce these
- Ability to handle emergency situations.

There have never been any known emergencies when transporting capybaras, but contingency plans should be made in case the worst happens. Alternative routes should be planned in case of inclement weather, road closures, and traffic bans to avoid unnecessary time on the road. Be aware of the locations of repair and refueling facilities in case of a vehicle breakdown. It is also important to have contact information for the proper authorities and other facilities en route in case an escape occurs or overnight stops are necessary. Additionally, the new facility should be provided with an estimated time of arrival in order to prepare for the new animal.

4.2 Protocols

Transport protocols should be well defined and clear to all animal care staff. Capybaras should be transported in individual crates and the transfer should be completed within 24–48 hours. The vehicle the animal is traveling in should have climate control to provide the appropriate temperature (8–27 °C (46–81 °F)) and low lighting that allows the animal to remain calm. The most effective and safest way to prepare a capybara for transport is to train the animal to enter the crate of its own volition. Animals do not usually require additional food or water during the first 24 hours following the time of dispatch, but if feeding becomes necessary due to an unforeseen delay or extended travel, fruit or grain should be offered. Detailed feeding instructions should be attached to the outside of the crate, in addition to paperwork relevant to the transport.

Once the animal has arrived at its destination, it should be brought directly to the quarantine location. Due to its size, a capybara should not be removed from the crate without a veterinarian present who can provide medical assistance to the animal should it become necessary. The animal should be released into a quarantine facility suitable for normal activity. The door of the shipping crate can be opened and the crate can remain in the enclosure, allowing the capybara to exit the crate on its own timetable. Some surveyed AZA-accredited institutions choose to leave the crate in the exhibit as an additional hiding space for the entire duration of the initial acclimation period.

Chapter 5. Social Environment

5.1 Group Structure and Size

Careful consideration should be given to ensure the social group size and structure meet the social, physical, and psychological well-being of the animals and facilitate species-appropriate behaviors.

Typical capybara herds are composed of one dominant male (subordinate males are peripheral members of the herd), several adult females, and their young. A herd's social order is normally very stable and the dominant position rarely changes. As a result, unfamiliar individuals are often attacked. The ideal social structure, in terms of age and sex, would result from developing a herd with a founder male and one or more females of approximately the same age. These individuals would then naturally reproduce to reach a herd size appropriate for the exhibit space (Herrera & McDonald, 1987). As such, the ideal number of capybaras at an institution will vary by exhibit size and institutional capabilities.

To prevent inbreeding and aggression, subordinate males and young females should be placed at another facility or in a separate exhibit once they reach sexual maturity, although contraceptive measures, such as castration and/or melengestrol acetate (MGA) administration, can be taken in an effort to continue to house the group together. Bachelor groups of capybaras are not advised, as males become aggressive towards one another once they reach sexual maturity (Moreira et al., 2013). Capybaras, both males and females, reach sexual maturity at approximately 15 months of age in managed care facilities (Moreira et al., 2013).

5.2 Influence of Others and Conspecifics

Animals cared for by AZA-accredited institutions are often found residing with conspecifics, but may also be found residing with animals of other species. Capybaras, being a gregarious species, should be housed in a social setting either with conspecifics or with other species.

Figure 4. Capybara and monkey
Photo courtesy of D. Cossaboon

Conspecific interactions: In nature, capybaras reside in groups varying in size depending on the season. Herd size often increases in the dry season as many individual groups unify (Lord, 2009). Since basic requirements for survival are provided by caretakers, capybaras in managed facilities should not be housed in groups containing multiple males. The social structure of capybaras in nature points to a society with one dominant male that may become aggressive toward subordinates. Social dominance is frequently linked to scent. Males possess a raised area on the top of the snout that contains a greatly enlarged sebaceous gland. The proportions of each chemical present in the secretions of individual capybaras are different, providing a potential for individual recognition via personal "olfactory fingerprints" (Macdonald et al., 1984). They also have two glandular pockets located on either side of the anus. Male anal glands are filled with easily detachable hairs abundantly coated with hard crystalline calcium salt (Macdonald et al., 1984). The dominance hierarchy of males tends to be extremely well defined by results of interactions between dyads, while there is no evidence of dominance relations among females (Herrera & MacDonald, 1994). Capybara hierarchy is strictly linear and results from stereotyped agonistic interactions; in instances where the removal of the dominant male occurs, the top-ranking subordinate male will take the dominant position (Herrera et al., 2011). If housed in groups that contain multiple males, dominant males will frequently interrupt courtships and potentially monopolize females in the group (Herrera et al., 2011). The complete social context of scent marking has not been described in detail and its function is unknown.

Conversely, group living for females could prove beneficial for multiple reasons including communal nursing and the formation of nursery groups, which is likely a form of protection from predators (in nature) and infanticidal males (a potential reality in a managed care facility) (Herrera et al., 2011).

Socially, capybaras tend to be gregarious and often vocalize using seven distinct vocalizations (Nogueira et al., 2012). The click call is used most often during aggregating and foraging behaviors, which suggests its use as a contact or monitoring call function aiding individuals in remaining in contact with their group (Nogueira et al., 2012). In addition, several other calls have been identified including barks, whistles, and chuckles. Each call appears to serve an independent function; barks indicate a predator, whistles are associated with infants calling to their mothers, and chuckles are associated with traveling together (Barros et al., 2011). The interactions mediated by vocalization suggest that species communication is an important mechanism to regulate social encounters and to alert members of the group about environmental cues, such as predator risk and juvenile isolation (Barros et al., 2011).

Interspecific interactions: Interspecies exhibits will provide a more naturalistic setting. Since capybaras live exclusively on the ground, they can cohabitate with many harmless arboreal species. Capybaras can also be easily housed with other ground-dwelling species. A list of species that are appropriate for housing with capybaras can be found in Table 3.

Mixed species exhibits require additional management considerations, specifically regarding feeding strategies. Potential strategies for managing the feeding of diets, so that each animal ingests only the appropriate diet, include: multiple feeding stations in the exhibit, utilizing holding spaces, separating species overnight, and providing diets off the ground for arboreal species. Additional management is also advised for monitoring interspecies interactions. In exhibits that contain multiple species of animals, it is important to note that a capybara territory is defended by all adult members and aggression may be exhibited by both males and females, particularly if neonates are present (Herrera & Macdonald, 1994).

Table 3. Possible species for mixed species exhibits

Species Name	Scientific Name
Agouti	*Dasyproctidae*
Bahama Pintail Duck	*Anas bahamensis*
Baird's Tapir	*Tapirus bairdii*
Black-Bellied Whistling Duck	*Dendrocygna autumnalis*
Brocket Deer	*Mazama spp.*
Brown Water Snake	*Nerodia taxispilota*
Bullfrog	*Lithobates catesbeianus*
Coscoroba Swan	*Coscoroba coscoroba*
Crested Screamer	*Chauna torquata*
Domestic Chicken	*Gallus gallus domesticus*
Domestic Rabbit	*Oryctolagus cuniculus*
Giant Anteater	*Myrmecophaga tridactyla*
Guanaco	*Lama guanicoe*
Howler Monkey	*Alouatta spp.*
King Vulture	*Sarcoramphus papa*
Llama	*Lama glama*
Lowland Tapir	*Tapirus terrestris*
Muscovy Duck	*Cairina moschata*
Mute Swan	*Cygnus olor*
Nene Goose	*Branta sandvicensis*
Ocellated Turkey	*Meleagris ocellata*
Orinoco Goose	*Neochen jubata*
Patagonian Cavy	*Dolichotis patagonum*

Prehensile-Tailed Porcupine	*Coendou spp.*
Red Eared Slider	*Trachemys scripta elegans*
Rhea	*Rhea spp.*
Rosy-Billed Pochard	*Netta peposaca*
Sandhill Crane	*Grus canadensis*
Scarlet Macaw	*Ara macao*
White-Faced Saki Monkey	*Pithecia pithecia*
White Ibis	*Eudocimus albus*
Wild Mallard	*Anas platyrhynchos*

5.3 Introductions and Reintroductions

Managed care for and reproduction of animals housed in AZA-accredited institutions are dynamic processes. Animals born in or moved between and within institutions require introduction and sometimes reintroductions to other animals. It is important that all introductions are conducted in a manner that is safe for all animals and caretakers involved.

As with any animal introduction, extreme caution is recommended. All new capybaras should be introduced to the facility as expressed in institutional standard operating procedures (SOPs). The current and most successful method involves introducing all members of the herd, including the new animal(s), should be simultaneously introduced to a neutral space. This can be done in the existing exhibit as long as all furniture, enrichment items, food and water bowls, and substrates are replaced. Before placing animals into the new exhibit, enrichment, visual barriers, preferred food items, etc. should be put in the exhibit. Then, all animals should be moved into the exhibit together. The animals should be able to develop compatible relationships while exploring the new territory. Initially, the new member(s) may be chased, but this should be short-lived.

If not all individuals can be introduced in the above manner, then introductions to new social groups can occur in a variety of ways. The most popular methods for introductions include using howdy cages or chain link dividers between stalls, introducing scents to all individuals involved by sharing exhibit furniture and enrichment items between the established group and the new member(s), and using unfamiliar enclosures. If the group shows no signs of accepting the new member(s) of the group, animal care staff should remove the new member(s) and attempt to make alternative housing arrangements, including contacting the SSP coordinator to rehome the animal(s).

Chapter 6. Nutrition

6.1 Nutritional Requirements

A formal nutrition program is required to meet the nutritional and behavioral needs of all capybaras (AZA Accreditation Standard 2.6.2). Diets should be developed using the recommendations of nutritionists, the AZA Nutrition Scientific Advisory Group (NAG) feeding guidelines: (https://nagonline.net/guidelines-aza-institutions/feeding-guidelines/), and veterinarians as well as AZA Taxon Advisory Groups (TAGs), and Species Survival Plan® (SSP) Programs. Diet formulation criteria should address the animal's nutritional needs, feeding ecology, as well as individual and natural histories to ensure that species-specific feeding patterns and behaviors are stimulated.

AZA Accreditation Standard

(2.6.2) The institution must follow a written nutrition program that meets the behavioral and nutritional needs of all species, individuals, and colonies/groups in the institution. Animal diets must be of a quality and quantity suitable for each animal's nutritional and psychological needs.

The AZA Rodent, Insectivore and Lagomorph TAG Nutrition Advisor is Michael Maslanka (maslankam@si.edu).

Dentition: The dentition of the capybara is typical of rodents with a large gap (diastema) between the incisors and the molars. In Figure 5, note the large incisors for ripping and cutting grasses and the molars designed for crushing forage as they chew. The incisors are continually growing; therefore, this species needs a complex plant-based diet that must be chewed in order to avoid the problems that result from over-grown teeth. The dentition pattern is incisors 1/1, canines 0/0, premolars 1/1, and molars 3/3 (Mones & Ojasti, 1986).

Figure 5. Capybara skull: rodent, herbivore
Photo courtesy of Skulls Unlimited

Nutritional niche and wild feeding ecology: The capybara is an herbivorous species indigenous to South America (Baldizán, Dixon, & Parra, 1983).

The wild diet of a capybara varies from the dry season (where food availability and quality are reduced) to the wet season (where there are abundant resources) and can include herbaceous plants, leaves, aquatic plants (reeds, water hyacinths), grasses, and crops (fruits, grains, seeds, vegetables). Although capybaras are classified as grazers, they selectively eat vegetation high in protein and energy whereas most grazers typically ingest diets low in protein and energy (Barreto & Herrera, 1998). Therefore, despite their grazer classification, capybaras' wild feeding ecology dictates that they should be fed as a browsing species in managed care.

During the rainy season when forage is abundant, capybaras selectively consume a large variety of plants (89 species) but 94% of the plants belong to Poaceae and Pontederiaceae (flowering plants), Cyperaceae (sedges), and Leguminosae (legumes) (Forero-Montaña, Bentancur, & Cavelier, 2003). Capybaras most often consume grasses and 60% of the grasses consist of Dahl grass (*Hymenachne amplexicauli*), crabgrass (*Digitaria bicornis*), and guinea grass (*Panicum maximum*) (Forero-Montaña, Bentancur, & Cavelier, 2003).

Table 4. Composition of some foods eaten by the capybara

Plant	Crude Protein (DM) %	Crude Fiber (DM) %	Crude Fat (DM) %	Neutral Detergent Fiber (NDF) %	Acid Detergent Fiber (ADF) %	Lignin %
Aquatic Plants						
Water hyacinth (*Eichhornia crassipes*)	22.6[1] 15.3[2] - 22.4[3]	15.2[2] 5.9[3]	1.6[2] 7.1[3]	58.0[7]	32.3[7]	8.6[7]
Waterlily (*Nymphaea sp.*)	21.8[7]	21.0[7]	3.5[7]	-	-	-

Plant	Crude Protein (DM) %	Crude Fiber (DM) %	Crude Fat (DM) %	Neutral Detergent Fiber (NDF) %	Acid Detergent Fiber (ADF) %	Lignin %
Waterpoppy (*Nymphoides sp.*)	21.9[7]	21.0[7]	3.5[7]	-	-	-
Water spinach (*Ipomoea aquatica*)	6.3[8]	17.7[8]	11.0[8]	-	-	-
Waterweed (*Elodea sp.*)	13.7–32.7[9]	4.0–26.0[9]	8.0–11.3[9]	38.0–63.3[9]	20.0–51.3[9]	4.2–27.2[9]
Grasses						
Crown grass (*Paspalum sp.*)	5.8–12.8[5a]	28.3–35.5[5a]	1.0–2.7[5a]	-	-	-
Dahl (Dal) grass (*Hymenachne amplexicaulis*)	7.5[4]	29.2[4]	1.4[4]	-	-	-
Guinea grass (*Panicum maximum*)	5.5–11.9[10]	31.7–40.1[10]	1.8–3.2[10]	-	-	-
Marsh grass (*Hymenachne amplexicaulis*)	8.9–22.6[5a]	32.4–36.7[5a]	1.9–2.8[5a]	64.7[4]	34.4[4]	4.2[4]
Rice grass (*Leersia hexandra*)	6.3[6]	31.4[6]	1.5[6]	66.8[6]	36.9[6]	4.7[6]
Water grass (*Echinochloa sp.)*	9.2–10.3[5a]		2.0-3.8[5a]			

[1]Virabalin, Kositsup, & Punnapayak, 1993
[2]Okoye, Daddy, & Ilseanmi, 2002
[3]Hontiveros & Serrano, 2015
[4]Talapatra & Dev Goswami, 1949
[5]Camaro et al., 2004
[6]Talapatra, 1950
[7]Biobaku & Ekpenyong, 1991
[8]Umar et al., 2007
[9]Ekunseitan, Yusuf, & Odesanmi, 2013
[10]Aganga & Tshwenyane, 2004
[a]Variable depending on whether stem, leaf, or flower

Gastrointestinal tract (GIT): Capybaras, as large hindgut fermenters, have the digestive efficiency of ruminant species when fed similar diets (Borges, Dominguez-Bello, & Herrera, 1996). The illustration in Figure 6 of the gastrointestinal tract (GIT) of the capybara shows a complex stomach (complex because of pouches) with an extremely complex (sacculated) cecum with four sections of smooth muscle (taenia) (Stevens & Hume, 2004; Vazquez, Senos, & William, 2012). This GIT complexity increases the area in the stomach and cecum that allows for the fermentation of ingested foods. The fermentation process is important to support a GIT bacterial population essential to capybara health.

GIT fermentation is also important for the digestion of carbohydrates, the synthesis of vitamins B and K, and for the production of short-chain fatty acids known as volatile fatty acids (VFAs) that provide energy. The major volatile fatty acids produced are acetic, proprionic, and butyric acids. A healthy GIT bacterial population for a capybara should also include cellulolytic and non-cellulolytic bacteria that enable them to digest the cellulose found in some foods. High concentrations of both bacteria and protozoa should be present in the caecum and colon (Borges, Dominguez-Bello, & Herrera, 1996).

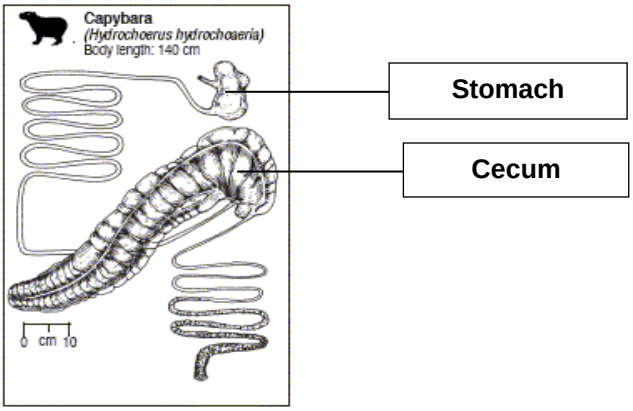

Figure 6. Capybara gastrointestinal tract (GIT). Herbivore, cecum (hindgut) fermenter
(Stevens & Hume, 1995)

Coprophagy: Capybaras are coprophagic; as such, they produce two types of feces when fed an appropriate diet (Borges, Dominguez-Bello, & Herrera, 1996; Vazquez, Senos, & William, 2012). The cecotrope is one type of feces and consists of matter formed by microbial digestion in the cecum. These feces are re-ingested by the animal after excretion. The other type of feces is digestive waste. Cecotrope pellets are soft, clear in color, and are usually produced early in the day (Borges, Dominguez-Bello, & Herrera, 1996; Mendes et al., 2000). Waste feces should be oval, firm, and green/brown in color (Borges, Dominguez-Bello, & Herrera, 1996; Mendes et al., 2000). As with most herbivores, if the fecal pellets are watery, then the diet of the animal does not provide enough dietary fiber and/or the correct fiber ratios.

The production and ingestion of cecotropes in species that normally practice cecotrophy is an indicator of a healthy GIT system populated by microbial species supported by an appropriate diet. Cecotropes provide dietary protein, B vitamins, and vitamin K. Cecotropes have 37% higher protein than waste feces (Mendes et al., 2000). Capybaras also can regurgitate food to chew it and this is normally done while resting (Lord, 1994).

6.2 Diets

The formulation, preparation, and delivery of all diets must be of a quality and quantity suitable to meet the animal's nutritional and psychological needs (AZA Accreditation Standard 2.6.2). Food should be purchased from reliable, sustainable, and well-managed sources. The nutritional analysis of the food should be regularly tested and recorded.

Nutritional problems related to managed care diets: Most diet-related health problems for capybaras are caused by diets too low in fiber and too high in simple carbohydrates like starch and sugar. As with most herbivorous species, simple dietary carbohydrates disrupt the cecal and gut microbial population. Some of the diet-related health problems (nutritional pathology) include dental caries, gastritis, malocclusion (over-grown teeth), obesity, colitis, cecitis and Type II diabetes. Furthermore, capybaras require a dietary source of vitamin C and can develop scurvy if the diet is vitamin C deficient (Cueto, Allekotte, & Kravetz, 2000).

The GIT microorganisms of herbivorous species—including the capybara—produce B vitamins, vitamin K, microbial proteins, and volatile fatty acids (VFAs). VFAs provide as much as 60–90% of the energy needed by herbivorous species. These gut microbes require a stable and specific GIT pH environment, in addition to an appropriate food supply, to survive and provide nutrients to the host animal. If GIT microorganisms are not within an appropriate range, animals can develop vitamin deficiencies which lead to low gut motility and diarrhea (Grant 2014).

Dietary recommendations:

Nogueira-Filho et al. (2013) have made some relevant recommendations for the nutritional maintenance of capybaras:

<u>Energy requirement (BMR):</u> 65.5 Kcal/kg $LW^{0.75}$ (Ojasti, 1991)
<u>Protein requirement:</u> 1.56 g of digestible protein/kg $LW^{0.75}$.
<u>Calcium requirement:</u> 0.30 g Ca/kg $LW^{0.75}$
<u>Phosphorus requirement:</u> 0.15 g P/kg $LW^{0.75}$

For animals that are still growing or reproducing: 3,500 kcal/kg with 12% crude protein are needed.

Supplementation with vitamin C: Capybaras need a dietary source of vitamin C and should receive 1 gram of ascorbic acid per day (Cueto, Allekotte, & Kravetz, 2000). Fruits are a typical ingredient in most capybara diets as fruits are high in vitamin C. However, fruit—especially our cultivated fruits—are high in sugars that will disrupt the gut microbial population. Therefore, fruit should not be used as the sole means of providing vitamin C to managed capybaras; vitamin C may be added to the diet in powder or liquid form as a supplement. Vitamin C may also be obtained in vegetables, like broccoli, known to have high levels or in guinea pig pellets.

Diets high in fiber: The capybara GIT requires a diet high in fiber to support fermentation and a healthy microbial population. Considering that an adult capybara should consume approximately 2,500 kilocalories (kcal) daily, and these kilocalories should come from foods that are high fiber but are relatively low in kilocalories, the capybara should occupy much of its day eating and chewing.

Fruits and vegetables: Fruits and vegetables should comprise less than 5% of the capybara diet. The fruit and vegetable varieties used should be low in sugars and vegetables should be emphasized over fruit. In other words, small amounts of such foods can be used for environmental enrichment and/or training purposes but should not be relied upon to comprise the bulk of the diet. Limiting fruits and vegetables will also encourage ingestion of appropriate foods such as pellets and forage (branches, hay) and will decrease dietary starch and sugars that can cause GIT inflammation, extreme fluctuations of GIT pH, and Type II diabetes. Fruits that have been fed to capybaras by AZA institutions include orange, pear, grape, apple, banana, melon, mango, peach, plum, kiwi, and cantaloupe. Vegetables include kale, mustard, turnip, dandelion, spinach, romaine, carrot, yam, bell pepper, squash, green beans, broccoli, sweet potato, corn, cucumber, lettuce, cabbage, endive, zucchini, pumpkin, onion, green pepper, cauliflower, collared greens, squash, celery, and tomato.

Pelleted feed: Pelleted feed should be based on alfalfa hay and should be low in sugar (less than 5%) and starches (less than 5%). The pelleted feed should not contain grains because of the high amounts of sugar and starch in grains. The feed pellets should not comprise more than 50% of the diet by weight as fed. Feeds that AZA institutions have used for their capybaras include: rodent chow, primate biscuits, rabbit chow, capybara pellets, to name a few. A diet should include:

Crude protein not less than…….. 15.0%	Calcium not less than……. 0.85%
Crude fat not less than……. 3.0%	Calcium not more than……. 1.35%
Crude fiber not more than……. 23.0%	Phosphorous not less than…… 0.75%
Acid Detergent Fiber not more than.. 26.0%	Sodium not more than……. 0.75%
Ash not more than……. 9.0%	Salt not less than……. 1.0%
	Salt not more than…….. 1.5%

Forage: At least 50% of the diet by weight as fed should be supplied by forage in the form of fresh browse (leaves, flowers, etc.) or alfalfa hay. Although capybaras are classed as grazers, they selectively ingest forage high in protein and energy. Recent studies have confirmed capybaras are actually browsers, but survive using a grazers diet. It is optimal to feed them a browsers diet. Mixed hay with at least 50% alfalfa hay content or higher can be considered. Some institutions have had great success with fresh, bagged hay products such as Chaffhaye (http://chaffhaye.com/). Other institutions have used the following plants as browse: canna (*Canna sp)*, bamboo (*Poaceae sp)*, mulberry (*Morus sp)*, maple (*Acer sp)*, dogwood (*Cornus sp)*, sassafras (*Sassafras sp)*, and timothy hay (*Phleum pretense sp)*.

Table 5. Comparison of composition of some wild foods eaten by capybaras to alfalfa and timothy hay (dry matter (DM) basis)

Analysis	**Wild Forage Range or Average %**	**Alfalfa Hay* %**	**Timothy Hay* %**
Crude Protein (DM) %	9.3–19.2	18.2	9.1
Crude Fiber (DM) %	10.7–30.3	28.9	35.6
Crude Fat (DM) %	4.0	2.1	2.3
Neutral Detergent Fiber (NDF) %	58.2	44.8	65.4
Acid Detergent Fiber (ADF) %	35.0	33.4	37.8
Lignin %	9.8	7.6	4.4

*www.feedepedia.com

Enrichment food options: Food items may also be used as sources of enrichment. Examples of suitable edible enrichment include peanuts, peanut butter, nuts, popcorn, Cheerios®, instant oatmeal, sunflower seeds, honey, bread, and dried pastas. Enrichment foods should never be more than 5% total diet and be low in sugar. All guidelines for regular diet item parameters remain in place for enrichment items.

Food preparation must be performed in accordance with all relevant federal, state, or local laws and/or regulations (AZA Accreditation Standard 2.6.1). The appropriate hazard analysis and critical control points (HACCP) food safety protocols for the diet ingredients, diet preparation, and diet administration should be established for capybaras. Diet preparation staff should remain current on food recalls, updates, and regulations per USDA/FDA. Remove food within a maximum of 24 hours of being offered unless state or federal regulations specify otherwise, and dispose of per USDA guidelines.

AZA Accreditation Standard

(2.6.1) Animal food preparation and storage must meet all applicable laws and/or regulations.

Feeding browse and forage to managed capybaras: All browse plants used within the animal's diet or for enrichment must be identified and assessed for safety. The responsibility for approval of plants and oversight of the program must be assigned to at least one qualified individual (AZA Accreditation Standard 2.6.3).

AZA Accreditation Standard

(2.6.3) The institution must assign at least one paid or unpaid staff member to oversee appropriate browse material for the animals (including aquatic animals).

Any browse or forage fed to capybaras should be clean and free of contaminants (dust, stones, plant detritus, animal matter, fecal matter, etc.) and pesticides. Avoid using plants grown along roads and highways because the plants can contain petroleum residues and chemicals used for road maintenance. Commercial suppliers should be asked about pesticide use for their products and/or edible plant materials should be purchased from organic producers. When possible, plants and produce should be washed before feeding them out.

There is a lack of research on the safety and suitability of browse and forage species for capybara. Therefore, we have to rely on historical safe use of plants. A comprehensive listing of browse species used in zoos across the United States and Canada can be found in Appendix G. Appendix H is a comprehensive guide for selecting quality alfalfa hay. Hay producers typically provide a detailed analysis of their alfalfa on a crop-by-crop basis, if it is not supplied institutions should conduct their own analysis prior to use. You will want to make sure that you perform routine nutrient analyses on your hay in order to know and adjust for the inherent variation in hay caused by a myriad of factors (location, cut number, prevailing weather patterns, fertilization and management regime, etc).

Ingestion of poisonous plants or chemicals: General symptoms caused by the ingestion of poisonous plants or chemicals are often sudden-onset and can be found on the mouth (frothing, blisters, burned tissue, panting), eyes (dilated pupils, glassy eyes), pulse (arrhythmia (erratic heartbeat) or a weak, rapid pulse), lack of muscle coordination (staggering, incoordination, inability to stand), excitability, and convulsions. Specific reactions to toxic plants can be found on reference sites found in Appendix H. ***If plant or chemical toxicity is suspected, veterinary professionals should be consulted immediately*.**

Sample Diets:

Table 6. Diet: Institution A, fed daily, AM & PM

Food Item	Form	Group Measure
Rodent Blocks	Pellet (large)	265g
Leaf Eater	Pellet(large)	200g
Cucumber	1/2" Slices	100g Cucumber
Banana	1/2" Slices	80g Bananas
Carrot	1/2" Slices	100g Carrot
Sweet Potato	Diced	650g Sweet Potato
Apple	Diced	90g Apples
Corn	1/2" Slices with husk	200g Ears

Table 7. Diet: Institution B, fed daily, PM

Food Item	Individual Weight (per animal)
Fruit	175 grams (Should include half an orange per animal every day) Fruits: orange, apple, pear, melon, grapes, banana (including peel), mango, peach, plum, and kiwi
Vegetable	750 grams Mostly root vegetables: carrot, yam, squash, sweet potato, corn, pumpkin, onion, broccoli (limited), cucumber, green peppers, zucchini, and cauliflower (limited)
Greens	1 large head of romaine lettuce (or equivalent)

Table 8. Diet: Institution C, fed daily

Food Item	Group Measure (1.1)
Guinea Pig Diet (Mazuri®)	600 grams
Apple	400 grams (XL chunks)
Sweet Potato	500 grams (XL chunks)
Carrot	450 grams
Fruit (pear or melon)	100 grams (XL chunks)
Corn w/ husk	1 ear, halved
Greens - choose any two	1100 grams
Alfalfa Hay	Handful in PM

6.3 Nutritional Evaluations

A portion of the health of capybaras can be evaluated by the assessment of their body condition score. This is not a complete measure to determine optimal nutritional health, but a quick way to alert a problem may be present. A breakdown of the 5 point scale can be viewed in Table 9. Currently, there is not a tested 9 point scale available for capybara. The only additional diet-related health concern for

managed capybaras is the development of scurvy because of insufficient vitamin C supplementation in the diet.

Table 9. Suggested body condition score (BCS) sheet (Carr, 1998)

Score Number	***Condition***	***Description***	***Shape of Body***
5	Overfat	Hips & backbone heavily covered	Bulbous
4	Fat	Hips & backbone cannot be felt	Tending to Bulge
3.5	Good Condition	Hips & backbone only felt with difficulty	Tube shape
3	Normal	Hips & backbone only felt with firm pressure	Tube shape
2.5	Somewhat Thin	Hips & backbone felt without palm pressure	Tube shaped but flat (slab) sides
2	Thin	Hips & backbone noticeable & easily felt	Ribs & spine can be felt
1	Emaciated	Hips & backbone visible	Bone structure apparent (ribs & backbone)

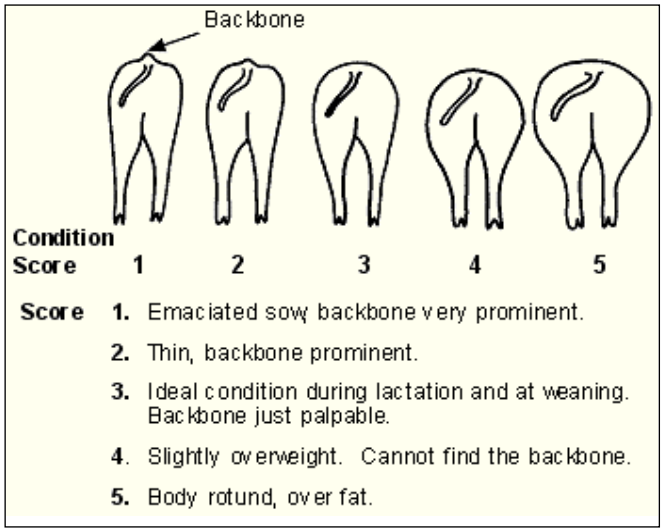

Figure 7. (Pig Body Condition Chart. Chosen due to similar body structure in Capybaras. #3 is an ideal condition for both males and females.)

Chapter 7. Veterinary Care

7.1 Veterinary Services

Veterinary services are a vital component of excellent animal care practices. A full-time staff veterinarian is recommended, however, in cases where this is not necessary, a consulting/part-time veterinarian must be under contract to make at least twice monthly inspections of the animal collection and to attend any emergencies (AZA Accreditation Standard 2.1.1). In some instances, because of their size or nature, exceptions may be made to the twice-monthly inspection requirement for certain institutions (e.g., insects only, etc.). Veterinary coverage must also be available at all times so that any indications of disease, injury, or stress may be responded to in a timely manner (AZA Accreditation Standard 2.1.2). All AZA-accredited institutions should adopt the guidelines for medical programs developed by the American Association of Zoo Veterinarians (AAZV), available at the AAZV website under "Publications", at https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazvveterinaryguidelines2016.pdf (AZA Accreditation Standard 2.0.1).

AZA Accreditation Standard

(2.1.1) A full-time staff veterinarian is recommended. In cases where such is not necessary because of the number and/or nature of the animals residing there, a consulting/part-time veterinarian must be under written contract to make at least twice monthly inspections of the animals and to respond as soon as possible to any emergencies.

AZA Accreditation Standard

(2.1.2) So that indications of disease, injury, or stress may be dealt with promptly, veterinary coverage must be available to the animals 24 hours a day, 7 days a week.

AZA Accreditation Standard

(2.0.1) The institution should adopt the *Guidelines for Zoo and Aquarium Veterinary Medical Programs and Veterinary Hospitals,* and policies developed or supported by the American Association of Zoo Veterinarians (AAZV). The most recent edition of the medical programs and hospitals booklet is available at the AAZV website, under "Publications", at https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazvveterinaryguidelines2016.pdf, and can also be obtained in PDF format by contacting AZA staff.

Kimberly Cook is the AZA SSP/TAG veterinary advisor for capybaras (k.cook@akronzoo.org). Protocols for health inspections vary by institution. Most operate on an as-needed basis based on keeper observation. Weights are typically monitored monthly, and fecal samples are collected and examined every 6 months. Those AZA-accredited facilities following a schedule for health inspections typically examine individuals once or twice a year. Comprehensive health exams necessitate that animals are restrained. Methods of restraint vary based on the exam but include crating, darting, netting, manual restraint, squeeze holding, and the use of baffle boards to administer injections. Chemical restraint of capybaras can be accomplished by the use of ketamine (3–4 mg/kg), medetomidine (0.03–0.04 mg/kg), and butorphanol (0.2–0.5 mg/kg) with isoflurane maintenance. Additional options include dexmedetomidine with ketamine (30 mg/kg), but check with veterinarians for proper dosage for specific capybara individuals.

AZA Accreditation Standard

(2.2.1) Written, formal procedures must be available to paid and unpaid animal care staff for the use of animal drugs for veterinary purposes, and appropriate security of the drugs must be provided.

Protocols for the use and security of drugs used for veterinary purposes must be formally written and available to animal care staff (AZA Accreditation Standard 2.2.1). Procedures should include, but are not limited to: a list of persons authorized to administer animal drugs, situations in which they are to be utilized, location of animal drugs and those persons with access to them, and emergency procedures in the event of accidental human exposure.

Capybaras occasionally need medication for preventative measures or treatment. Medications that work to deworm them include ivermectin (0.2 mg/kg) and metronidazole (25 mg/kg). Biosponges can be used to treat diarrhea; check with a veterinarian for proper dosage. Aggression can be treated using diazepam (10–15 mg depending on aggression level). For joint pain, carprofen (4–5 mg/kg) can be used.

Practitioners are required to follow all state and federal guidelines regarding controlled substances. Safety precautions should always be followed from both a human and an animal perspective. *Guidelines for Zoo and Aquarium Veterinary Medical Programs and Veterinary Hospitals* (2016) has the most current standards.

Veterinary recordkeeping is an important element of animal care and ensures that information about individual animals and their treatment is always available. A designated staff member should be responsible for maintaining accurate animal veterinary records. Health records should be maintained for

all animals and should include health concerns, examinations, anesthetic episodes, prescribed medications/dosages/side effects/outcome of therapy, vaccination history, and diagnostic tests and results. The most effective method of managing this data would be an electronic format like that of ZIMS which is part of the Species360 program (Joslin et al., 1998).

7.2 Transfer Examination and Diagnostic Testing Recommendations

Animals may be transferred between AZA-accredited institutions or certified related facilities as the result of individual institutional wishes or, in the case of SSP animals, due to AZA Animal Program recommendations as part of a concerted effort to preserve this species. These transfers should be done as altruistically as possible and the costs associated with specific examination and diagnostic testing for determining the health of these animals should be considered. A complete physical examination of the capybara should be conducted before any transfer.

Blood collected for plasma banking should be stored in a -70 °C (-94 °F) freezer OR in a -20 °C (-4 °F) freezer. The freezer used for storing plasma should NOT be a frost-free freezer.

Table 10. Physiological reference intervals for captive *Hydrochoerus hydrochaeris (*Teare, 2013)

Test	Units	Reference Interval	Mean	Median	Low Sample[a]	High Sample[b]	Sample Size[c]	Animals[d]
Glucose	mg/dL	63 - 314	152	143	25	349	255	168
Blood Urea Nitrogen	mg/dL	5 - 24	14	13	4	29	253	169
Creatinine	mg/dL	0.9 - 2.8	1.8	1.7	0.1	3.0	255	172
BUN/Cr ratio		3.4 - 16.3	8.3	7.7	2.1	19.3	247	163
Uric Acid	mg/dL	0.0 - 2.9	1.4	1.3	0.3	3.5	44	39
Calcium	mg/dL	9.7 - 13.1	11.2	11.1	8.6	13.6	237	160
Phosphorus	mg/dL	2.9 - 10.7	6.2	5.9	1.2	14.3	196	143
Ca/Phos ratio		1.0 - 3.6	2.0	1.9	0.6	4.1	192	140
Sodium	mEq/L	127 - 145	135	135	123	147	202	136
Potassium	mEq/L	3.7 - 7.5	4.9	4.7	2.5	8.2	207	140
Na/K ratio		16.8 - 36.8	28.1	28.6	11.8	44.5	204	138
Chloride	mEq/L	89 - 106	96	96	78	115	193	133
Total Protein	g/dL	4.5 - 7.7	6.2	6.3	2.9	8.8	236	154
Albumin	g/dL	1.5 - 4.1	3.0	3.1	1.3	4.8	207	139
Globulin	g/dL	1.5 - 4.9	3.2	3.2	0.0	6.0	211	141
Alkaline Phosphatase	IU/L	58 - 1074	367	307	4	1395	234	154
Lactate Dehydrogenase	IU/L	*	444	269	57	1718	35	28
Aspartate Aminotransferase	IU/L	12 - 102	37	32	7	110	220	158
Alanine Aminotransferase	IU/L	15 - 69	37	36	10	100	234	160
Creatine Kinase	IU/L	70 - 1315	384	273	11	1471	128	96
Gamma-glutamyltransferase	IU/L	0 - 9	4	4	0	14	78	66
Amylase	IU/L	324 - 6074	3222	3199	107	6645	70	54
Lipase	IU/L	0 - 68	37	32	14	91	55	43
Total Bilirubin	mg/dL	0.0 - 0.8	0.2	0.2	0.0	1.0	211	142
Direct Bilirubin	mg/dL	*	0.1	0.0	0.0	0.5	32	26
Cholesterol	mg/dL	36 - 124	72	69	12	135	173	125
Triglyceride	mg/dL	0 - 244	112	89	21	341	74	63
Bicarbonate	mEq/L	*	22.4	22.5	9.0	39.0	34	26
Magnesium	mg/dL	1.71 - 3.30	2.51	2.51	1.70	3.50	42	35
Carbon Dioxide	mEq/L	11.8 - 36.3	24.1	24.1	12.0	36.0	51	33
Test	Units	Reference Interval	Mean	Median	Low Sample[a]	High Sample[b]	Sample Size[c]	Animals[d]

[a] Lowest sample value used to calculate the reference interval.
[b] Highest sample value used to calculate the reference interval.
[c] Number of samples used to calculate the reference interval.

[d] Number of different individuals contributing to the reference interval.

Table 11. Physiological reference intervals for *Hydrochoerus hydrochaeris* (Teare, 2013)

Test	Units	Reference Interval	Mean	Median	Low Sample[a]	High Sample[b]	Sample Size[c]	Animals[d]
White Blood Cell Count	*10^3 cells/µL	2.93 - 13.41	6.85	6.68	1.81	16.30	261	186
Red Blood Cell Count	*10^6 cells/µL	2.18 - 4.83	3.40	3.40	1.76	5.46	246	172
Hemoglobin	g/dL	9.8 - 18.0	14.0	14.1	8.7	20.4	250	172
Hematocrit	%	29.6 - 56.4	42.3	42.1	21.0	62.0	305	209
MCV	fL	88.0 - 157.8	123.3	123.6	80.6	168.3	238	168
MCH	pg	32.6 - 48.6	41.3	41.2	28.5	54.2	227	158
MCHC	g/dL	29.8 - 38.2	33.7	33.6	27.4	40.2	230	158
Segmented Neutrophils	*10^3 cells/µL	0.09 - 8.70	3.83	3.62	0.02	10.50	261	186
Neutrophilic Band Cells	*10^3 cells/µL	0.01 - 0.07	0.03	0.03	0.00	0.11	254	180
Lymphocytes	*10^3 cells/µL	0.50 - 5.73	2.15	1.72	0.35	7.50	258	184
Monocytes	cells/µL	39 - 1071	372	310	1	1231	245	173
Eosinophils	cells/µL	0 - 534	222	183	31	912	107	87
Basophils	cells/µL	*	84	73	32	273	32	31
Platelet Count	*10^3 cells/µL	85 - 429	252	249	47	600	140	106
Test	Units	Reference Interval	Mean	Median	Low Sample[a]	High Sample[b]	Sample Size[c]	Animals[d]

[a] Lowest sample value used to calculate the reference interval.
[b] Highest sample value used to calculate the reference interval.
[c] Number of samples used to calculate the reference interval.
[d] Number of different individuals contributing to the reference interval.

7.3 Quarantine

AZA institutions must have holding facilities or procedures for the quarantine of newly arrived animals and isolation facilities or procedures for the treatment of sick/injured animals. Quarantine duration should be assessed and determined by the pathogen risk and best practice for animal welfare (AZA Accreditation Standard 2.7.1). All quarantine, hospital, and isolation areas should be in compliance with AZA standards/guidelines (AZA Accreditation Standard 2.7.3; Appendix E). All quarantine procedures should be supervised by a veterinarian and formally written and available to paid and unpaid staff working with quarantined animals (AZA Accreditation Standard 2.7.2). If a specific facility is not present, then newly acquired animals should be kept separate from the established collection to prohibit physical contact, prevent disease transmission, and avoid aerosol and drainage contamination. If the receiving institution lacks appropriate facilities for quarantine, pre-shipment quarantine at an AZA or American Association for Laboratory Animal Science (AALAS) accredited institution may be applicable. Local, state, or federal regulations that are more stringent than AZA Standards and recommendation have precedence.

Quarantine is typically only for new individuals, although individuals that are sick, injured, or that have been off grounds may be required to go through an additional quarantine at the discretion of the facility. Most individuals have no issues going

AZA Accreditation Standard

(2.7.1) The institution must have holding facilities or procedures for the quarantine of newly arrived animals and isolation facilities or procedures for the treatment of sick/injured animals. Quarantine duration should be assessed and determined by the pathogen risk and best practice for animal welfare.

AZA Accreditation Standard

(2.7.3) Quarantine, hospital, and isolation areas should be in compliance with standards/guidelines contained within the *Guidelines for Zoo and Aquarium Veterinary Medical Programs and Veterinary Hospitals* developed by the American Association of Zoo Veterinarians (AAZV), which can be obtained at: https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazvveterinaryguidelines2016.pdf.

AZA Accreditation Standard

(2.7.2) Written, formal procedures for quarantine must be available and familiar to all paid and unpaid staff working with quarantined animals.

through the quarantine process, but some individuals can become stressed and skittish. Based on the 2014 survey of AZA-accredited institutions with capybaras, these behaviors tend to decrease and even disappear once the individual is acclimated to the exhibit.

Quarantine facilities should be remote from main exhibits, preferably in a separate building. If a facility is unable to provide a separate building for the purpose of quarantine, then quarantine animals should be prevented from making contact with collection animals by housing them in separate holdings and using separate feeding, cleaning, and enrichment equipment. Keepers should attend to quarantine animals last and the equipment used to attend to quarantine animals should be separated from the equipment used for the collection animals. All animal holding areas should be separate from human areas. The holding areas should have nonporous walls and floors that can be cleaned by water and disinfectants. These surfaces also need to be non-absorbent as well as resistant to impact, hot water, steam, and biological agents (e.g. urine and feces). They should also have good acoustics in order to keep noise to a minimum. The floors in these areas should be sloped towards the drains and have no cracks. Drains cannot be connected and should have anti-back flow devices and auto-disinfectant systems (Joslin et al., 1998).

AZA institutions must have zoonotic disease prevention procedures and training protocols established to minimize the risk of transferable diseases (AZA Accreditation Standard 11.1.2) with all animals, including those newly acquired in quarantine. Keepers should be designated to care only for quarantined animals, if possible. If keepers must care for both quarantined and resident animals of the same class, they should care for the quarantined animals only after caring for the resident animals. Care should be taken to ensure that these keepers are “decontaminated” before caring for the healthy resident animals again. Equipment used to feed, care for, and enrich animals in quarantine should be used only with these animals. If this is not possible, then all items must be appropriately disinfected, as designated by the veterinarian supervising quarantine, before use with resident animals.

AZA Accreditation Standard

(11.1.2) Training and procedures must be in place regarding zoonotic diseases.

To prevent the transmission of zoonotic diseases to healthy collection animals, quarantine animals should have a keeper dedicated solely to their care. Keepers in these areas should take care with their personal hygiene, taking measures such as washing hands and changing clothes before interacting with collection animals. Footwear should be disinfected using a bacterialcidal and virocidal footbath. The footbath should be changed regularly to prevent deactivation due to organic compounds. In place of a footbath, a disinfectant spray may be used. Equipment should be dedicated to the quarantine facility and should be disinfected regularly. Waste should be picked up last and disposed of right away (Joslin et al., 1998).

Typical length for quarantine is a minimum of 30 days for capybaras (longer if they are shipped internationally or unless otherwise directed by the staff veterinarian). If additional mammals of the same order are introduced into their corresponding quarantine areas, the minimum quarantine period must begin over again. However, the addition of mammals of a different order to those already in quarantine will not require the re-initiation of the quarantine period.

A complete physical exam, as detailed in Chapter 7.2, should be performed on each animal at the beginning of the quarantine period and a visual examination should be conducted at the end of this period. During quarantine a regular schedule of fecals, as determined by the veterinarian, should be completed. During the quarantine period, specific diagnostic tests should be conducted on each animal (see Appendix E). A dental examination, if applicable, should be performed. Animals should be evaluated for ectoparasites and treated accordingly. Blood should be collected, analyzed, and then sera banked in either a -70 °C (-94 °F) freezer or a non-frost-free -20 °C (-4 °F) freezer for retrospective evaluation. Fecal samples should be collected and analyzed for gastrointestinal parasites and the animals should be treated accordingly. Vaccinations should be updated as appropriate and, if the vaccination history is not known, the animal should be treated as immunologically naive and given the appropriate series of vaccinations. During hospitalization, capybaras need to be protected from injury by being held in a quiet area with smooth walls.

Animals should be permanently identified by intrascapular transponder, ear tag, tattoo, or ear notch. Intrascapular transponder is the most frequently used and preferred method of identification. Medical records for each animal should be accurately maintained and easily available during the quarantine period. Release from quarantine should be contingent upon normal results from complete blood counts (CBCs), serum chemistry, and three negative fecals (direct and float) that are spaced a minimum of two

weeks apart. Some facilities choose to require three negative fecals in addition to the 30 days in quarantine, while others require whichever is longer, 30 days or three negative fecals (Joslin et al., 1998).

Individuals in quarantine have occasionally been known to exhibit signs of stress. Some signs of stress include pacing, prolonged vocalization, sudden movements (running, jumping, etc.) and a general display of unease. Capybara are typically calm and docile spending most of the day resting in a pool or on land. In the case of an individual who becomes severely stressed by quarantine, the facility's veterinarian, in conjunction with keeper and curatorial staff, can make the decision for modifications to the facility for quarantine (Joslin et al., 1998).

If a capybara should die in quarantine, a necropsy should be performed on it to determine cause of death in order to strengthen the program of veterinary care and meet SSP-related requests (AZA Accreditation Standard 2.5.1). If the animal is on loan, the owning institution must be informed and should provide consent for the necropsy. The institution should have an area dedicated to performing necropsies, and the subsequent disposal of the body must be done in accordance with any local or federal laws (AZA Accreditation Standards 2.5.2 and 2.5.3). Necropsies should include a detailed external and internal gross morphological examination and representative tissue samples from the body organs should be submitted for histopathological examination (see Chapter 7.6).

AZA Accreditation Standard

(2.5.1) Deceased animals should be necropsied to determine the cause of death for tracking morbidity and mortality trends to strengthen the program of veterinary care and meet SSP-related requests.

AZA Accreditation Standard

(2.5.2) The institution should have an area dedicated to performing necropsies.

AZA Accreditation Standard

(2.5.3) Cadavers must be kept in a dedicated storage area before and after necropsy. Remains must be disposed of in accordance with local/federal laws.

There are no necropsy procedures specifically for capybaras, however, the necropsy should be conducted on zoo grounds whenever possible and should occur no longer than 24 hours after death. If the institution does not have facilities to conduct the necropsy, then the animal should be transported to a location where the necropsy can be conducted. Once the necropsy has been completed, the animal should be incinerated or disposed of according to local or state ordinances. The zoo should maintain collections of 10% buffered formalin fixed tissues, paraffin blocks, and slides from the postmortem examination for future reference (Joslin et al., 1998).

7.4 Preventive Medicine

AZA-accredited institutions should have an extensive veterinary program that must emphasize disease prevention (AZA Accreditation Standard 2.0.2). AZA institutions should be aware of and prepared for periodic disease outbreaks in other animal populations that might affect the institution's animals, and should develop plans to protect the institution's animals in these situations (AZA Accreditation Standard 2.0.3). The American Association of Zoo Veterinarians (AAZV) has developed an outline of an effective preventative veterinary medicine program that should be implemented to ensure proactive veterinary care for all animals (https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazv veterinaryguidelines2016.pdf).

AZA Accreditation Standard

(2.0.2) The veterinary care program must emphasize disease prevention.

AZA Accreditation Standard

(2.0.3) Institutions should be aware of and prepared for periodic disease outbreaks in wild or other domestic or exotic animal populations that might affect the institution's animals (ex – Avian Influenza, Eastern Equine Encephalitis Virus, etc.). Plans should be developed that outline steps to be taken to protect the institution's animals in these situations.

Currently there are no mandatory regulations for capybara health exams. Most managed care facilities choose to conduct full health exams on their capybaras at regular intervals ranging from once every six months to every other year. Full Neonate exams should be conducted in the first few months. This is to determine overall health, gender, and to assign long term identification (transponder, tattoo, etc.). However, brief exams may be conducted within a few days of birth. This is to determine if animals are nursing and growing sufficiently and to apply temporary identification (color on fur, nail polish, etc.) In order to conduct a full health exam, the veterinarian should have access to all necessary equipment that is included in a properly outfitted clinic.

There are no specific veterinary standards for capybaras. Pregnant capybaras typically do not require any keeper intervention, although some facilities choose to remove pregnant females as they get close to parturition. Capybaras are born precocial, and care for neonates typically does not differ from the care of

other animals in the social group. Occasionally, geriatric animals may require veterinary intervention for issues related to arthritis (Bernal et al., 2011).

Animals that are taken off zoo/aquarium grounds for any purpose have the potential to be exposed to infectious agents that could spread to the rest of the institution's healthy population. AZA-accredited institutions must have adequate protocols in place to avoid this (AZA Accreditation Standard 1.5.5).

> **AZA Accreditation Standard**
>
> **(1.5.5)** For animals used in offsite programs and for educational purposes, the institution must have adequate protocols in place to protect the rest of the animals at the institution from exposure to infectious agents.

Two AZA-accredited facilities currently use capybaras as ambassador animals. To protect the institution's animal collection from exposure to infectious agents that might be carried back by these ambassador animals, they should be housed separately from the rest of the collection. In addition, ambassador animals should be cared for by solely dedicated staff. These animals are participants in on-grounds programs only and do not directly contact the public, which limits the opportunities the ambassador capybaras have to exposure of infectious agents.

A tuberculin testing and surveillance program must be established for paid and unpaid animal care staff, as appropriate, to protect the health of both staff and animals (AZA Accreditation Standard 11.1.3). To prevent specific disease transmission, vaccinations should be updated as appropriate for the species.

> **AZA Accreditation Standard**
>
> **(11.1.3)** A tuberculin (TB) testing/surveillance program must be established for appropriate paid and unpaid staff in order to assure the health of both the paid and unpaid staff and the animals.

Capybaras are not known to be commonly susceptible to *Mycobacterium tuberculosis*. There has been a single report of a capybara infected with a non-tuberculous mycobacterium, which occurred in 2013 (Pezzone et al., 2013). No routine tuberculin testing for capybaras or their keepers is called for.

No vaccinations are currently recommended for rodents, although tetanus and/or killed rabies prophylaxis may be appropriate in some areas. In addition, according to a survey of AZA-accredited facilities conducted in 2014, some facilities are beginning to vaccinate capybaras for leptospirosis. The type, serial, and/or lot number of the vaccine should be noted in the individuals' medical records (Backues et al., 2009).

7.5 Capture, Restraint, and Immobilization

The need for capturing, restraining and/or immobilizing an animal for normal or emergency husbandry procedures may be required. All capture equipment must be in good working order and available to authorized and trained animal care staff at all times (AZA Accreditation Standard 2.3.1).

> **AZA Accreditation Standard**
>
> **(2.3.1)** Capture equipment must be in good working order and available to authorized, trained personnel at all times.

Methods of restraint vary based on the exam, but include crating, darting, netting, manually restraining, squeeze holding, and using baffle boards to administer injections.

Chemical restraint of capybaras include the use of ketamine (3–4 mg/kg), medetomidine (0.03–0.04 mg/kg), or butorphanol (0.2–0.5 mg/kg) with isoflurane maintenance. Additional options include dexmedetomidine with ketamine (30 mg/kg); check with veterinarians for proper dosage for specific capybara individuals. Several institutions, including those using ambassador animals, have had success with crate training animals, which allows hand injections to occur and, when surveyed in 2014, some institutions indicated that they were interested in beginning injection training. Animals have also been harness-trained in the past (not limited to ambassador animals). Neonates can be physically restrained by corralling them and picking them up using gloves.

7.6 Management of Diseases, Disorders, Injuries and/or Isolation

AZA-accredited institutions should have an extensive veterinary program that manages animal diseases, disorders, or injuries and has the ability to isolate these animals in a hospital setting for treatment if necessary. The owner of an animal on loan at a facility is to be consulted prior to any elective invasive procedures, including permanent contraception.

> **AZA Accreditation Standard**
>
> **(2.4.2)** Keepers should be trained to recognize abnormal behavior and clinical signs of illness and have knowledge of the diets, husbandry (including enrichment items and strategies), and restraint procedures required for the animals under their care. However, keepers should not diagnose illnesses nor prescribe treatment.

Capybara keepers should be trained for meeting the animal's

dietary, husbandry, and enrichment needs, as well as in restraint techniques, and recognizing behavioral indicators animals may display if their health becomes compromised (AZA Accreditation Standard 2.4.2). Protocols should be established for reporting these observations to the veterinary department. Hospital facilities for capybaras must have radiographic equipment or access to radiographic services (AZA Accreditation Standard 2.3.2), contain appropriate equipment and supplies on hand for treatment of diseases, disorders or injuries, and have staff available that are trained to address health issues, manage short and long-term medical treatments and control for zoonotic disease transmission.

> **AZA Accreditation Standard**
>
> **(2.3.2)** Institution facilities must have radiographic equipment or have access to radiographic services.

Some of the more common disease conditions that capybaras in zoos face include GI parasitism, dystocia, dental malocclusions, trauma from conspecifics, degenerative arthritic conditions of the limbs, and upper respiratory infections (Moreira et al., 2013). Abnormal social separation, fluffed hair/coat, teeth grinding, a hunched back, and decreased appetite are all signs of a potential health issue. If these signs are observed, a health inspection is recommended.

Viruses and bacterial infections: Capybaras are capable of carrying several types of viruses and bacterial infections, including rabies (in rare cases), foot-and-mouth disease (can contract and transmit between capybaras and cattle), brucellosis, and leptospirosis. Brucellosis is an infectious, intracellular bacterium mainly found in cattle, although it can affect humans. It has been reported in wild capybaras and it is known that capybaras can be a host to the brucellosis bacteria. Leptospirosis can cause an enlarged spleen, jaundice, and nephritis. It is the widest spread zoonotic disease and can be transmitted through soil and water contaminated by urine and secretions from infected animals, or through the ingestion of contaminated food. Rats are the universal reservoir of leptospirosis; however, almost all species of mammals can carry and excrete leptospirosis (Meites et al., 2004). There are currently no recommended vaccines for capybaras, although, according to the AZA survey conducted in 2014, some facilities are beginning to vaccinate for rabies, tetanus, and leptospirosis.

Ectoparasites: Capybaras can be affected by both sarcoptic scabies and ticks. Scabies (mites) can cause dermatitis of varying severity. Animals carrying a high load of mites can become anorexic, losing weight to the point of death. Sarcoptic mites are highly contagious, but are readily treatable with ivermectin. Ticks can and should be controlled via ivermectin as they can cause anemia of varying severity (Moreira et al., 2013). Check with veterinarians for proper dosage for specific capybara individuals

Endoparasites: Endoparasites are more common in the capybara's native habitat, although managed animals may still be affected by helminthes (nematodes, cestodes, and trematodes). Common signs of infection include coarse/pointed hair, inactivity, weight loss, anemia, a distended abdomen, dragging of the hindquarters, and larvae in the feces. Endoparasites can be diagnosed via fecal analysis in which larvae and/or eggs are visible either to the naked eye or microscopically, depending on the species. Many can be treated using ivermectin; check with veterinarians for proper dosage for specific capybara individuals. In addition to helminthes, capybaras may also be host to some species of protozoans, including *Eimeria* and *Cryptosporidium parvum*. Cases involving strains of *Eimeria* tend to be asymptomatic or subclinical, although individuals may exhibit diarrhea and/or stunted growth. Coccidiosis can occur in close confinement when there is a disregard for hygiene as well as in places of high humidity. It can be diagnosed via fecal floatation or direct examination. At low parasite loads treatment is not recommended. In acute cases of hematochezia, capybara can be treated with sulfonamides in the drinking water for 10 days; check with veterinarians for proper dosage for specific capybara individuals. Although many of these parasites are common to capybaras in their native habitat or in farm situations, some intermediate host species may be absent *ex situ*, preventing the continuation of the parasite/host cycle in managed care facilities (Moreira et al., 2013).

If keeper staff notices any behavioral indicators that an individual may be experiencing health problems, they should contact veterinary staff. Isolation facilities should be equipped with the items listed in Chapter 7.4. Isolation facilities should be remote from the main exhibits and should follow the design of the quarantine facilities described in Chapter 7.3. Individuals that are housed in isolation or in the hospital for long-term treatment may become skittish, show signs of stress, seem nervous, or stop eating. If an individual shows signs of these behaviors, every effort should be made to eliminate the stressful stimulus. In addition, offering the individual's favorite food and/or enrichment may help reduce stress. There are no

special requirements for the treatment facility when dealing with capybaras, however, a hospital holding with a pool that is easy for the capybara to get in and out of is ideal.

AZA Accreditation Standard

(1.5.8) The institution must develop and implement a clear and transparent process for identifying, communicating, and addressing animal welfare concerns from paid or unpaid staff within the institution in a timely manner, and without retribution.

AZA-accredited institutions must have a clear process for identifying and addressing capybara animal welfare concerns within the institution (AZA Accreditation Standard 1.5.8) and should have an established Institutional Animal Welfare Committee. This process should identify the protocols needed for animal care staff members to communicate animal welfare questions or concerns to their supervisors, their Institutional Animal Welfare Committee or if necessary, the AZA Animal Welfare Committee. Protocols should be in place to document the training of staff about animal welfare issues, identification of any animal welfare issues, coordination and implementation of appropriate responses to these issues, evaluation (and adjustment of these responses if necessary) of the outcome of these responses, and the dissemination of the knowledge gained from these issues.

Given the wide variety of zoos and aquariums that house capybaras, the AZA Rodent, Insectivore and Lagomorph TAG cannot provide specific recommendations for the best approaches to take in order to communicate animal welfare issues effectively within every institution. However, there are basic principles regarding animal welfare of which keepers and other staff should be aware. Good welfare is composed of many facets but includes an individual's health, comfort, nutrition, safety, ability to develop and express species-typical relationships, behaviors, cognitive abilities, and the lack of pain, fear, or distress (Miller & Priest, 2013). Welfare should be measured on a continuum from good to poor and should be based on objective, evidence-based physical and psychological measures. Stress is a factor that it is important to consider when measuring animal welfare. Stress is defined by the AZA as the body's response to a perceived or actual threat that results in physiological and behavioral changes. Acute stress is the reaction to an immediate threat. Once the threat is gone, the physiological or behavioral changes return to normal. Several experiences of acute and severe stress that affect an animal within a short time can have long-term negative impacts (Miller & Priest, 2013). Likewise, chronic stress, or prolonged exposure to a threat that the animal does not have the ability to control or change, can have negative long-term impacts. Chronic stress results in the activation of the autonomic and endocrine responses. Chronic activation of these systems can result in the dysregulation of the immune function.

In order to evaluate welfare, staff needs to understand animal management, an individual animal's history, and the natural history of the species. Consideration should be given to the stream of inputs created by animal management which include, but are not limited to, environmental enrichment, training, veterinary care, nutrition, social structure, exhibit size and complexity, staff interactions, visitor viewing areas, neighboring exhibits, multispecies exhibits, and exhibit deficiencies. In addition to these inputs, the individual animal's history also influences its welfare. The conditions of how the individual was raised (e.g., hand-reared vs. mother-reared), any injury or trauma the animal experienced, its personality/behavioral type, and its history with staff all affect the welfare continuum. Another area of consideration is the natural history of the animal including habitat type, activity budgets, social systems, and diet. Using these inputs, staff can then make observations regarding outputs that will help to evaluate overall welfare. Outputs can fall into several categories including behaviors, physiology, physical, sustainability, and health and can be classified as either positive or negative (see Table 12 below).

Table 12. Positive and negative outputs (Miller & Priest, 2013)

Outputs	Positive	Negative
Behaviors	Play and exploration	Lethargy and low levels of behavioral diversity
Physiology	Adrenal system and immune function	Chronically elevated adrenal activity, compromised immune function
Physical	Good body condition and appropriate weight	Poor body condition and decreased appetite
Sustainability	High reproductive success and normal infant mortality rate	Low reproductive success and high infant morality
Health	Absence of disease and long life span	High prevalence of disease and short life span

Keepers should use observations and records to help evaluate the animal welfare continuum. When making observations staff should aim to collect only objective facts and valid and reliable scientific

evidence. Facts should speak for themselves and should not be interpreted. Observations should not be influenced by opinions, subjectivity, or bias. Any individual bias should be acknowledged to ensure accurate evidence-based documentation (Miller & Priest, 2013).

Animals should be allowed to make choices and exert control in their environment. Opportunities for the animals to engage in a diverse array of behaviors based on their natural history should be provided whenever possible. There are four additional opportunities to thrive that should be provided to individuals: (1) the opportunity for a well-balanced diet, (2) the opportunity to self-maintain, (3) the opportunity for optimal health, and (4) the opportunity to express species-specific behavior (Miller & Priest, 2013). As part of providing these opportunities, keepers should conduct preference assessments to determine animal needs and preferences.

If an individual begins to exhibit signs of negative welfare, the suspected stressor should be removed if possible. If the removal of the stressor is not possible, then efforts should be made to work with the individual using principles of counter-conditioning, systematic desensitization, and habituation.

If euthanasia becomes necessary, it is important that it be performed by a skilled person. The euthanasia procedure should begin with a sedative or anesthetic to relieve anxiety and pain. The premedication should only be administered by intramuscular injection or orally. Drugs acceptable for use as an anesthesia prior to euthanasia include: Acepromazine (0.1 IM), Diazepam (0.1–1.0 IM,PO), Midazolam (0.1–0.5 IM), Xylazine (1–5 IM,SC), Atropine (0.03 SC,IM), Glycopyrrolate (0.01 SC,IM), Telazol (200 mg/ml), Ketamine/dexmedetomidine (30 mg/kg), Ketamine/xylazine, Medetomidine (0.03–0.04 mg/kg), Ketamine (3 mg/kg), and Butorphanol (0.2–0.5 mg/kg). No standard anesthetic protocol has been developed for this species, but there are recommended immobilization protocols for minimally invasive procedures when inhalant anesthetics are unavailable (King et. al., 2010). Check with veterinarians for proper dosage for specific capybara individuals. Animals should be restrained and handled in a way that reduces stress and undue pain as per USDA guidelines. The method used to achieve euthanasia should not interfere with the postmortem inspection when possible. Once the chosen method of euthanasia has been administered, there needs to be a lack of a heartbeat and respiration for 10 minutes before declaring the animal dead.

AZA-accredited zoos and aquariums provide superior daily care and husbandry routines, high quality diets, and regular veterinary care, to support capybara longevity. In the occurrence of death however, information obtained from necropsies is added to a database of information that assists researchers and veterinarians in zoos and aquariums to enhance the lives of capybara both in their care and in the wild. As stated earlier, necropsies should be conducted on deceased capybara to determine their cause of death, and the subsequent disposal of the body must be done in accordance with local, state, or federal laws (AZA Accreditation Standard 2.5.1). If the animal is on loan from another facility, the loan agreement should be consulted as to the owner's wishes for disposition of the carcass; if nothing is stated, the owner should be consulted. Necropsies should include a detailed external and internal gross morphological examination and representative tissue samples from the body organs should be submitted for histopathological examination. Many institutions utilize private labs, partner with Universities, or have their own in-house pathology department to analyze these samples. The AZA and American Association of Zoo Veterinarians (AAZV) website should be checked for any AZA Capybara SSP Program approved active research requests that could be filled from a necropsy.

Common causes of death can be found in Table 13. After euthanasia (or natural death), the individual should be necropsied according to the guidelines in Chapter 7.3.

Table 13. Death of Capybaras from 1995–2009 (CDC)

Causes of Death	Description	Number of Instances from 1995–2009
Chronic Nephritis (including glomerulonephropathy)	Infection of the kidneys caused by a buildup of toxins or autoimmune disease.	21
Ulcerative Colitis (Inflammatory Bowel Disease and Colitis)	Genetic condition in the colon that causes compaction to occur.	20
Gastroenteritis	Result of bacterial toxins or viral infection in the stomach or intestines.	17

Myocardial fibrosis	Abnormal thickening of the heart valves due to inappropriate proliferation of cardiac fibroblast in the cardiac muscle. Fibrotic cardiac muscle is stiffer and less compliant and is seen in the progression to heart failure.	15
Shock	Reduces blood flow to the body and includes heart problems such as heart failure, low blood volume, or changes in the blood vessels.	14
Degenerative Cardiomyopathy	Weakening of the myocardium in the heart that causes the heart to be unable to pump blood through the body.	12
Pneumonia	Infection of the lungs, caused by multiple microbes.	11

Chapter 8. Reproduction

8.1 Reproductive Physiology and Behavior

It is important to have a comprehensive understanding of the reproductive physiology and behaviors of the animals in our care. This knowledge facilitates all aspects of reproduction, artificial insemination, birthing, rearing, and even contraception efforts that AZA-accredited zoos and aquariums strive to achieve.

Reproduction differs slightly between capybaras in managed care facilities and their wild counterparts. In a managed care facility, females reach sexual maturity at 15 months of age while their wild counterparts tend to reach sexual maturity between 6–12 months of age. Otherwise, the physiology is the same for both managed and wild capybaras. At the onset of puberty, the uterus doubles in weight and the ovaries undergo follicular differentiation. The first luteal bodies are accompanied by a significant increase in progestogens in the blood (Moreira et al., 2013). A study completed on capybaras in a managed care setting showed that in capybaras that have yet to undergo puberty (~4 months old), the progestogens in the blood flux from 3.5–4.3 nmol/l. At the onset of puberty (~6 months), the blood progestogen level averages 9.45 nmol/l and when sexually mature (~1 year), the average level of is 13.36 nmol/l (Lopez-Barbella, 1993). During estrus, in the capybaras being studied, the blood progesterone level increased from 0.8 mg/ml to 3.5 mg/ml on day three of the cycle and then fell to 0.9 mg/ml on day four of the cycle. Likewise, the body temperature rose from 36–36.2 °C (96.8–97.2 °F) to 36.6 °C (97.9 °F) (Lopez-Barbella, 1982). Males appear to reach sexual maturity at 18 months of age (Moreira et al., 1997) and experience no seasonal changes. The testosterone concentration has been measured in male capybaras at 25.07 +/- 14.73 ng/ml (Costa & Paula, 2006).

The estrous cycle is approximately 7.5 +/- 1.2 days and ovulation typically lasts 8 hours. When estrus begins, a female capybara will typically exhibit vaginal smears with parabasal dark cells and some polymorphic cells followed by mucous secretion with few leucocytes. About 24–30 hours after the onset of estrus, the vaginal epithelium peels off keratinized cells, which may have pyknotic nuclei or no nuclei. Subsequently, these keratinized cells disappear from the vaginal smear, and there is a considerable increase in the keratin content in polymorphic cells for approximately 2–5 days. When the estrous cycle restarts, parabasal cells appear again in the vaginal cytology (Moreira et al., 2013).

Capybaras produce no external signs of being in estrus. During this period, females become receptive to the male and copulation—this usually occurs in the water. The female moves in and out of the water, always followed by the male, until she demonstrates receptivity. The male will initiate the courtship by scent marking and sniffing the female's external sex organ. The pair will then circle each other for approximately 5–10 minutes, culminating in the first initiation of coition, and this is when the female will first expose her vulva. The male will make about six attempts before introducing his penis, and ultimately ejaculates for about 3 seconds (Moreira et al., 2013).

8.2 Pregnancy & Parturition

It is extremely important to understand the physiological and behavioral changes that occur throughout an animal's pregnancy. Capybaras have the longest gestation of all rodents. Gestation length ranges from 147–156 days, and capybaras can produce litters of 4–8 young, with older females generally producing larger litters. Capybaras are born precocial, which is unusual for rodents.

Most facilities that have had success with a breeding program have not noticed any behavioral, psychological, or physical issues during mating. In fact, most facilities surveyed indicated that their capybaras often bred in view of all. In one accredited facility, breeding was observed in a recently introduced (within 6 months) capybara pair. There appears to be no separation necessary to facilitate mating and conception, as this typically occurs naturally with no interference from staff. However, during breeding behaviors, some superficial or minor bite wounds have occurred. Breeding in managed care facilities to this point has always been achieved by natural insemination as capybaras breed with little to no problems. No research has been completed on the use or merits of artificial insemination in capybaras. Institutions participating in the AZA Capybara Species Survival Plan® (SSP) have reported that there have been no special husbandry requirements or limitations.

Most facilities do not provide materials strictly for nesting. Capybaras are usually provided with hay, straw, or mulch on exhibit, and are simply provided with an increased amount of materials as parturition nears. Signs of parturition can be physical or behavioral, and facilities that have had success in breeding

capybaras record that as parturition neared the female separated herself from the rest of the herd. Females nearing parturition may also start to vocalize more. Pregnant females should not be completely isolated from the herd. The den or sheltered area should have a section that is heavily bedded with hay or straw for her comfort.

Potential problems that can arise during parturition include stillborns, infanticide, and dystocia. In several instances in managed care facilities, conception was achieved inadvertently while the female was being given MGA for contraception. In these instances, the females did not carry the pregnancy to term due to either natural miscarriage or veterinary intervention. In the event of complications during parturition, cesarean sections can be used to birth the pups.

Lactation in capybaras lasts for approximately 10 weeks. In the wild, capybaras have been observed utilizing nursery groups; at times one female has been seen with up to 10 pups of different ages and sizes (Grzimek et al., 2004). It is believed that the lactating females in the group take turns caring for all the young in the group. At least one of the surveyed institutions reported seeing females sharing pup-rearing duties on exhibit. Due to this natural behavior, females should not be separated from the herd during this time, however if it becomes necessary, then the female in question should remain within smelling distance of the herd to prevent any reintroduction issues.

Figure 8. Capybara pups
Photos courtesy of C. Atherton

8.3 Birthing Facilities

As parturition approaches, animal care staff should ensure that the mother is comfortable in the area where the birth will take place, and that this area is “baby-proofed.”

The capybara is a precocial species. As such, few changes need to be made to prepare an exhibit for young. Facilities may decide to lower the level of their pools; however, capybaras are known to swim as early as the day they are born. Based on the design of the capybara exhibit, adjustments may need to be made so neonates can access (or be prevented from accessing) different areas of the exhibit. Culverts that are associated with ponds or pools need to be grated to prevent neonates from swimming through them. Exhibits that contain mixed species require additional monitoring for negative interactions between species, and it is necessary to separate species for a short period. Facilities may also choose to offer a smaller pelleted food for neonates.

When nearing parturition, females generally prefer to be alone, returning to the herd shortly after parturition. Physical separation by keepers is unnecessary as long as access to a holding area is provided, allowing females some privacy. Keepers should use caution when working around newborns since the mothers and other females in the herd become very protective of their young. Once a female has given birth, usually in the early morning hours, the area should be kept quiet with as little activity as possible. Because the young are precocial, this does not have to last longer than a day or two.

Most facilities only perform an initial introduction. As births typically occur in the exhibit or holding areas, re-introductions are not usually required. Typically, no special procedures are associated with re-introduction.

8.4 Assisted Rearing

Although mothers may successfully give birth, there are times when they are not able to properly care for their offspring, both in the wild and in *ex situ* populations. Fortunately, animal care staff in AZA-accredited institutions are able to assist with the rearing of these offspring when deemed necessary. A mother may not be able to care adequately for her offspring due to maternal aggression, refusal to nurse, poor lactation, or the health of the offspring. Neonates that are obviously injured or too weak to survive on their own should be removed from the herd for hand-rearing. The hand-rearing techniques for capybaras mentioned in this section were developed by Kimberly Cook, DVM.

A syringe or baby bottle can be used for feeding milk. Puppy milk replacer Esbilac® can be used for formula. If diarrhea develops, the formula may need to be cut with ~25% water. Check the nipple hole of the bottle to prevent a fast flow of formula into the neonate's mouth, which could cause aspiration in the lungs and result in pneumonia. Any formula that is open to the air should be used within 24 hours. Bottles should be washed with warm water and scrubbed with a brush to remove milk residue from the bottle, followed by a disinfectant bath to soak the bottle and nipple. A solution of 28 g (1 oz.) Nolvalsan® or bleach to 1 gallon of warm water is adequate. Before filling the bottle with fresh formula, rinse the bottle and nipple well to remove residue from disinfectants.

Of the AZA-accredited facilities surveyed in 2014, only one facility has had cause to hand-rear a capybara offspring. In that circumstance, the pup was kept at 24 °C (75.2 °F), weighed every day at the first feeding, and detailed records were kept of its progress. The pup was fed Esbilac®, and feeding occurred every 3 hours, day and night, for the first 3 weeks. During weeks four and five, the volume per feeding was increased while the number of feedings was decreased, until night feedings were eventually discontinued. From 3 days of age onward, the pup was offered solid food (grasses, fruits, grains, and romaine lettuce) in addition to the Esbilac®. The pup was weaned at 16 weeks. In addition to feedings, staff also oversaw visits with the pup's mother and siblings in an effort to get the mother to nurse. The visits were tailored to the level of interest the animals displayed in each other with the end goal of placing the pup back into her mother's care.

8.5 Contraception

Many animals cared for in AZA-accredited institutions breed so successfully that contraception techniques are implemented to ensure that the population remains at a healthy size. In the case of an animal on loan from another facility, consult the loan agreement or owner regarding authority to contracept. In the case of permanent contraception, prior permission of the animal's owner must be obtained.

Recommended methods of reversible contraception include Melengestrol acetate (MGA) implants, MGA liquid, Depo-Provera® injections, and Suprelorin® (deslorelin). The individual's health, immobilization risk, social situation, and facility design (i.e., constraints on delivery methods) are factors to consider in determining the most appropriate contraceptive method for a particular capybara.

Progestins: Melengestrol acetate (MGA) implants are the most widely used contraceptive in zoos and are the most commonly used product in capybaras. The MGA implant consists of a silastic rod containing 20% by weight of MGA. Implants can be ordered from Wildlife Pharmaceuticals, with dosages provided by the AZA Wildlife Contraception Center (WCC). MGA implants are considered effective for a minimum of 2 years, but may be active longer. Thus, for continued contraception, implants should be replaced every 2 years. If reversal is desired, the implant should be removed even if 2 years have elapsed. Progestin contraceptives should not be used during pregnancy in capybaras because they may delay parturition, resulting in stillbirth or death. Progestins are considered safe to use in lactating females.

MGA Liquid is an oral contraceptive containing MGA dissolved in a special propylene glycol formulation. MGA Liquid is easy to administer and clears rapidly from the system once discontinued, but the full dose should be consumed every day in order to be effective. Institutions wanting to use MGA Liquid should contact the WCC for dosage recommendations and will need to register with the WCC prior to ordering from Wildlife Pharmaceuticals.

Depo-Provera® is an injectable contraceptive that contains the synthetic progestin medroxyprogesterone acetate. The recommended dose is 5 mg/kg of body weight every 2 months. There is individual variation in time to reversal, but this can be as long as 2 years. It may be best used as an interim contraceptive method or in individuals for which immobilization poses a health risk. Depo-Provera® can be ordered

from a human or veterinary pharmacy. Depo-Provera® injections are a good short-term option. It has been used most often in seasonal species, species in which anesthesia for implant insertion is problematic, and as an immediately available interim contraceptive. Individuals should be separated for 1 - 2 weeks following their injection, if you feel confident that they can be easily reunited. Depo-Provera® has been linked to mood changes in some taxa. Depo-Provera® is highly reversible and, being progestin, is thought to be fine to use for lactating mothers.

Gonadotropin releasing hormone (GnRH) agonist: GnRH agonist (e.g., Suprelorin® (deslorelin)) implants achieve contraception by suppressing the reproductive endocrine system, preventing production of pituitary (FSH and LH) and gonadal steroid hormones (estradiol and progesterone in females and testosterone in males). The observed effects are similar to those following either ovariectomy in females or castration in males. GnRH agonists first stimulate the reproductive system, which can result in estrus and ovulation in females and temporary enhancement of testosterone and semen production in males. Then, down-regulation follows the initial stimulation. The stimulation phase can be suppressed by daily Ovaban® (megestrol acetate) administration for one week before and one week after implant placement (Wright et al., 2001); however, this protocol has not yet been tested in capybaras. Alternatively, it is considered safe to separate female capybaras from males for two to three weeks following implantation.

Suprelorin® can also be an effective contraceptive in males. However, it is more commonly used in females, because monitoring efficacy in females by suppression of estrous behavior or gonadal steroids in feces is usually easier than ensuring continued absence of sperm in males, since most institutions cannot perform regular semen collections. If used in males, disappearance of sperm from the ejaculate following down-regulation of testosterone may take an additional six weeks, as with a vasectomy. Suprelorin® should not be used during pregnancy, since it may cause spontaneous abortion or prevent mammary development necessary for lactation. Institutions wanting to order Suprelorin® should submit a registration form that can be found on the WCC's website.

A drawback of Suprelorin® is that time of reversal can be quite variable. The most widely used formulations are designed to be effective for a minimum of either 6 or 12 months, but the actual time of suppression can be longer in some individuals. Preliminary results indicate that removing the implants can hasten reversal. More information for placing implants to facilitate removal can be found at www.stlzoo.org/contraception.

Chapter 9. Behavior Management

9.1 Animal Training

Classical and operant conditioning techniques have been used to train animals for over a century. Classical conditioning is a form of associative learning demonstrated by Ivan Pavlov. Classical conditioning involves the presentation of a neutral stimulus that will be conditioned (CS) along with an unconditioned stimulus (US) that evokes an innate, often reflexive, response. If the CS and the US are repeatedly paired, eventually the two stimuli become associated and the animal will begin to produce a conditioned behavioral response to the CS.

Operant conditioning uses the consequences of a behavior to modify the occurrence and form of that behavior. Reinforcement and punishment are the core tools of operant conditioning. Positive reinforcement occurs when a behavior is followed by a favorable stimulus to increase the frequency of that behavior. Negative reinforcement occurs when a behavior is followed by the removal of an aversive stimulus to also increase the frequency of that behavior. Positive punishment occurs when a behavior is followed by an aversive stimulus to decrease the frequency of that behavior. Negative punishment occurs when a behavior is followed by the removal of a favorable stimulus also to decrease the frequency of that behavior.

AZA-accredited institutions are expected to utilize reinforcing conditioning techniques to facilitate husbandry procedures and behavioral research investigations. Institutions should follow a formal written animal training program that facilitates husbandry, science, and veterinary procedures and enhances the health and well-being of the animals (AZA Accreditation Standard 1.6.4).

AZA Accreditation Standard

(1.6.4) The institution should follow a formal written animal training program that facilitates husbandry, science, and veterinary procedures and enhances the overall health and well-being of the animals.

Capybaras are able to be trained, and behaviors that have been successfully trained include: crate training, scale training, target training, tactile, halter, hand feeding, belly rubs, shifting, station training, standing for microchip reading, standing on hind legs, voluntary injections, following, and opening their mouth. Although not all facilities engage in a training program, training is recommended to help with necessary procedures. Capybaras are not typically used as ambassador animals, but they may be used for guest feedings and donor experiences. Training sessions may also be done for guests at the capybara exhibit.

Capybaras are not classified as dangerous animals, and as such, most facilities allow free contact. Additionally, keepers (and even guests and donors) may hand-feed capybaras. Most facilities that do not engage in a training program reported that their capybaras tend to stay away from keepers and never approach. To work with capybaras, most facilities simply require new keeper training, usually performed by current keeper staff. Common institutional rules for working with capybaras include moving slowly, refraining from sudden movements, taking care not to be surrounded, and carrying barriers with you.

9.2 Environmental Enrichment

Environmental enrichment, also called behavioral enrichment, refers to the practice of providing a variety of stimuli to the animal's environment, or changing the environment itself to increase physical activity, stimulate cognition, and promote natural behaviors. Stimuli, including natural and artificial objects, scents, and sounds, are presented in a safe way for the capybara to interact with. Some suggestions include providing food in a variety of ways (i.e., frozen in ice or in a manner that requires an animal to solve simple puzzles to obtain it), using the presence or scent/sounds of other animals of the same or different species, and incorporating an animal training regime (husbandry or behavioral research) into the daily schedule.

Enrichment programs for capybaras should take into account the natural history of the species, individual needs of the animals, and facility constraints. The capybara enrichment plan should include the following elements: goal setting, planning and approval process, implementation, documentation/record-keeping, evaluation, and subsequent program refinement. The capybara enrichment program should ensure that all environmental enrichment devices (EEDs) are "capybara" safe and are presented on a variable schedule to prevent habituation. AZA-accredited institutions must follow a formal written enrichment program that promotes capybara-appropriate behavioral opportunities (AZA Accreditation Standard 1.6.1). Enrichment activities must be documented and evaluated, and the program should be

refined based on the results, if appropriate. Records must be kept current (AZA Accreditation Standard 1.6.3).

Capybara enrichment programs should be integrated with veterinary care, nutrition, and animal training programs to maximize the effectiveness and quality of animal care provided. AZA-accredited institutions must have a specific paid staff member(s) assigned to oversee, implement, assess, and coordinate interdepartmental enrichment programs (AZA Accreditation Standard 1.6.2).

Examples of potential capybara enrichment include: olfactory items (perfumes, lotions, spices, scents from other animals), novel foods (see feeding section), puzzle feeders, tree stumps, browse, ice blocks, logs, boxes, various feeders, wood, tubes, phone books, laser pointers, mirrors, radios, paints, and mud wallows. There are no enrichment recordkeeping methods specific to capybaras; however, records of enrichment should be kept.

The range of species-appropriate behaviors for capybaras is not overly large and includes sleeping, eating, foraging, walking, swimming, chewing, and aggressive and non-aggressive behaviors (courting behaviors, dominance-determining behaviors). Animals typically respond to these behaviors socially using bark and click calls (two of several variations of vocalizations performed by capybaras).

AZA Accreditation Standard

(1.6.1) The institution must follow a formal written enrichment program that promotes species-appropriate behavioral opportunities.

AZA Accreditation Standard

(1.6.3) Enrichment activities must be documented and evaluated, and program refinements should be made based on the results, if appropriate. Records must be kept current.

AZA Accreditation Standard

(1.6.2) The institution must have a specific staff member(s) or committee assigned for enrichment program oversight, implementation, assessment, and interdepartmental coordination of enrichment efforts.

Husbandry training provides an opportunity for enrichment for capybaras. Human interaction gives animals the opportunity to interact with their primary keepers (building relationships with their keepers), novel keepers, and even the public (donor feeds). Operant conditioning training engages animals cognitively, requires the animal to think about what the trainer is asking, and allows them to work for rewards; both the cognitive stimulation and rewards can result in an enriching experience.

Formal written enrichment plans should include goal-setting, planning and approval process, implementation plans, documentation and record-keeping examples, the method of evaluation, and subsequent refinement methods. An example from an AZA accredited facility can be seen below in Table 14. Keepers should keep a calendar and provide the animal with the appropriate enrichment items for the day. Activity logs should be evaluated every 6 months (at least) to check the activity levels of the animals. All enrichment items should be approved based on the institution's policy. Items should be evaluated for safety prior to each use and the activity should be monitored during the first few times the animal interacts with the enrichment. This monitoring should include the primary keeper for the animals, the curator, and the vet. Items should be of materials and shapes that the animal cannot chew through or swallow. Animals should be slowly desensitized to the item if necessary to prevent the animal from developing a stress response to new items. After each use, any enrichment item needs to be evaluated for choking hazards (loose string, sharp edges, and obvious damage) and less obvious damage that can cause entrapment. If an enrichment item is hung in the exhibit, it should be done with the least amount of rope possible. For safety reasons, enrichment items and goals should be reviewed from a veterinary, nutritional, and training perspective. Consideration should be given for the potential for chewing and ingestion, and items that may become a hazard in a pool. No new enrichment items should be introduced during training sessions.

Enrichment opportunities should occur multiple times throughout the week and should be divided into multiple categories. Categories of enrichment for capybaras include:

- Human interaction (training, public interaction, novel keepers)
- Novel food (changes in diet presentation, feeding in a nontraditional manner (freezing, cooking, size of food, forage box) or items not normally fed)
- Occupational (non-food items that capybaras can manipulate and are meant to increase activity)
- Physical (semi-permanent exhibit furniture that is moved or removed and changed as part of the animals surroundings)
- Sensory (non-food that stimulates olfactory, visual, tactile, or auditory senses, such as bubbles, scents, or movies)
- Social (introduction of other animals either conspecific or counter specific)

Table 14. Sample capybara enrichment schedule

Report Date	Accession #	Keeper	Schedule	Item Given	Direct Evidence of use	Indirect Evidence of use	Category	Comments
1/8/2015				Strawberry extract scent	2		Senses - olfactory	
1/17/2015			1/17/2015	Outside for 30 mins	3		Senses - visual	
1/17/2015			1/17/2015	Cardboard		2	Senses - tactile	
1/21/2015			1/21/2015	Japanese cherry blossom scent on posts	3		Senses - olfactory	
1/23/2015			1/23/2015	Crafts paper		3	Occupational	

9.3 Staff and Animal Interactions

Animal training and environmental enrichment protocols and techniques should be based on interactions that promote safety for all involved.

In an effort to facilitate an animal training and enrichment program for capybaras, their exhibit should contain (or give access to) an easily cleanable pool with a drain, space to change out chewing material without compromising the rest of the exhibit design, and, if necessary, a restraint chute (helpful for veterinary procedures and training sessions).

Capybaras are non-dangerous animals and as such almost all surveyed facilities allow the keeper staff to work with capybaras with no barriers between them. Some precautionary measures (barriers, spray water bottle, etc.) may be required depending on the facility. As part of working the animals free contact, interactions may include hand feeding and tactile sessions. Keepers and any guests should always be cognizant as to where their hands are in relation to the animals' teeth. In addition, capybaras can be skittish and may become spooked during interactions.

Training animals to shift from one area to another becomes helpful in terms of separating animals for training sessions to prevent distractions, as well as separating animals when enrichment is being offered in the form of food to prevent overeating.

Figure 9. Capybara interacting with staff member
Photo courtesy of S. Curtis

9.4 Staff Skills and Training

Capybara staff members should be trained in all areas of capybara behavior management. Funding should be provided for AZA continuing education courses, related meetings, conference participation, and other professional opportunities. A reference library appropriate to the size and complexity of the institution should be available to all staff and volunteers to provide them with accurate information on the behavioral needs of the animals with which they work.

To meet the behavioral needs of capybaras, staff should have basic animal skills and knowledge, prior experience handling and caring for capybaras, and a basic understanding of operant conditioning. If a keeper does not have prior capybara experience, they should be given the opportunity to shadow another more experienced keeper and to participate in continuing education.

Chapter 10. Ambassador Animals

AZA recognizes many public education and, ultimately, conservation benefits from ambassador animal presentations. AZA's Conservation Education Committee's Ambassador (previously called Program) Animal Position Statement (Appendix E) summarizes the value of ambassador animal presentations. For the purpose of this policy, an ambassador animal is described as an animal presented either within or outside of its normal exhibit or holding area that is intended to have regular proximity to or physical contact with trainers, handlers, the public, or will be part of an ongoing conservation education/outreach program. Capybaras can make great ambassador animals if they begin desensitization to handling and training at as young an age as possible.

Ambassador animal presentations bring a host of responsibilities, including the welfare of the animals involved, the safety of the animal handler and public, and accountability for the take-home, educational messages received by the audience. Therefore, AZA requires all accredited institutions that give ambassador animal presentations to develop an institutional ambassador animal policy that clearly identifies and justifies those species and individuals approved as ambassador animals and details their long-term management plan and educational program objectives. If an animal on loan from another facility is used as an ambassador animal, the owner's permission is to be obtained prior to program use.

AZA Accreditation Standard

(1.5.4) If ambassador animals are used, a written policy on the use of live animals in programs must be on file and incorporate the elements contained in AZA's "Recommendations For Developing an Institutional Ambassador Animal Policy" (see policy in the current edition of the *Accreditation Standards and Related Policies* booklet). An education, conservation, and welfare message must be an integral component of all programs. Animals in education programs must be maintained and cared for by paid and/or unpaid trained staff, and housing conditions must meet standards required for the remainder of the animals in the institution. While outside their primary enclosure, although the conditions may be different, animal safety and welfare need to be assured at all times.

10.1 Ambassador Animal Husbandry

AZA's accreditation standards require that the conditions and treatment of animals in education programs must meet standards set for the remainder of the animal collection, including species-appropriate shelter, exercise, sound and environmental enrichment, access to veterinary care, nutrition, and other related standards (AZA Accreditation Standard 1.5.4). All record-keeping requirements noted previously apply to ambassador animals (AZA Accreditation Standards 1.4.1, 1.4.2, 1.4.3, 1.4.4, 1.4.5, 1.4.6, and 1.4.7). In addition, providing ambassador animals with options to choose among a variety of conditions within their environment is essential to ensuring effective care, welfare, and management. Some of these requirements can be met outside of the primary exhibit enclosure while the animal is involved in a program or is being transported. For example, housing may be reduced in size compared to a primary enclosure as long as the animal's physical and psychological needs are being met during the program; upon return to the facility the animal should be returned to its species-appropriate housing as described above.

Enclosure design and size: Providing ambassador animals with options to choose among a variety of conditions within their environment is essential to ensuring effective care, welfare, and management (AZA Accreditation Standard 1.5.2.2). Careful consideration must be given to the design and size of all ambassador animal enclosures, including exhibit, off-exhibit holding, hospital, quarantine, and isolation areas, such that the physical, social, behavioral, and psychological needs of the species are met and species-appropriate behaviors are facilitated (AZA Accreditation Standard 10.3.3, 1.5.2, 1.5.2.1).

Ambassador capybaras are typically housed in separate enclosures from exhibit animals. However, they still have the same physical and psychological needs as non-ambassador capybaras and their enclosures must help maintain physical and behavioral normality. The needs of ambassador capybaras can be consistently maintained by housing them in a social setting and providing them access to pools and open spaces that will allow them the opportunity to run, play, and forage.

With only two facilities utilizing capybaras for education programs, procedures for housing vary widely. The average space the capybaras are kept in is 5.6 m^2 (60 ft^2), with one space being 9.3 m^2 (100 ft^2) and the other being 1.9 m^2 (20 ft^2). One facility houses two capybaras together where they are trained together for shows, while the other facility has only one capybara, which is housed with a king crane (*B. regulorum*) and domesticated ducks for social enrichment. Guidelines for the overall size of ambassador capybara housing can be found in Table 15 below.

Table 15. Criteria for evaluating ambassador capybara housing

Good	Better	Best
18.288 m^2	30.48 m^2 with regular access to outdoor exercise yard	>30.48 m^2 including both indoor and outdoor holding

Ambassador capybaras require access to water in their enclosures for drinking and encouraging other natural behaviors, such as wallowing and defecation. Pools can vary in size and style, ranging from 88.9 cm (35 in.) x 88.9 cm (35 in.) x 15.24 cm (6 in.) to 214.48 cm (84 in.) at the two facilities with ambassador capybaras, and should be large enough to allow the animal to comfortably roll around and lie down. Water should be accessible year-round, particularly as a cooling option seasonally on warmer days; both facilities had pools outdoors for use in hot weather.

Temperature guidelines: Air temperature determines if ambassador animals have outside access, although the temperature guidelines should be defined by individual facilities. Identify limitations/restrictions regarding ambient temperatures/weather conditions based on exhibit elements such as shelter, substrate, and water (i.e., heated pool). Capybaras are larger rodents and not overly susceptible to cold temperatures unless extreme conditions exist. One of the facilities that houses ambassador capybaras keeps the enclosure at 26.6 °C (80 °F) and if the temperature falls below 15.6 °C (60 °F) then the capybaras are denied access to their outside space. The second facility has a heated pool kept at 22.2 °C (72 °F) and therefore the temperature threshold for denying outside access is lower at 7.2 °C (45 °F). When capybaras are transported for programs in extreme heat, cooling elements such as fans and misters can be provided.

Diet: A formal nutrition program is recommended to meet the nutritional and behavioral needs of any species (AZA Accreditation Standard 2.6.2). Diets should be developed using the recommendations of nutritionists, including the Nutrition Scientific Advisory Group (NAG) feeding guidelines: (https://nagonline.net/guidelines-aza-institutions/feeding-guidelines/), and veterinarians as well as AZA Taxon Advisory Groups (TAGs), and Species Survival Plan® (SSP) Programs. Diet formulation criteria should address the animal's nutritional needs, feeding ecology, as well as individual and natural histories to ensure that species-specific feeding patterns and behaviors are stimulated.

Capybaras trained as ambassadors have shown preferences for corn, sweet potato, and apple. Carrots, grapes, a variety of greens, bamboo, and rodent block may also be used as training reinforcers. For detailed information about the diets of capybaras in managed care, see Chapter 6.

Enrichment: Capybaras are gregarious and curious creatures, so opportunities for social interaction and exploration of new environments are both important. At one of the facilities that house ambassador capybaras, there is a play yard for the program animal where there are enrichment items like puzzle feeders and boomer balls, and social interactions with ducks and a king crane can take place. Exhibit elements should include water for soaking and mud for wallowing. Browse is another important kind of enrichment, as capybaras eat a variety of plants and tree bark in the wild. Finally, scent marking is another common natural behavior for capybara, so offering new and diverse scents can also be enriching.

Animal training: To enrich the lives of ambassador capybaras, trainers hold maintenance sessions for behaviors they want to see. Some of these behaviors are shifting when called, walking behind the trainers to get to kennels, and stationing in the holding area behind the stage. Positive reinforcement has also been used to train capybaras to station on stage, walk with the trainer on stage, and go into the pool and swim backstage. Successfully trained husbandry behaviors include scale, crate, shifting, and coming when called/recall. Capybaras can be trained to respond to either their names or another indiscriminate sound. As with many larger mammals, capybaras may also be trained for a voluntary blood draw.

Social grouping: Ambassador capybaras can be housed alone if given enough opportunities for social interactions with trained staff. As long as spatial considerations are appropriate, capybaras might also be housed in pairs (same or different sex) or small groups (same sex). Due to their social nature, it is possible to train parent-reared capybaras from a young age.

Signs of stress: It is important to avoid stress when dealing with animals in managed care. Defining stressors can be accomplished by communicating with other trainers in the holding area and on stage during the shows. Beyond physically viewing the behavior at the time of the incident, trainers can check video footage to find behavioral changes overnight. Capybara stress may manifest its presence in a

number of ways; it is important to identify behavioral changes such as huffing, freezing in place, and not eating. Once something has been identified as a stressor, it should be removed from the animal's surroundings. If the capybara shows continued stress, the animal should be returned it to its exhibit or enclosure as soon as possible and supervised.

The welfare of capybaras can also be judged through veterinary checks including fecal samples, body condition scoring, and weight evaluation. Trainers should constantly evaluate willingness to participate in various behaviors during stage routines and training sessions, as well as their confidence levels when on stage and in the kennel. Signs that an individual's welfare may need to be further evaluated include negative or stereotypical behaviors (pacing, head bobbing, etc.) and aggression toward other animals or trainers. Area managers should be alerted to the situation and medical staff should be contacted, if warranted.

10.2 Institutional Ambassador Animal Programs

AZA's policy on the presentation of animals is as follows: AZA is dedicated to excellence in animal care and welfare, conservation, education, research, and the presentation of animals in ways that inspire respect for wildlife and nature. AZA's position is that animals should always be presented in adherence to the following core principles:

- Animal and human health, safety, and welfare are never compromised.
- Education and a meaningful conservation message are integral components of the presentation.
- The individual animals involved are consistently maintained in a manner that meets their social, physical, behavioral, and nutritional needs.

AZA-accredited institutions that have designated ambassador animals are required to develop their own Institutional Ambassador Animal Policy that articulates and evaluates the program benefits (see Appendix G for recommendations). Ambassador animals should be consistently maintained in a manner that meets their social, physical, behavioral, and nutritional needs. Education and conservation messaging must be an integral component of any ambassador animal demonstration (AZA Accreditation Standard 1.5.3).

AZA Accreditation Standard

(1.5.3) If animal demonstrations are a part of the institution's programs, an educational/conservation message must be an integral component.

Program types: Ambassador animals are utilized in many ways and many settings to engage, educate, and create connections with zoo visitors. Capybaras may be presented both on and off-grounds in formal (encounters or wildlife shows) and informal (chats or displays) programs. While program presentations typically occur outdoors, they can take place in a variety of environments with proper training and attention to the safety of the animals.

Figure 10. Capybara on stage during program
Photo courtesy of S. Curtis

Messaging:
The Conservation Education Committee recommends that facilities design educational experiences with ambassador animals with one or more the following outcomes in mind:

1. Species information: Understanding of the species natural history, role in the ecosystem, and/or status in the wild.
2. Animals in human care: Understanding of the commitment of AZA facilities to excellence in animal care and conservation and appropriate pet choices, where applicable.
3. Empathy development: Foster a sense of empathy and wonder by connecting visitors and audiences to the individual animal.
4. Conservation action: Empower audiences and visitors to take action to protect the species and wildlife in general.

Capybara presented in an educational setting provide an opportunity to achieve these outcomes in a number of specific ways. Recommendations for messaging with capybara are listed below.

Outcome 1: Species information

- Capybara are the world's largest rodents. Native to South America, they are social animals that live in family groups of a mated pair and their offspring. Pups are precocial and are eating solid food within weeks of birth. During the dry season, when food is more scarce, they can be found in larger groups.
- Capybara are semi-aquatic and spend a lot of time in the water. They eat grasses and reeds along waterways and have several adaptations for this lifestyle including webbed feet and the ability to hold their breath for up to five minutes. Their eyes, ears, and nose are all located near the top of their head, giving them the ability to stay nearly submerged and avoid predators, which include jaguar, anaconda, and caiman.
- Capybara are currently listed by IUCN as Least Concern. There are areas of their range where populations have dwindled such as Ecuador. Overall, however, they are well represented.

Outcome 2: Animals in human care

- Capybara do not make good pets. They require a large source of clean water, fresh produce, space, and they can get aggressive. As they are rodents, their incisors are large and can cause damage when they bite. They are also social animals and humans are not the same as conspecifics.
- In contact and behind-the-scenes programs, there is an opportunity to explain more thoroughly the relationship that trainers have with the capybaras. The trainer should discuss the benefits of training, positive reinforcement training, and the time spent developing the animal/human relationship, in order to ensure the capybara is comfortable in its program role.

Outcome 3: Empathy development

- Every presentation should begin by setting expectations, explaining rules and sharing what to expect. Empathy development can be encouraged by presenting the capybara in a way that helps the audience connect on an individual level. Some suggestions include: Sharing information about the individual animal, such as name, age, weight, preferences, and temperament, and discussion of how the capybara is like us.
- A presentation of any capybara, regardless of temperament, is an opportunity to explain that prey animals are very alert to signs of danger. Loud noises or sudden movements, for example, could startle or scare them, they same way they might scare one of us, especially if we are on the lookout for danger.
- Allowing the capybara to display natural behaviors, such as swimming, and positive interactions with the presenter is also an opportunity to foster empathy.

Outcome 4: Conservation action

- Capybara are semi-aquatic and are dependent on healthy aquatic ecosystems to survive. Encouraging visitors to conserve water and protect aquatic habitats can help to protect capybara and many other species. Programs such as "Cans for Corridors" can directly help capybaras and other species in the ecosystem such as tamarins and jaguars by helping to reconnect fragmented forests in Brazil.

Social, physical, behavioral, and nutritional needs: The needs of a capybara being used as an ambassador animal can be consistently maintained by keeping them housed in an appropriate social setting and providing them access to pools and open spaces that will allow them the opportunity to run, play, and forage. All nutritional needs should be monitored by a nutritionist or veterinary staff. Trained behaviors should be maintained by having consistent sessions where the individual is encouraged to perform the behavior for a form of positive reinforcement (clicker, food, etc.). Detailed information about effective care of ambassador capybaras can be found in Chapter 10.2.

Animal care and education staff should be trained in ambassador animal-specific handling protocols, conservation, and education messaging techniques, and public interaction procedures. Paid and/or unpaid staff assigned to handle animals during demonstrations or educational programs must be trained in accordance with the institution's written animal handling protocols. Such training must take place before handling may occur (Accreditation Standard 1.5.12). These staff members should be competent in recognizing stress or discomfort behaviors exhibited by the ambassador animals and be able to address any safety issues that arise. Additionally, when in operation, animal contact areas must be supervised by trained paid and/or unpaid staff (AZA Accreditation Standard 1.5.13).

AZA Accreditation Standard

(1.5.12) Paid and/or unpaid staff assigned to handle animals during demonstrations or educational programs must be trained in accordance with the institution's written animal handling protocols. Such training must take place before handling may occur.

AZA Accreditation Standard

(1.5.13) When in operation, animal contact areas (petting zoos, touch tanks, etc.) must be supervised by trained, paid and/or unpaid staff.

Display options and contact: Careful consideration should be given to the presentation of ambassador animals, including safety of the animal, handler, and public. When using capybaras as ambassador animals, there should be a visitor management strategy in place. Currently, facilities using capybaras in education programs do not permit visitor interaction with the animals. As there is typically a barrier between the animal and guests, except during special private tours, no official guidelines on touching exist. Guests participating in such an events should remain seated to prevent the animal from startling, and the capybara should approach the guest of their own accord. Children under the age of five should not participate in these interactions, as they typically do not possess the ability to interact calmly with the capybara.

10.3 Handling and Staff Training

Handling limits: Consideration should be given as to appropriate times for handling ambassador animals during presentations, and rest breaks should be scheduled accordingly. Program handlers should maintain the animal's basic husbandry needs and a medical protocol should be in place in case concerns arise. Ambassador animals should be rotated whenever possible; if this is not possible, then the individual should be given ample rest periods during/between programs.

Individual animals should be allowed to choose whether or not participate in programming; this can be signaled by a voluntary crate behavior. Capybaras should be trained to participate in programming as appropriate for the setting, and can typically stay motivated and focused for up to five minutes at a time, although individual animals may have variable limits. The capybaras currently being used as program animals have a high level of motivation and work well for food.

Handlers training and evaluation: Handling consistency and uniformity in technique for those who work with ambassador capybaras can have a profound impact on the welfare of these animals. Although this is best achieved through careful and species-specific training, handlers and their requirements will differ between institutions depending on many variables including staff size, program demands, and size of ambassador animal collection. AZA accreditation standards require that all handlers, whether staff or

volunteers, must be properly trained in the institution's protocols before handling occurs (AZA Accreditation Standard 1.5.12.)

It is generally recommended to keep a smaller number of staff members working with an ambassador capybara due to the type of training required to maintain positive, safe, and healthy interactions. Training for these types of programs requires the staff to build lasting relationships with each individual animal to produce the best results. At AZA-accredited facilities with ambassador capybaras, staff members new to the area are trained by managers or supervisors. Only certified handlers may have contact with the animals.

Each institution should create an ambassador animal handling policy that conforms to AZA guidelines as well as any local legislation. The program, including species/individual animals, program types/messaging, and all handlers, should be reviewed regularly. Handler competency should be evaluated, and concerns with training performance should be addressed.

The performance of capybaras used in an education program should be reviewed after each show in an effort to identify any problems or things that need to be modified prior to the next show. To keep trainers accountable for their work, detailed training records should be kept. If a problem arises, incident reports should be used, and if an incident reoccurs, the staff member should be retrained on appropriate techniques for working with capybaras.

Ambassador animals that are taken off zoo or aquarium grounds for any purpose have the potential to be exposed to infectious agents that could spread to the rest of the institution's healthy population. AZA-accredited institutions must have adequate protocols in place to avoid this (AZA Accreditation Standard 1.5.5).

Safety and disease: Injuries to the capybaras and staff are not common. However, if an animal is injured, the vet staff should be called and the incident should be reported to the area manager of the zoo. If an injury to a trainer occurs, a safety officer and a manager should file an incident report, and additional institutional policies should be followed. In the two AZA-accredited facilities that use capybaras as ambassador animals, the public has no direct physical contact with the capybaras, so no associated injuries can occur and there should be minimal zoonotic disease risk for visitors.

Proper hygiene and cleaning protocols should be followed to prevent the spread of disease. Keepers should follow basic hygiene procedures including washing hands and disinfecting footwear, immediately after handling any ambassador animal and/or their food items. Enclosures used by ambassador animals, including pools and crates, should be disinfected based on the normal protocols of the facility.

Both facilities utilizing capybaras in their programs have similar guidelines for disinfecting the enclosures though the frequency differs. One facility's disinfectant schedule mandates that enclosure disinfecting occurs three times a week and the crates get disinfected once a week, while the second facility disinfects the enclosure every day and the pools in the enclosure are cleaned three times a day due to the size of the pool.

Careful consideration must be given to the design and size of all ambassador animal enclosures, including exhibit, off-exhibit holding, hospital, quarantine, and isolation areas, such that the physical, social, behavioral, and psychological needs of the species are met and species-appropriate behaviors are facilitated (AZA Accreditation Standards 10.3.3, 1.5.2, 1.5.2.1, 1.5.2.2).

Similar consideration needs to be given to the means in which an animal will be transported both within the institution's grounds, and to/from an off-grounds program. Animal transportation must be conducted in a manner that is lawful, safe, well planned, and coordinated, and minimizes risk to the animal(s), employees, and general public (AZA Accreditation Standard 1.5.11).

Transport: Capybaras can be trained to voluntarily enter a transport carrier to move between their enclosure and the stage or program area. Alternatively, in the appropriate setting, the capybara may be trained to just walk to the kennel behind the stage and wait to go on. These behaviors are trained using positive reinforcement techniques.

Transport for capybaras may vary from commercial dog carriers to custom crates. If the animal is crated, adequate ventilation should be ensured and the carrier should be secured closed to prevent animal escape. Substrate is optional when crated for short periods, but hay may be used if they are to be in the carriers for more than a few minutes. When capybaras are transported for programs in extreme heat, cooling elements such as fans and misters can also be provided. As capybaras are rodents, be sure to inspect carriers regularly for signs of chewing.

The use of chemical sanitation is important for all transport carriers, presentation surfaces and maintenance tools. There is a variety of sanitation chemicals available for proper hygiene. A facility's animal management team and/or medical staff should be consulted to identify the best chemical compounds for the situation.

If a capybara is taken off grounds (this only occurs at one of the two AZA-accredited facilities housing ambassador capybaras), the animal should be quarantined for 30 days after it comes into contact with animals that are not from its facility. For more information on the transportation of capybaras, see Chapter 4.

10.4 Program Evaluation

AZA-accredited institutions that have an Institutional Ambassador Animal Plan are required to evaluate the efficacy of the plan routinely (see Appendix G for recommendations). Education and conservation messaging content retention, animal health and well-being, guest responses, policy effectiveness, and accountability and ramifications of policy violations should be assessed and revised as needed.

The welfare of capybaras can be judged through veterinary checks where fecal samples, body condition scoring, and weight are assessed. The trainers can check for behavioral changes like excitement to participate in various behaviors (e.g., stage routines and training sessions), as well as their confidence levels when on stage and in the kennel. Signs that an individual's welfare may need to be evaluated include negative or stereotypical behaviors (pacing, head bobbing, etc.) and aggression toward other animals and trainers.

The performance of capybaras used in an education program should be reviewed after each show in an effort to identify any problems or things that need to be modified prior to the next show. To keep trainers accountable for their work, detailed training records should be kept. If a problem arises, incident reports should be used, and if an incident reoccurs, the staff member should be retrained on appropriate techniques for working with capybaras. Recommendations to check the overall value and effectiveness of the program can be done through meet and greets, surveys, and questionnaires for the patrons of the show. In addition, changing the behaviors and facts presented during programs at least once a year encourages guests to return to see the program.

Chapter 11. Research

11.1 Known Methodologies

AZA believes that contemporary capybara management, husbandry, veterinary care and conservation practices should be based in science, and that a commitment to scientific research, both basic and applied, is a trademark of the modern zoological park and aquarium. AZA-accredited institutions have the invaluable opportunity, and are expected, to conduct or facilitate research in both *in situ* and *ex situ* settings to advance scientific knowledge of the animals in our care and enhance the conservation of wild populations. Participating in AZA Taxon Advisory Group (TAG) or Species Survival Plan® (SSP) Program sponsored research when applicable, conducting and publishing original research projects, affiliating with local universities, and/or employing staff with scientific credentials could help achieve this (AZA Accreditation Standard 5.3). An AZA institution must demonstrate a commitment to scientific study that is in proportion to the size and scope of its facilities, staff, and animals (AZA Accreditation Standard 5.0).

AZA Accreditation Standard

(5.3) The institution should maximize the generation and dissemination of scientific knowledge gained. This might be achieved by participating in AZA TAG/SSP sponsored studies when applicable, conducting and publishing original research projects, affiliating with local universities, and/or employing staff with scientific credentials.

AZA Accreditation Standard

(5.0) The institution must have a demonstrated commitment to scientific study that is in proportion to the size and scope of its facilities, staff (paid and unpaid), and animals.

All record-keeping requirements noted previously apply to most research animals, especially those which are part of the exhibit collection. When an animal on loan to a facility is subject to an invasive research procedure, including when done as part of a routine health exam, the owner's prior permission is to be obtained.

Research investigations, whether observational, behavioral, physiological, or genetically based, should have a clear scientific purpose with the reasonable expectation that they will increase our understanding of the species being investigated and may provide results which benefit the health or welfare of animals in wild populations. Many AZA-accredited institutions incorporate superior positive reinforcement training programs into their routine schedules to facilitate sensory, cognitive, and physiological research investigations and these types of programs are strongly encouraged by the AZA.

AZA-accredited institutions are required to follow a clearly written research policy that includes a process for the evaluation of project proposals and identifies the types of research being conducted, methods used, staff involved, evaluations of the projects, animals included, and guidelines for the reporting or publication of any findings (AZA Accreditation Standard 5.2). Institutions must designate a qualified staff member or committee to oversee and direct its research program (AZA Accreditation Standard 5.1).

AZA Accreditation Standard

(5.2) The institution must follow a formal written policy that includes a process for the evaluation and approval of scientific project proposals, and outlines the type of studies it conducts, methods, staff (paid and unpaid) involvement, evaluations, animals that may be involved, and guidelines for publication of findings.

AZA Accreditation Standard

(5.1) Scientific studies must be under the direction of a paid or unpaid staff member or committee qualified to make informed decisions.

An Institutional Animal Care and Use Committee (IACUC) should be established within the institution if animals are included in research or instructional programs. The IACUC should be responsible for reviewing all research protocols and conducting evaluations of the institution's animal care and use.

If institutions are not able to conduct in-house research investigations, they are strongly encouraged to provide financial, personnel, logistical, and other support for priority research and conservation initiatives identified by Taxon Advisory Groups (TAGs) or Species Survival Plans® (SSP) Programs.

11.2 Future Research Needs

This Animal Care Manual is a dynamic document that will need to be updated as new information is acquired. Knowledge gaps have been identified throughout the manual and are included in this section to promote future research investigations. Knowledge gained from these areas will maximize AZA-accredited institutions' capacity for excellence in animal care and welfare as well as enhance conservation initiatives for the species.

Items that may be of relevance and should be researched in the future include:

- If variation in light cycles plays an important role in affecting capybara behavior (Chapter 1.2)
- How well capybaras can tolerate vibrations (Chapter 1.4)
- The complete social context of scent marking and its function (Chapter 2.1)
- The safety and suitability of browse and forage species for capybaras (Chapter 6.2)
- The use of artificial insemination in capybara breeding (Chapter 8.2)
- Suppression of the stimulation phase caused by GnRH agonists via daily Ovaban® (megestrol acetate) administration for one week before and one week after implant placement in capybaras (Chapter 8.5)

Chapter 12. Other Considerations

12.1 Surplus Animals

All SSP species held by institutions should be reported to the SSP Program Leader. The SSP Program Leader should be responsible for making the decision as to whether or not specific animals are to be included in the managed population (e.g., over-represented animals or animals beyond reproductive age). Those animals not included in the managed population should be considered surplus to the managed population, but records still must be maintained on them to the same degree as those in the managed population.

Instances where individuals cannot be included in the managed breeding program are private holdings, institutions that are not approved Sustainability Partners, castrated individuals, abnormally aggressive animals, geriatric individuals, or animals with medical problems. Individuals with unknown pedigrees will be considered on an individual basis. The SSP Program Leader in conjunction with the PMC will make these determinations.

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Appendix A: Accreditation Standards by Chapter

The following specific standards of care relevant to capybara are taken from the AZA Accreditation Standards and Related Policies (AZA, 2011) and are referenced fully within the chapters of this animal care manual:

General Information

(1.1.1) The institution must comply with all relevant local, state/provincial, and federal wildlife laws and/or regulations. It is understood that, in some cases, AZA accreditation standards are more stringent than existing laws and/or regulations. In these cases the AZA standard must be met.

Chapter 1

(1.5.7) The animals must be protected or provided accommodation from weather or other conditions clearly known to be detrimental to their health or welfare.

(10.2.1) Critical life-support systems for the animals, including but not limited to plumbing, heating, cooling, aeration, and filtration, must be equipped with a warning mechanism, and emergency backup systems must be available. Warning mechanisms and emergency backup systems must be tested periodically.

(1.5.9) The institution must have a regular program of monitoring water quality for fish, marine mammals, and other aquatic animals. A written record must be maintained to document long-term water quality results and chemical additions.

Chapter 2

(1.5.1) All animals must be well cared for and presented in a manner reflecting modern zoological practices in exhibit design, balancing animals' welfare requirements with aesthetic and educational considerations.

(1.5.2) All animals must be housed in enclosures which are safe for the animals and meet their physical and psychological needs.

(1.5.2.1) All animals must be kept in appropriate groupings which meet their social and welfare needs.

(1.5.2.2) All animals should be provided the opportunity to choose among a variety of conditions within their environment.

(10.3.3) All animal enclosures (exhibits, holding areas, hospital, and quarantine/isolation) must be of a size and complexity sufficient to provide for the animal's physical, social, and psychological well-being. AZA housing guidelines outlined in the Animal Care Manuals should be followed.

(10.3.4) When sunlight is likely to cause overheating of or discomfort to the animals, sufficient shade (in addition to shelter structures) must be provided by natural or artificial means to allow all animals kept outdoors to protect themselves from direct sunlight.

(11.3.3) Special attention must be given to free-ranging animals so that no undue threat is posed to either the institution's animals, the free-ranging animals, or the visiting public. Animals maintained where they will be in contact with the visiting public must be carefully monitored, and treated humanely at all times.

(11.3.1) All animal exhibits and holding areas must be secured to prevent unintentional animal egress.

(1.5.15) All animal exhibit and holding area air and water inflows and outflows must be securely protected to prevent animal injury or egress.

(2.8.1) Pest control management programs must be administered in such a manner that the animals, paid and unpaid staff, the public, and wildlife are not threatened by the pests, contamination from pests, or the control methods used.

(11.3.6) There must be barriers in place (for example, guardrails, fences, walls, etc.) of sufficient strength and/or design to deter public entry into animal exhibits or holding areas, and to deter public contact with animals in all areas where such contact is not intended.

(11.2.4) All emergency procedures must be written and provided to appropriate paid and unpaid staff. Appropriate emergency procedures must be readily available for reference in the event of an actual emergency.

(11.2.5) Live-action emergency drills (functional exercises) must be conducted at least once annually for each of the four basic types of emergency (fire; weather or other environmental emergency appropriate to the region; injury to visitor or paid/unpaid staff; and animal escape). Four separate drills are required. These drills must be recorded and results evaluated for compliance with emergency procedures, efficacy of paid/unpaid staff training, aspects of the emergency response that are deemed adequate are reinforced, and those requiring improvement are identified and modified. (See 11.5.2 and 11.7.4 for other required drills).

(11.6.2) Security personnel, whether employed by the institution, or a provided and/or contracted service, must be trained to handle all emergencies in full accordance with the policies and procedures of the institution. In some cases, it is recognized that Security personnel may be in charge of the respective emergency (i.e. shooting teams).

(11.2.6) The institution must have a communication system that can be quickly accessed in case of an emergency.

(11.2.0) A paid staff member or a committee must be designated as responsible for ensuring that all required emergency drills are conducted, recorded, and evaluated in accordance with AZA accreditation standards (see 11.2.5, 11.5.2, and 11.7.4).

(11.2.7) A written protocol should be developed involving local police or other emergency agencies and include response times to emergencies.

(11.5.3) Institutions maintaining potentially dangerous animals must have appropriate safety procedures in place to prevent attacks and injuries by these animals. Appropriate response procedures must also be in place to deal with an attack resulting in an injury. These procedures must be practiced routinely per the emergency drill requirements contained in these standards. Whenever injuries result from these incidents, a written account outlining the cause of the incident, how the injury was handled, and a description of any resulting changes to either the safety procedures or the physical facility must be prepared and maintained for five years from the date of the incident.

(11.5.1) Institutions maintaining venomous animals must have appropriate antivenin readily available, and its location must be known by all paid and unpaid staff working in those areas. An individual must be responsible for inventory, disposal/replacement, and storage of antivenin.

Chapter 3

(1.4.0) The institution must show evidence of having a zoological records management program for managing animal records, veterinary records, and other relevant information.

(1.4.6) A paid or unpaid staff member must be designated as being responsible for the institution's animal record-keeping system. That person must be charged with establishing and maintaining the institution's animal records, as well as with keeping all paid and unpaid animal care staff members, apprised of relevant laws and regulations regarding the institution's animals.

(1.4.7) Animal and veterinary records must be kept current.

(1.4.4) Animal records and veterinary records, whether in electronic or paper form, must be duplicated and stored in a separate location. Animal records are defined as data, regardless of physical form or medium, providing information about individual animals, or samples or parts thereof, or groups of animals.

(1.4.5) At least one set of the institution's historical animal and veterinary records must be stored and protected. Those records should include permits, titles, declaration forms, and other pertinent information.

(1.4.1) An animal inventory must be compiled at least once a year and include data regarding acquisition, transfer, euthanasia, release, and reintroduction.

(1.4.2) All species owned by the institution must be listed on the inventory, including those animals on loan to and from the institution.

(1.4.3) Animals must be identifiable, whenever practical, and have corresponding ID numbers. For animals maintained in colonies/groups or other animals not considered readily identifiable, the institution must provide a statement explaining how record keeping is maintained.

Chapter 4

(1.5.11) Animal transportation must be conducted in a manner that is safe, well-planned and coordinated, and minimizes risk to the animal(s), employees, and general public. All applicable laws and/or regulations must be adhered to.

(1.5.10) Temporary, seasonal and traveling live animal exhibits, programs, or presentations (regardless of ownership or contractual arrangements) must be maintained at the same level of care as the institution's permanent resident animals, with foremost attention to animal welfare considerations, both onsite and at the location where the animals are permanently housed.

Chapter 6

(2.6.2) The institution must follow a written nutrition program that meets the behavioral and nutritional needs of all species, individuals, and colonies/groups in the institution. Animal diets must be of a quality and quantity suitable for each animal's nutritional and psychological needs.

(2.6.1) Animal food preparation and storage must meet all applicable laws and/or regulations.

(2.6.3) The institution must assign at least one paid or unpaid staff member to oversee appropriate browse material for the animals (including aquatic animals).

Chapter 7

(2.1.1) A full-time staff veterinarian is recommended. In cases where such is not necessary because of the number and/or nature of the animals residing there, a consulting/part-time veterinarian must be under written contract to make at least twice monthly inspections of the animals and to respond as soon as possible to any emergencies.

(2.1.2) So that indications of disease, injury, or stress may be dealt with promptly, veterinary coverage must be available to the animals24 hours a day, 7 days a week.

(2.0.1) The institution should adopt the *Guidelines for Zoo and Aquarium Veterinary Medical Programs and Veterinary Hospitals,* and policies developed or supported by the American Association of Zoo Veterinarians (AAZV). The most recent edition of the medical programs and hospitals booklet is available at the AAZV website, under "Publications", at https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazvveterinaryguidelines2016.pdf, and can also be obtained in PDF format by contacting AZA staff.

(2.2.1) Written, formal procedures must be available to paid and unpaid animal care staff for the use of animal drugs for veterinary purposes, and appropriate security of the drugs must be provided.

(2.7.1) The institution must have holding facilities or procedures for the quarantine of newly arrived animals and isolation facilities or procedures for the treatment of sick/injured animals. Quarantine duration should be assessed and determined by the pathogen risk and best practice for animal welfare.

(2.7.3) Quarantine, hospital, and isolation areas should be in compliance with standards/guidelines contained within the *Guidelines for Zoo and Aquarium Veterinary Medical Programs and Veterinary Hospitals* developed by the American Association of Zoo Veterinarians (AAZV), which can be obtained at: https://cdn.ymaws.com/www.aazv.org/resource/resmgr/files/aazvveterinaryguidelines2016.pdf.

(2.7.2) Written, formal procedures for quarantine must be available and familiar to all paid and unpaid staff working with quarantined animals.

(11.1.2) Training and procedures must be in place regarding zoonotic diseases.

(11.1.3) A tuberculin (TB) testing/surveillance program must be established for appropriate paid and unpaid staff in order to assure the health of both the paid and unpaid staff and the animals.

(2.5.1) Deceased animals should be necropsied to determine the cause of death for tracking morbidity and mortality trends to strengthen the program of veterinary care and meet SSP-related requests.

(2.5.2) The institution should have an area dedicated to performing necropsies.

(2.5.3) Cadavers must be kept in a dedicated storage area before and after necropsy. Remains must be disposed of in accordance with local/federal laws.

(2.0.2) The veterinary care program must emphasize disease prevention.

(2.0.3) Institutions should be aware of and prepared for periodic disease outbreaks in wild or other domestic or exotic animal populations that might affect the institution's animals (ex – Avian Influenza, Eastern Equine Encephalitis Virus, etc.). Plans should be developed that outline steps to be taken to protect the institution's animals in these situations.

(1.5.5) For animals used in offsite programs and for educational purposes, the institution must have adequate protocols in place to protect the rest of the animals at the institution from exposure to infectious agents.

(11.1.3) A tuberculin (TB) testing/surveillance program must be established for appropriate staff in order to ensure the health of both the employees and the animals. Each institution must have an employee occupational health and safety program.

(2.3.1) Capture equipment must be in good working order and available to authorized, trained personnel at all times.

(2.1.3) Paid and unpaid animal care staff should be trained to assess welfare and recognize abnormal behavior and clinical signs of illness and have knowledge of the diets, husbandry (including enrichment items and strategies), and restraint procedures required for the animals under their care. However, animal care staff (paid and unpaid) must not diagnose illnesses nor prescribe treatment.

(2.3.2) Institution facilities must have radiographic equipment or have access to radiographic services.

(1.5.8) The institution must develop and implement a clear and transparent process for identifying, communicating, and addressing animal welfare concerns from paid or unpaid staff within the institution in a timely manner, and without retribution.

Chapter 9

(1.6.4) The institution should follow a formal written animal training program that facilitates husbandry, science, and veterinary procedures and enhances the overall health and well-being of the animals.

(1.6.1) The institution must follow a formal written enrichment program that promotes species-appropriate behavioral opportunities.

(1.6.3) Enrichment activities must be documented and evaluated, and program refinements should be made based on the results, if appropriate. Records must be kept current.

(1.6.2) The institution must have a specific paid staff member(s) or committee assigned for enrichment program oversight, implementation, assessment, and interdepartmental coordination of enrichment efforts.

Chapter 10

(1.5.4) If ambassador animals are used, a written policy on the use of live animals in programs must be on file and incorporate the elements contained in AZA's "Recommendations For Developing an Institutional Ambassador Animal Policy" (see policy in the current edition of the *Accreditation Standards and Related Policies* booklet). An education, conservation, and welfare message must be an integral component of all programs. Animals in education programs must be maintained and cared for by paid and/or unpaid trained staff, and housing conditions must meet standards required for the remainder of the animals in the institution. While outside their primary enclosure, although the conditions may be different, animal safety and welfare need to be assured at all times.

(1.5.3) If animal demonstrations are a part of the institution's programs, an educational/conservation message must be an integral component.

(1.5.12) Paid and/or unpaid staff assigned to handle animals during demonstrations or educational programs must be trained in accordance with the institution's written animal handling protocols. Such training must take place before handling may occur.

(1.5.13) When in operation, animal contact areas (petting zoos, touch tanks, etc.) must be supervised by trained, paid and/or unpaid staff.

(1.5.5) For animals used in offsite programs and for educational purposes, the institution must have adequate protocols in place to protect the rest of the animals at the institution from exposure to infectious agents.

Chapter 11

(5.3) The institution should maximize the generation and dissemination of scientific knowledge gained. This might be achieved by participating in AZA TAG/SSP sponsored studies when applicable, conducting and publishing original research projects, affiliating with local universities, and/or employing staff with scientific credentials.

(5.0) The institution must have a demonstrated commitment to scientific study that is in proportion to the size and scope of its facilities, staff (paid and unpaid), and animals

(5.2) The institution must follow a formal written policy that includes a process for the evaluation and approval of scientific project proposals, and outlines the type of studies it conducts, methods, staff (paid and unpaid) involvement, evaluations, animals that may be involved, and guidelines for publication of findings.

(5.1) Scientific studies must be under the direction of a paid or unpaid staff member or committee qualified to make informed decisions.

Appendix B: Guidelines for Creating and Sharing Animal and Collection Records

Developed by the AZA Institutional Data Management Scientific Advisory Group
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The goal of maintaining a centralized, compiled record for each animal cared for in a zoo or aquarium is ideal, however, oftentimes, information belonging in an animal record is spread across many departments and may originate with any member of the animal care staff. Therefore, it is important for zoos and aquariums to have a formal method for collecting or linking various pieces of information into the official records and that the roles and responsibilities for each named record type are clearly defined in written protocols for the reporting, recording, distribution, storage, and retrieval processes; there should also be a stated process of review for the accuracy and completeness of these records. For example, a recording/reporting protocol would state who reports births or deaths, to whom they are reported, in what manner and in what time frame they are reported, who officially records the information, and who reviews the resulting record for accuracy and completeness. Then, the maintenance and archiving protocol would state where the record is to be filed, who may have access, and how long the record is to be maintained before being archived or disposed of.

Information contained in animal records is essential not only to the immediate care of the individual animal but also as pooled data to manage larger concerns (e.g., providing norms for species-related veterinary and population management decisions, evidence of compliance with laws and regulations, showing trends in populations on every level from institutional to global, etc.). No matter what its use, it is critical for the information contained in an animal record to be factual, clear, complete, and documented. Because zoos and aquariums vary greatly in size and organizational structure, it is impossible to set defined procedures that would be applicable to all; therefore the following guidelines for creating and sharing animal records have been developed to assist with the establishment of written policies that best fit their own internal structure and protocols.

Animal and Collection Records – Definitions and Examples

The AZA Institutional Data Management Scientific Advisory Group (IDMAG) defines an animal record as: *"data, regardless of physical form or medium, providing information about individual animals, groups of animals, or samples or parts thereof".* An animal's record may include, but is not limited to, information about its provenance, history, daily care, activities, and condition; some may originate in non-animal care departments. Some examples of animal records are:

- transaction documents (including proof of legal ownership, purchase contracts, etc.)
- identification information
- reports of collection changes (including in-house moves)
- pedigrees/lineages
- veterinary information, including images, test results, etc.
- nutrition and body condition information
- information on sampling and parts/products distribution

In addition, the IDMAG defines collection records as: *"information, evidence, rationalizations about an animal collection as a whole that may supplement or explain information contained in an animal record"*. Collection records may include, but are not limited to, documentation of collection decisions and changes, evidence of structural change at the institution, evidence of building name changes, and documentation of institution level or unit level husbandry protocols and changes. Some examples of collection records are:

- collection plans
- permits
- annual inventories (which include reconciliation with the previous year)
- area journals/notebooks (including information to/from/between other animal care staff)
- keeper reports

- animal management protocols (e.g., species hand-rearing protocols, special care or treatments, etc.)
- enclosure maps/trees
- enclosure/exhibit information (monitoring, maintenance, modifications, etc.)
- research plans and published papers

Animal and Collection Records - Development

It is recommended that each zoo and aquarium develop written policies and procedures, applicable to all staff involved with animal care, that:

- define the types of records that are required.
  For example, daily keeper reports might be required from the keeper staff and weekly summaries of activities might be required from the animal curator and senior veterinary.
- define the information that is to be included in each type of record.
  Following the example above, the institution would state the specific types of information to be recorded on the daily keeper report and the weekly summaries.
- define the primary location where each record can be found.
  For example, if a zoo does not employ a nutritionist, the policy or procedures might state that animal diet information will be found in keeper daily reports, curator-developed daily diets, and/or veterinary-prescribed treatment diets.
- assign responsibility for the generation of each record type and set time limits for the their creation.
  For example, keepers might be held responsible for producing daily reports by the start of the next day and curators might be held responsible for producing weekly summaries by the Tuesday of the following week.
- define a process to review the accuracy of each record type and assign responsibility for that review process.
  For example, the identity of who will review each type of record, the date of reviews, and the review/correction processes might be included in the policy.
- define a process to identify official records and assign responsibility for the recording of, or linking of, information into these records.
  For example, the identity of who will be responsible for placing information into the official records and the processes of how to identify official records might be included in the policy.
- ensure entries in official records are never erased or deleted.
  For example, if an entry is determined to be erroneous, rather than deleting it, the entry should be amended and an audit trail should be created that identifies what data was changed, who made the change, the date it was changed, and the reason for the change.
- ensure records relating to specific animals in the collection, including the records of non--animal care departments, are permanently archived as part of the animal's record.
  For example, if your zoo or aquarium's records retention schedules differ from this recommendation every attempt should be made to exempt these records from schedules requiring their destruction.

Animal and Collection Records – Sharing of Information

Each zoo and aquarium should assess the ownership of their animal and collection records and determine the rights of employees and outside entities to the information contained in them. It is recommended that each zoo and aquarium develop written policies and procedures for the distribution and/or availability of the animal and collection records that:

- identify who has access to animal and collection records and under what conditions.
  For example, animal care staff whose duties require a direct need for information about specific animals or collection of animals should be identified as individuals who are allowed access to any or specified records, regardless of who created them or when they were created.
- assign responsibility for the distribution, archiving and retrieval of each record type.

For example, the recordkeeper or registrar might be held responsible for maintaining all past and current transaction documents and the curator might be held responsible for maintaining the daily keeper reports from his/her section.

- define a notification system that specifies what information will be provided in the notification, who will be notified, the date they will be notified by, and the mechanism that will be used to ensure the notification is communicated appropriately.

  For example, the shipment of an animal might require that written notice be made to the senior keeper in the animal's area, the curator, and the veterinary at least 30 days prior to the move, and identifies the animal by group or individual identification/accession number, sex, and tag/transponder number, etc.
- define where each record type (stored or archived) is available and what format (paper or digital) it is in.

  For example, all original animal transaction documents might be kept in the registrar's office in fire-proof file cabinets but copies of the Animal Data Transfer Forms are kept in the appropriate keeper area.
- define a system for obtaining necessary information such that the information is available regardless of department and regardless of staffing issues

  For example, keeper daily reports might be maintained in an electronic database run on the institution's network, to which all animal care staff members have at least read-only access.

Implementation of these Recommendations

Well-written, consistent data-recording protocols and clear lines of communication will increase the quality of animal records and should be implemented by all institutions, regardless of technical resources. While the best option for availability of information is an electronic database system run on a computer network (intranet) to which all animal care staff members have unrestricted access, the above recommendations may also be adopted by zoos and aquariums without full electronic connections.

Appendix C: AZA Policy on Responsible Population Management

PREAMBLE

The stringent requirements for AZA accreditation, and high ethical standards of professional conduct, are unmatched by similar organizations and far surpass the United States Department of Agriculture's Animal and Plant Health Inspection Service's requirements for licensed animal exhibitors. Every AZA member must abide by a Code of Professional Ethics (https://www.aza.org/code-of-ethics). In order to continue these high standards, AZA-accredited institutions and certified related facilities should make it a priority, when possible, to acquire animals from and transfer them to other AZA member institutions, or members of other regional zoo associations that have professionally recognized accreditation programs.

AZA-accredited institutions and certified related facilities cannot fulfill their important missions of conservation, education, and science without live animals. Responsible management and the long-term sustainability of living animal populations necessitates that some individuals be acquired and transferred, reintroduced or even humanely euthanized at certain times. The acquisition and transfer of animals should be prioritized by the long-term sustainability needs of the species and AZA-managed populations among AZA-accredited and certified related facilities, and between AZA member institutions and non-AZA entities with animal care and welfare standards aligned with AZA. AZA member institutions that acquire animals from the wild, directly or through commercial vendors, should perform due diligence to ensure that such activities do not have a negative impact on species in the wild. Animals should only be acquired from non-AZA entities that are known to operate legally and conduct their business in a manner that reflects and/or supports the spirit and intent of the AZA Code of Professional Ethics as well as this Policy.

I. INTRODUCTION

This AZA Policy on Responsible Population Management provides guidance to AZA members to:

1. Assure that animals from AZA member institutions and certified related facilities are not transferred to individuals or organizations that lack the appropriate expertise or facilities to care for them [*see taxa specific appendices (in development)*],

2. Assure that the health and conservation of wild populations and ecosystems are carefully considered as appropriate,

3. Maintain a proper standard of conduct for AZA members during acquisition and transfer/reintroduction activities, including adherence to all applicable laws and regulations,

4. Assure that the health and welfare of individual animals is a priority during acquisition and transfer/reintroduction activities, and

5. Support the goals of AZA's cooperatively managed populations and associated Animal Programs [Species Survival Plans® (SSPs), Studbooks, and Taxon Advisory Groups (TAGs)].

This AZA Policy on Responsible Population Management will serve as the default policy for AZA member institutions. Institutions should develop their own Policy on Responsible Population Management in order to address specific local concerns. Any institutional policy must incorporate and not conflict with the AZA acquisition and transfer/transition standards.

II. LAWS, AUTHORITY, RECORD-KEEPING, IDENTIFICATION AND DOCUMENTATION

The following must be considered with regard to the acquisition or transfer/management of all living animals and specimens (their living and non-living parts, materials, and/or products):

1. Any acquisitions, transfers, euthanasia and reintroductions must meet the requirements of all applicable local, state, federal, national, and international laws and regulations. Humane euthanasia

must be performed in accordance with the established euthanasia policy of the institution and follow the recommendations of current AVMA Guidelines for the Euthanasia of Animals (2013 Edition https://www.avma.org/KB/Policies/Documents/euthanasia.pdf) or the AAZV's Guidelines on the Euthanasia of Non-Domestic Animals. Ownership and any applicable chain-of-custody must be documented. If such information does not exist, an explanation must be provided regarding such animals and specimens. Any acquisition of free-ranging animals must be done in accordance with all local, state, federal, national, and international laws and regulations and must not be detrimental to the long-term viability of the species in the wild.

2. The Director/Chief Executive Officer of the institution must have final authority for all acquisitions, transfers, and euthanasia.

3. Acquisitions or transfers/euthanasia/reintroductions must be documented through institutional record keeping systems. The ability to identify which animal is being transferred is very important and the method of identifying each individual animal should be documented. Any existing documentation must accompany all transfers. Institutional animal records data, records guidelines have been developed for certain species to standardize the process (https://www.aza.org/idmag-documents-and-guidelines).

4. For some colonial, group-living, or prolific species, it may be impossible or highly impractical to identify individual animals when these individuals are maintained in a group. These species can be maintained, acquisitioned, transferred, and managed as a group or colony, or as part of a group or colony.

5. If the intended use of specimens from animals either living or non-living is to create live animal(s), their acquisition and transfer should follow the same guidelines. If germplasm is acquired or transferred with the intention of creating live animal(s), ownership of the offspring must be clearly defined in transaction documents (e.g., breeding loan agreements).

   Institutions acquiring, transferring or otherwise managing specimens should consider current and possible future uses as new technologies become available. All specimens from which nuclear DNA could be recovered should be carefully considered for preservation as these basic DNA extraction technologies already exist.

6. AZA member institutions must maintain transaction documents (e.g., confirmation forms, breeding agreements) which provide the terms and conditions of animal acquisitions, transfers and loans, including documentation for animal parts, products and materials. These documents should require the potential recipient or provider to adhere to the AZA Policy on Responsible Population Management, and the AZA Code of Professional Ethics, and must require compliance with the applicable laws and regulations of local, state, federal, national, and international authorities.

7. In the case of animals (living or non-living) and their parts, materials, or products (living or non-living) held on loan, the owner's written permission should be obtained prior to any transfer and documented in the institutional records.

8. AZA SSP and TAG necropsy and sampling protocols should be accommodated.

9. Some governments maintain ownership of the species naturally found within their borders. It is therefore incumbent on institutions to determine whether animals they are acquiring or transferring are owned by a government entity, foreign or domestic, and act accordingly by reviewing the government ownership policies available on the AZA website. In the case of government owned animals, proposals for and/or notifications of transfers must be sent to the species manager for the government owned species.

III. ACQUISITION REQUIREMENTS

A. General Acquisitions

1. Acquisitions must be consistent with the mission of the institution, as reflected in its Institutional Collection Plan, by addressing its exhibition/education, conservation, and/or scientific goals regarding the individual or species.

2. Animals (wild, feral, and domestic) may be held temporarily for reasons such as assisting governmental agencies or other institutions, rescue and/or rehabilitation, research, propagation or headstarting for reintroduction, or special exhibits.

3. Any receiving institution must have the necessary expertise and resources to support and provide for the professional care and management of the species, so that the physical, psychological, and social needs of individual animals and species are met.

4. If the acquisition involves a species managed by an AZA Animal Program, the institution should communicate with the Animal Program Leader and, in the case of Green SSP Programs, must adhere to the AZA Full Participation Policy (https://www.aza.org/board-approved-policies-and-position-statements).

5. AZA member institutions should consult AZA Wildlife Conservation and Management Committee (WCMC)-approved TAG Regional Collection Plans (RCPs), Animal Program Leaders, and AZA Animal Care Manuals (ACMs) when making acquisition decisions.

6. AZA member institutions that work with commercial vendors that acquire animals from the wild, must perform due diligence to assure the vendors' collection of animals is legal and using ethical practices. Commercial vendors should have conservation and animal welfare goals similar to those of AZA institutions.

7. AZA member institutions may acquire animals through public donations and other non-AZA entities when it is in the best interest of the animal and/or species.

B. Acquisitions from the Wild

Maintaining wild animal populations for exhibition, education and wildlife conservation purposes is a core function of AZA-member institutions. AZA zoos and aquariums have saving species and conservation of wildlife and wildlands as a basic part of their public mission. As such, the AZA recognizes that there are circumstances where acquisitions from the wild are needed in order to maintain healthy, diverse animal populations. Healthy, sustainable populations support the objectives of managed species programs and the core mission of AZA members. In some cases, acquiring individuals from the wild may be a viable option in addition to, or instead of, relying on breeding programs with animals already in human care.

Acquiring animals from the wild can result in socioeconomic benefit and environmental protection and therefore the AZA supports environmentally sustainable/beneficial acquisition from the wild when conservation is a positive outcome.

1. Before acquiring animals from the wild, institutions are encouraged to examine alternative sources including other AZA institutions and other regional zoological associations or other non-AZA entities.

2. When acquiring animals from the wild, both the long-term health and welfare impacts on the wild population as well as on individual animals must be considered. In crisis situations, when the survival of a population is at risk, rescue decisions will be made on a case-by-case basis by the appropriate agency and institution.

3. AZA zoos and aquariums may assist wildlife agencies by providing homes for animals born in nature if they are incapable of surviving on their own (e.g., in case of orphaned or injured animals) or by euthanizing the animals because they pose a risk to humans or for humane reasons.

4. Institutions should only accept animals from the wild after a risk assessment determines the zoo/aquarium can mitigate any potential adverse impacts on the health, care and maintenance of the existing animals already being housed at the zoo or aquarium, and the new animals being acquired.

IV. TRANSFER, EUTHANASIA AND REINTRODUCTION REQUIREMENTS

A. Living Animals

Successful conservation and animal management relies on the cooperation of many entities, both AZA and non-AZA. While preference is given to placing animals with AZA-accredited institutions or certified related facilities, it is important to foster a cooperative culture among those who share AZA's mission of saving species and excellence in animal care.

1. AZA members should assure that all animals in their care are transferred, humanely euthanized and/or reintroduced in a manner that meets the standards of AZA, and that animals are not transferred to those not qualified to care for them properly. Refer to IV.12, below, for further requirements regarding euthanasia.

2. If the transfer of animals or their specimens (parts, materials, and products) involves a species managed by an AZA Animal Program, the institution should communicate with that Animal Program Leader and, in the case of Green SSP Programs must adhere to the AZA Full Participation Policy (https://www.aza.org/board-approved-policies-and-position-statements).

3. AZA member institutions should consult WCMC-approved TAG Regional Collection Plans, Animal Program Leaders, and Animal Care Manuals when making transfer decisions.

4. Animals acquired solely as a food source for animals in the institution's care are not typically accessioned. There may be occasions, however, when it is appropriate to use accessioned animals that exceed population carrying capacity as feeder animals to support other animals. In some cases, accessioned animals may have their status changed to "feeder animal" status by the institution as part of their program for long-term sustained population management of the species.

5. In transfers to non-AZA entities, AZA members must perform due diligence and should have documented validation, including one or more letters of reference, for example from an appropriate AZA Professional Fellow or other trusted source with expertise in animal care and welfare, who is familiar with the proposed recipient and their current practices, and that the recipient has the expertise and resources required to properly care for and maintain the animals. Any recipient must have the necessary expertise and resources to support and provide for the professional care and management of the species, so that the physical, psychological, and social needs of individual animals and species are met within the parameters of modern zoological philosophy and practice. Supporting documentation must be kept at the AZA member institution (see #IV.9 below).

6. Domestic animals should be transferred in accordance with locally acceptable humane farming practices, including auctions, and must be subject to all relevant laws and regulations.

7. AZA members must not send any non-domestic animal to auction or to any organization or individual that may display or sell the animal at an animal auction. *See certain taxa-specific appendices to this Policy (in development) for information regarding exceptions*.

8. Animals must not be sent to organizations or individuals that allow the hunting of these individual animals; that is, no individual animal transferred from an AZA institution may be hunted. For purposes of maintaining genetically healthy, sustainable zoo and aquarium populations, AZA-accredited institutions and certified related facilities may send animals to non-AZA organizations or individuals (refer to #IV.5 above). These non-AZA entities (for instance, ranching operations) should follow

appropriate ranch management practices and other conservation minded practices to support population sustainability.

9. Every loaning institution must annually monitor and document the conditions of any loaned specimen(s) and the ability of the recipient(s) to provide proper care (refer to #IV.5 above). If the conditions and care of animals are in violation of the loan agreement, the loaning institution must recall the animal or assure prompt correction of the situation. Furthermore, an institution's loaning policy must not be in conflict with this AZA Policy on Responsible Population Management.

10. If living animals are sent to a non-AZA entity located in the U.S. for research purposes, it must be a registered research facility by the U.S. Department of Agriculture and accredited by the Association for the Assessment & Accreditation of Laboratory Animal Care, International (AAALAC), if eligible. For international transactions, the receiving facility must be registered by that country's equivalent body having enforcement over animal welfare. In cases where research is conducted, but governmental oversight is not required, institutions should do due diligence to assure the welfare of the animals during the research.

11. Reintroductions and release of animals into the wild must meet all applicable local, state, and international laws and regulations. Any reintroduction requires adherence to best health and veterinary practices to ensure that non-native pathogens are not released into the environment exposing naive wild animals to danger. Reintroductions may be a part of a recovery program and must be compatible with the IUCN Reintroduction Specialist Group's Reintroduction Guidelines (http://www.iucnsscrsg.org/index.php).

12. Humane euthanasia may be employed for medical reasons to address quality of life issues for animals or to prevent the transmission of disease. AZA also recognizes that humane euthanasia may be employed for managing the demographics, genetics, and diversity of animal populations. Humane euthanasia must be performed in accordance with the established euthanasia policy of the institution and follow the recommendations of current AVMA Guidelines for the Euthanasia of Animals (2013 Edition https://www.avma.org/KB/Policies/Documents/euthanasia.pdf) or the AAZV's Guidelines on the Euthanasia of Non-Domestic Animals.

B. Non-Living Animals and Specimens

AZA members should optimize the use and recovery of animal remains. All transfers must meet the requirements of all applicable laws and regulations.

1. Optimal recovery of animal remains may include performing a complete necropsy including, if possible, histologic evaluation of tissues which should take priority over specimens' use in education/exhibits. AZA SSP and TAG necropsy and sampling protocols should be accommodated. This information should be available to SSP Programs for population management.

2. The educational use of non-living animals, parts, materials, and products should be maximized, and their use in Animal Program sponsored projects and other scientific projects that provide data for species management and/or conservation must be considered.

3. Non-living animals, if handled properly to protect the health of the recipient animals, may be utilized as feeder animals to support other animals as deemed appropriate by the institution.

4. AZA members should consult with AZA Animal Program Leaders prior to transferring or disposing of remains/samples to determine if existing projects or protocols are in place to optimize use.

5. AZA member institutions should develop agreements for the transfer or donation of non-living animals, parts, materials, products, and specimens and associated documentation, to non-AZA entities such as universities and museums. These agreements should be made with entities that

have appropriate long term curation/collections capacity and research protocols, or needs for educational programs and/or exhibits.

DEFINITIONS

Acquisition: Acquisition of animals can occur through breeding (births, hatchings, cloning, and division of marine invertebrates = "fragging"), trade, donation, lease, loan, transfer (inter- and intra-institution), purchase, collection, confiscation, appearing on zoo property, or rescue and/or rehabilitation for release.

Annual monitoring and Due diligence: Due diligence for the health of animals on loan is important. Examples of annual monitoring and documentation include and are not limited to inventory records, health records, photos of the recipient's facilities, and direct inspections by AZA professionals with knowledge of animal care. The level of due diligence will depend on professional relationships.

AZA member institution: In this Policy "AZA member institutions" refers to AZA-accredited institutions and certified related facilities (zoological parks and aquariums). "AZA members" may refer to either institutions or individuals.

Data sharing: When specimens are transferred, the transferring and receiving institutions should agree on data that must be transferred with the specimen(s). Examples of associated documentation include provenance of the animal, original permits, tags and other metadata, life history data for the animal, how and when specimens were collected and conserved, etc.

Dispose: "Dispose/Disposing of" in this document is limited to complete and permanent removal of an individual via incineration, burying or other means of permanent destruction

Documentation: Examples of documentation include ZIMS records, "Breeding Loan" agreements, chain-of-custody logs, letters of reference, transfer agreements, and transaction documents. This is documentation that maximizes data sharing.

Domestic animal: Examples of domestic animals may include certain camelids, cattle, cats, dogs, ferrets, goats, pigs, reindeer, rodents, sheep, budgerigars, chickens, doves, ducks, geese, pheasants, turkeys, and goldfish or koi.

Ethics of Acquisition/Transfer/Euthanasia: Attempts by members to circumvent AZA Animal Programs in the acquisition of animals can be detrimental to the Association and its Animal Programs. Such action may also be detrimental to the species involved and may be a violation of the Association's Code of Professional Ethics. Attempts by members to circumvent AZA Animal Programs in the transfer, euthanasia or reintroduction of animals may be detrimental to the Association and its Animal Programs (unless the animal or animals are deemed extra in the Animal Program population by the Animal Program Coordinator). Such action may be detrimental to the species involved and may be a violation of the Association's Code of Professional Ethics.

"Extra" or Surplus: AZA's scientifically-managed Animal Programs, including SSPs, have successfully bred and reintroduced critically endangered species for the benefit of humankind. To accomplish these critical conservation goals, populations must be managed within "carrying capacity" limits. At times, the number of individual animals in a population exceeds carrying capacity, and while meaning no disrespect for these individual animals, we refer to these individual animals as "extra" within the managed population.

Euthanasia: Humane death. This act removes an animal from the managed population. Specimens can be maintained in museums or cryopreserved collections. Humane euthanasia must be performed in accordance with the established euthanasia policy of the institution and follow the recommendations of current AVMA Guidelines for the Euthanasia of Animals (2013 Edition https://www.avma.org/KB/Policies/Documents/euthanasia.pdf) or the AAZV's Guidelines on the Euthanasia of Non-Domestic Animals.

Feral: Feral animals are animals that have escaped from domestication or have been abandoned to the wild and have become wild, and the offspring of such animals. Feral animals may be acquired for temporary or permanent reasons.

Group: Examples of colonial, group-living, or prolific species include and are not limited to certain terrestrial and aquatic invertebrates, fish, sharks/rays, amphibians, reptiles, birds, rodents, bats, big herds, and other mammals,

Lacey act: The Lacey Act prohibits the importation, exportation, transportation, sale, receipt, acquisition or purchase of wildlife taken or possessed in violation of any law, treaty or regulation of the United States or any Indian tribal law of wildlife law. In cases when there is no documentation accompanying an acquisition, the animal(s) may not be transferred across state lines. If the animal was illegally acquired at any time then any movement across state or international borders would be a violation of the Lacey Act.

Museum: It is best practice for modern zoos and aquariums to establish relationships with nearby museums or other biorepositories, so that they can maximize the value of animals when they die (e.g., knowing who to call when they have an animal in necropsy, or specimens for cryopreservation). Natural history museums that are members of the Natural Science Collections Alliance (NSCA) and frozen biorepositories that are members of the International Society of Biological and Environmental Repositories (ISBER) are potential collaborators that could help zoos find appropriate repositories for biological specimens.

Non-AZA entity: Non – AZA entities includes facilities not accredited or certified by the AZA, facilities in other zoological regions, academic institutions, museums, research facilities, private individuals, etc.

Reintroduction: Examples of transfers outside of a living zoological population include movements of animals from zoo/aquarium populations to the wild through reintroductions or other legal means.

Specimen: Examples of specimens include animal parts, materials and products including bodily fluids, cell lines, clones, digestive content, DNA, feces, marine invertebrate (coral) fragments ("frags"), germplasm, and tissues.

Transaction documents: Transaction documents must be signed by the authorized representatives of both parties, and copies must be retained by both parties*. In the case of loans, the owner's permission for appropriate activities should be documented in the institutional records. This document(s) should be completed prior to any transfer. In the case of rescue, confiscation, and evacuation due to natural disasters, it is understood that documents may not be available until after acceptance or shipping. In this case documentation (e.g., a log) must be kept to reconcile the inventory and chain of custody after the event occurs. (*In the case of government owned animals, notification of transfers must be sent to species manager for the government owned species).

Transfer: Transfer occurs when an animal leaves the institution for any reason. Reasons for transfer or euthanasia may include cooperative population management (genetic, demographic or behavioral management), animal welfare or behavior management reasons (including sexual maturation and individual management needs). Types of transfer include withdrawal through donation, trade, lease, loan, inter- and intra-institution transfers, sale, escape, theft. Reintroduction to the wild, humane euthanasia or natural death are other possible individual animal changes in a population.

RECIPIENT PROFILE EXAMPLE

Example questions for transfers to non-AZA entities (from AZA-member Recipient Profile documents):

Has your organization, or any of its officers, been indicted, convicted, or fined by a State or Federal agency or any national agency for any statute or regulation involving the care or welfare of animals housed at your facility? (If yes, please explain on a separate sheet).

Recipients agree that the specimen(s) or their offspring will not be utilized, sold or traded for any purpose contrary to the Association of Zoos and Aquariums (AZA) Code of Ethics (enclosed)

References, other than (LOCAL ZOO/AQUARIUM) employees, 2 minimum (please provide additional references on separate sheet):

Reference Name		Phone	
Facility		Fax	
Address		E-mail	
City	State		Zip
Country		AZA Member?	

Reference Name		Phone	
Facility		Fax	
Address		E-mail	
City	State		Zip
Country		AZA Member?	

Veterinary Information:

Veterinarian		Phone	
Clinic/Practice		Fax	
Address		E-mail	
City	State		Zip
Country			

How are animals identified at your facility? If animals are not identified at your facility, please provide an explanation about why they are not here:

Where do you acquire and send animals? (Select all that apply)

AZA Institutions	Non-AZA Institutions	Exotic Animal Auctions	Pet Stores
Hunting Ranches	Dealers	Private Breeders	Non-hunting Game

Entertainment Industry Hobbyists Research Labs Ranches Wild
Other

What specific criteria are used to evaluate if a facility is appropriate to receive animals from you?

Please provide all of the documents listed below:
Required:
1. Please provide a brief statement of intent for the specimens requested.
2. Resumes of primary caretakers and those who will be responsible for the husbandry and management of animals.
3. Description (including photographs) of facilities and exhibits where animals will be housed.
4. Copy of your current animal inventory.

Only if Applicable:
5. Copies of your last two USDA inspection reports (if applicable).
6. Copies of current federal and state permits.
7. Copy of your institutional acquisition/disposition policy.

(in-house use only) In-Person Inspection of this facility (Staff member/Date, attach notes):

(Local institution: provide Legal language certifying that the information contained herein is true and correct)

(Validity of this: This document and all materials associated will be valid for a period of 2 years from date of signature.)

Example agreement for Receiving institution (agrees to following condition upon signing):
RECIPIENT AGREES THAT THE ANIMAL(S) AND ITS (THEIR) OFFSPRING WILL NOT BE UTILIZED, SOLD OR TRADED FOR THE PURPOSE OF COMMERCE OR SPORT HUNTING, OR FOR USE IN ANY STRESSFUL OR TERMINAL RESEARCH OR SENT TO ANY ANIMAL AUCTION. RECIPIENT FURTHER AGREES THAT IN THE EVENT THE RECIPIENT INTENDS TO DISPOSE OF AN ANIMAL DONATED BY (INSITUTION), RECIPIENT WILL FIRST NOTIFY (INSTITUTION) OF THE IDENTITY OF THE PROPOSED TRANSFEREE AND THE TERMS AND CONDITIONS OF SUCH DISPOSITION AND WILL PROVIDE (INSTITUTION) THE OPPORTUNITY TO ACQUIRE THE ANIMAL(S) WITHOUT CHARGE. IF (INSTITUTION) ELECTS NOT TO RECLAIM THE ANIMAL WITHIN TEN (10) BUSINESS DAYS FOLLOWING SUCH NOTIFICATION, THEN, IN SUCH EVENT, (INSTITUTION) WAIVES ANY RIGHT IT MAY HAVE TO THE ANIMAL AND RECIPIENT MAY DISPOSE OF THE ANIMAL AS PROPOSED.

Institutional note: The text above is similar to the language most dog breeders use in their contracts when they sell a puppy. If people can provide that protection to the puppies they place, zoos/aquariums can provide it for animals that we place too! Some entities have been reluctant to sign it, and in that case we revert to a loan and our institution retains ownership of the animal. Either way, we are advised of the animal's eventual placement and location.

Appendix D: Recommended Quarantine Procedures

Quarantine facility: A separate quarantine facility, with the ability to accommodate mammals, birds, reptiles, amphibians, and fish should exist. If a specific quarantine facility is not present, then newly acquired animals should be isolated from the established collection in such a manner as to prohibit physical contact, to prevent disease transmission, and to avoid aerosol and drainage contamination.

Such separation should be obligatory for primates, small mammals, birds, and reptiles, and attempted wherever possible with larger mammals such as large ungulates and carnivores, marine mammals, and cetaceans. If the receiving institution lacks appropriate facilities for isolation of large primates, pre-shipment quarantine at an AZA or American Association for Laboratory Animal Science (AALAS) accredited institution may be applied to the receiving institutions protocol. In such a case, shipment must take place in isolation from other primates. More stringent local, state, or federal regulations take precedence over these recommendations.

Quarantine length: Quarantine for all species should be under the supervision of a veterinary and consist of a minimum of 30 days (unless otherwise directed by the staff veterinary). Mammals: If during the 30-day quarantine period, additional mammals of the same order are introduced into a designated quarantine area, the 30-day period must begin over again. However, the addition of mammals of a different order to those already in quarantine will not have an adverse impact on the originally quarantined mammals. Birds, Reptiles, Amphibians, or Fish: The 30-day quarantine period must be closed for each of the above Classes. Therefore, the addition of any new birds into a bird quarantine area requires that the 30-day quarantine period begin again on the date of the addition of the new birds. The same applies for reptiles, amphibians, or fish.

Quarantine personnel: A keeper should be designated to care only for quarantined animals or a keeper should attend quarantined animals only after fulfilling responsibilities for resident species. Equipment used to feed and clean animals in quarantine should be used only with these animals. If this is not possible, then equipment must be cleaned with an appropriate disinfectant (as designated by the veterinary supervising quarantine) before use with post-quarantine animals.

Institutions must take precautions to minimize the risk of exposure of animal care personnel to zoonotic diseases that may be present in newly acquired animals. These precautions should include the use of disinfectant foot baths, wearing of appropriate protective clothing and masks in some cases, and minimizing physical exposure in some species; e.g., primates, by the use of chemical rather than physical restraint. A tuberculin testing/surveillance program must be established for zoo/aquarium employees in order to ensure the health of both the employees and the animal collection.

Quarantine protocol: During this period, certain prophylactic measures should be instituted. Individual fecal samples or representative samples from large numbers of individuals housed in a limited area (e.g., birds of the same species in an aviary or frogs in a terrarium) should be collected at least twice and examined for gastrointestinal parasites. Treatment should be prescribed by the attending veterinary. Ideally, release from quarantine should be dependent on obtaining two negative fecal results spaced a minimum of two weeks apart either initially or after parasiticide treatment. In addition, all animals should be evaluated for ectoparasites and treated accordingly.

Vaccinations should be updated as appropriate for each species. If the animal arrives without a vaccination history, it should be treated as an immunologically naive animal and given an appropriate series of vaccinations. Whenever possible, blood should be collected and sera banked. Either a -70 °C (-94 °F) frost-free freezer or a -20 °C (-4 °F) freezer that is not frost-free should be available to save sera. Such sera could provide an important resource for retrospective disease evaluation.

The quarantine period also represents an opportunity to, where possible, permanently identify all unmarked animals when anesthetized or restrained (e.g., tattoo, ear notch, ear tag, etc.). Also, whenever animals are restrained or immobilized, a complete physical, including a dental examination, should be performed. Complete medical records should be maintained and available for all animals during the quarantine period. Animals that die during quarantine should have a necropsy performed under the supervision of a veterinary and representative tissues submitted for histopathologic examination.

Quarantine procedures: The following are recommendations and suggestions for appropriate quarantine procedures for capybara:

Required:

1. Direct and floatation fecals
2. Vaccinate as appropriate

Strongly recommended:

1. CBC/sera profile
2. Urinalysis
3. Appropriate serology (FIP, FeLV, FIV)
4. Heartworm testing

Appendix E: Ambassador (Program) Animal Policy and Position Statement

Ambassador (Program) Animal Policy

Originally approved by the AZA Board of Directors—2003
Updated and approved by the Board—July 2008 & June 2011

The Association of Zoos & Aquariums (AZA) recognizes many benefits for public education and, ultimately, for conservation in ambassador animal presentations. AZA's Conservation Education Committee's *Ambassador Animal Position Statement* summarizes the value of ambassador animal presentations (see pages 42–44).

For the purpose of this policy, an Ambassador animal is defined as "an animal whose role includes handling and/or training by staff or volunteers for interaction with the public and in support of institutional education and conservation goals." Some animals are designated as Ambassador Animals on a full-time basis, while others are designated as such only occasionally. Ambassador Animal-related Accreditation Standards are applicable to all animals during the times that they are designated as Ambassador Animals.

There are three main categories of Ambassador Animal interactions:

1. On Grounds with the Ambassador Animal Inside the Exhibit/Enclosure:
    a. Public access outside the exhibit/enclosure. Public may interact with animals from outside the exhibit/enclosure (e.g., giraffe feeding, touch tanks).
    b. Public access inside the exhibit/enclosure. Public may interact with animals from inside the exhibit/enclosure (e.g., lorikeet feedings, 'swim with' programs, camel/pony rides).
2. On Grounds with the Ambassador Animal Outside the Exhibit/Enclosure:
    a. Minimal handling and training techniques are used to present Ambassador Animals to the public. Public has minimal or no opportunity to directly interact with Ambassador Animals when they are outside the exhibit/enclosure (e.g., raptors on the glove, reptiles held "presentation style").
    b. Moderate handling and training techniques are used to present Ambassador Animals to the public. Public may be in close proximity to, or have direct contact with, Ambassador Animals when they're outside the exhibit/enclosure (e.g., media, fund raising, photo, and/or touch opportunities).
    c. Significant handling and training techniques are used to present Ambassador Animals to the public. Public may have direct contact with Ambassador Animals or simply observe the in-depth presentations when they're outside the exhibit/enclosure (e.g., wildlife education shows).
3. Off Grounds:
    a. Handling and training techniques are used to present Ambassador Animals to the public outside of the zoo/aquarium grounds. Public may have minimal contact or be in close proximity to and have direct contact with Ambassador Animals (e.g., animals transported to schools, media, fund raising events).

These categories assist staff and accreditation inspectors in determining when animals are designated as Ambassador Animals and the periods during which the Ambassador Animal-related Accreditation Standards are applicable. In addition, these Ambassador Animal categories establish a framework for understanding increasing degrees of an animal's involvement in Ambassador Animal activities.

Ambassador Animal presentations bring a host of responsibilities, including the safety and welfare of the animals involved, the safety of the animal handler and public, and accountability for the take-home, educational messages received by the audience. Therefore, AZA requires all accredited institutions that make Ambassador Animal presentations to develop an institutional Ambassador Animal policy that clearly identifies and justifies those species and individuals approved as Ambassador Animals and details their long-term management plan and educational program objectives.

AZA's accreditation standards require that education and conservation messages must be an integral component of all Ambassador Animal presentations. In addition, the accreditation standards require that the conditions and treatment of animals in education programs must meet standards set for the remainder of the animal collection, including species-appropriate shelter, exercise, appropriate environmental enrichment, access to veterinary care, nutrition, and other related standards. In addition, providing Ambassador Animals with options to choose among a variety of conditions within their environment is essential to ensuring effective care, welfare, and management. Some of these requirements can be met outside of the primary exhibit enclosure while the animal is involved in a program or is being transported. For example, free-flight birds may receive appropriate exercise during regular programs, reducing the need for additional exercise. However, the institution must ensure that in such cases, the animals participate in programs on a basis sufficient to meet these needs or provide for their needs in their home enclosures; upon return to the facility the animal should be returned to its species-appropriate housing as described above.

Ambassador Animal Position Statement

Last revision 1/28/03
Re-authorized by the Board June 2011

The Conservation Education Committee (CEC) of the Association of Zoos and Aquariums supports the appropriate use of Ambassador Animals as an important and powerful educational tool that provides a variety of benefits to zoo and aquarium educators seeking to convey cognitive and affective (emotional) messages about conservation, wildlife and animal welfare.
Utilizing these animals allows educators to strongly engage audiences. As discussed below, the use of Ambassador Animals has been demonstrated to result in lengthened learning periods, increased knowledge acquisition and retention, enhanced environmental attitudes, and the creation of positive perceptions concerning zoo and aquarium animals.

Audience Engagement
Zoos and aquariums are ideal venues for developing emotional ties to wildlife and fostering an appreciation for the natural world. However, developing and delivering effective educational messages in the free-choice learning environments of zoos and aquariums is a difficult task.
Zoo and aquarium educators are constantly challenged to develop methods for engaging and teaching visitors who often view a trip to the zoo as a social or recreational experience (Morgan & Hodgkinson, 1999). The use of Ambassador Animals can provide the compelling experience necessary to attract and maintain personal connections with visitors of all motivations, thus preparing them for learning and reflection on their own relationships with nature.

Ambassador Animals are powerful catalysts for learning for a variety of reasons. They are generally active, easily viewed, and usually presented in close proximity to the public. These factors have proven to contribute to increasing the length of time that people spend watching animals in zoo exhibits (Bitgood, Patterson & Benefield, 1986, 1988; Wolf & Tymitz, 1981).

In addition, the provocative nature of a handled animal likely plays an important role in captivating a visitor. In two studies (Povey, 2002; Povey & Rios, 2001), visitors viewed animals three and four times longer while they were being presented in demonstrations outside of their enclosure with an educator than while they were on exhibit. Clearly, the use of Ambassador Animals in shows or informal presentations can be effective in lengthening the potential time period for learning and overall impact.

Ambassador Animals also provide the opportunity to personalize the learning experience, tailoring the teaching session to what interests the visitors. Traditional graphics offer little opportunity for this level of personalization of information delivery and are frequently not read by visitors (Churchman, 1985; Johnston, 1998). For example, Povey (2001) found that only 25% of visitors to an animal exhibit read the accompanying graphic; whereas, 45% of visitors watching the same animal handled in an educational presentation asked at least one question and some asked as many as seven questions. Having an animal accompany the educator allowed the visitors to make specific inquiries about topics in which they were interested.

Knowledge Acquisition
Improving our visitors' knowledge and understanding regarding wildlife and wildlife conservation is a fundamental goal for many zoo educators using Ambassador Animals. A growing body of evidence supports the validity of using Ambassador Animals to enhance delivery of these cognitive messages as well.

- MacMillen (1994) found that the use of live animals in a zoomobile outreach program significantly enhanced cognitive learning in a vertebrate classification unit for sixth grade students.
- Sherwood and his colleagues (1989) compared the use of live horseshoe crabs and sea stars to the use of dried specimens in an aquarium education program and demonstrated that students made the greatest cognitive gains when exposed to programs utilizing the live animals.
- Povey and Rios (2002) noted that in response to an open-ended survey question ("Before I saw this animal, I never realized that . . . "), visitors watching a presentation utilizing a Ambassador Animal provided 69% cognitive responses (i.e., something they learned) versus 9% made by visitors viewing the same animal in its exhibit (who primarily responded with observations).
- Povey (2002) recorded a marked difference in learning between visitors observing animals on exhibit versus being handled during informal presentations. Visitors to demonstrations utilizing a raven and radiated tortoises were able to answer questions correctly at a rate as much as eleven times higher than visitors to the exhibits.

Enhanced Environmental Attitudes
Ambassador Animals have been clearly demonstrated to increase affective learning and attitudinal change.

- Studies by Yerke and Burns (1991), and Davison and her colleagues (1993) evaluated the effect live animal shows had on visitor attitudes. Both found their shows successfully influenced attitudes about conservation and stewardship.
- Yerke and Burns (1993) also evaluated a live bird outreach program presented to Oregon fifth-graders and recorded a significant increase in students' environmental attitudes after the presentations.
- Sherwood and his colleagues (1989) found that students who handled live invertebrates in an education program demonstrated both short and long-term attitudinal changes as compared to those who only had exposure to dried specimens.
- Povey and Rios (2002) examined the role Ambassador Animals play in helping visitors develop positive feelings about the care and well-being of zoo animals.
- As observed by Wolf and Tymitz (1981), zoo visitors are deeply concerned with the welfare of zoo animals and desire evidence that they receive personalized care.

Conclusion
Creating positive impressions of aquarium and zoo animals, and wildlife in general, is crucial to the fundamental mission of zoological institutions. Although additional research will help us delve further into this area, the existing research supports the conclusion that Ambassador Animals are an important tool for conveying both cognitive and affective messages regarding animals and the need to conserve wildlife and wild places.

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Appendix F: Developing an Institutional Ambassador Animal Policy

Last revision 2003
Re-authorized by the Board, June 2011

Rationale

Membership in AZA requires that an institution meet the AZA Accreditation Standards collectively developed by our professional colleagues. Standards guide all aspects of an institution's operations; however, the accreditation commission has asserted that ensuring that member institutions demonstrate the highest standards of animal care is a top priority. Another fundamental AZA criterion for membership is that education be affirmed as core to an institution's mission. All accredited public institutions are expected to develop a written education plan and to regularly evaluate program effectiveness.

The inclusion of animals (native, exotic, and domestic) in educational presentations, when done correctly, is a powerful tool. CEC's **Ambassador Animal Position Statement** describes the research underpinning the appropriate use of Ambassador Animals as an important and powerful educational tool that provides a variety of benefits to zoo and aquarium educators seeking to convey cognitive and affective messages about conservation and wildlife.

Ongoing research, such as AZA's Multi-Institutional Research Project (MIRP) and research conducted by individual AZA institutions will help zoo educators to determine whether the use of Ambassador Animals conveys intended and/or conflicting messages and to modify and improve programs accordingly and to ensure that all Ambassador Animals have the best possible welfare.

When utilizing Ambassador Animals our responsibility is to meet both our high standards of animal care and our educational goals. Additionally, as animal management professionals, we must critically address both the species' conservation needs and the welfare of the individual animal. Because "wild creatures differ endlessly," in their forms, needs, behavior, limitations and abilities (Conway, 1995), AZA, through its Animal Welfare Committee, has recently given the responsibility to develop taxon- and species-specific animal welfare standards and guidelines to the Taxon Advisory Groups (TAG) and Species Survival Plan® Program (SSP). Experts within each TAG or SSP, along with their education advisors, are charged with assessing all aspects of the taxons' and/or species' biological and social needs and developing Animal Care Manuals (ACMs) that include specifications concerning their use as Ambassador Animals.

However, even the most exacting standards cannot address the individual choices faced by each AZA institution. Therefore, each institution is required to develop a Ambassador Animal policy that articulates and evaluates program benefits. The following recommendations are offered to assist each institution in formulating its own Institutional Ambassador Animal Policy, which incorporates the AZA Ambassador Animal Policy and addresses the following matters.

The Policy Development Process

Within each institution, key stakeholders should be included in the development of that institution's policy, including, but not limited to representatives from:

- The Education Department
- The Animal Husbandry Department
- The Veterinary and Animal Health Department
- The Conservation & Science Department
- The Behavioral Husbandry Department
- Any animal show staff (if in a separate department)
- Departments that frequently request special Ambassador Animal situations (e.g., special events, development, marketing, zoo or aquarium society, administration)

Additionally, staff from all levels of the organization should be involved in this development (e.g., curators, keepers, education managers, interpreters, volunteer coordinators).

To develop a comprehensive Ambassador Animal Policy, we recommend that the following components be included:

I. Philosophy

In general, the position of the AZA is that the use of animals in up close and personal settings, including animal contact, can be extremely positive and powerful, as long as:

1. The use and setting is appropriate.
2. Animal and human welfare is considered at all times.
3. The animal is used in a respectful, safe manner and in a manner that does not misrepresent or degrade the animal.
4. A meaningful conservation message is an integral component. Read the AZA Board-approved Conservation Messages.
5. Suitable species and individual specimens are used.

Institutional Ambassador Animal policies should include a philosophical statement addressing the above, and should relate the use of Ambassador Animals to the institution's overall mission statement.

II. Appropriate Settings

The Ambassador Animal Policy should include a listing of all settings both on and off site, where Ambassador Animal use is permitted. This will clearly vary among institutions. Each institution's policy should include a comprehensive list of settings specific to that institution. Some institutions may have separate policies for each setting; others may address the various settings within the same policy. Examples of settings include:

1. On-site programming
    a. Informal and non-registrants:
        i. On-grounds programming with animals being brought out (demonstrations, lectures, parties, special events, and media)
        ii. Children's zoos and contact yards
        iii. Behind-the-scenes open houses
        iv. Shows
        v. Touch pools
    b. Formal (registration involved) and controlled settings
        i. School group programs
        i. Summer camps
        ii. Overnights
        iii. Birthday parties
        iv. Animal rides
        v. Public animal feeding programs
    c. Offsite and outreach
        i. PR events (TV, radio)
        ii. Fundraising events
        iii. Field programs involving the public
        iv. School visits
        v. Library visits
        vi. Nursing home visits (therapy)
        vii. Hospital visits
        viii. Senior centers
        ix. Civic group events

In some cases, policies will differ from setting to setting (e.g., on-site and off-site use with media). These settings should be addressed separately, and should reflect specific animal health issues, assessment of distress in these situations, limitations, and restrictions.

III. Compliance with Regulations

All AZA institutions housing mammals are regulated by the USDA's Animal Welfare Act. Other federal regulations, such as the Marine Mammal Protection Act, may apply. Additionally, many states, and some cities, have regulations that apply to animal contact situations. Similarly, all accredited institutions are bound by the AZA Code of Professional Ethics. It is expected that the Institution Ambassador Animal Policy address compliance with appropriate regulations and AZA Accreditation Standards.

IV. Collection Planning

AZA accredited institutions should have a collection planning process in place. Ambassador Animals are part of an institution's overall collection and must be included in the overall collection planning process. The AZA Guide to Accreditation contains specific requirements for the institution collection plan. For more information about collection planning in general, please see the Collection Management pages in the Members Only section.

The following recommendations apply to Ambassador Animals:

1. Listing of approved Ambassador Animals (to be periodically amended as collection changes). Justification of each species should be based upon criteria such as:
    a. Temperament and suitability for program use
    b. Husbandry requirements
    c. Husbandry expertise
    d. Veterinary issues and concerns
    e. Ease and means of acquisition / disposition according to the AZA code of ethics
    f. Educational value and intended conservation message
    g. Conservation Status
    h. Compliance with TAG and SSP guidelines and policies
2. General guidelines as to how each species (and, where necessary, for each individual) will be presented to the public, and in what settings
3. The collection planning section should reference the institution's acquisition and disposition policies.

V. Conservation Education Message

As noted in the AZA Accreditation Standards, if animal demonstrations are part of an institution's programs, an educational and conservation message must be an integral component. The Ambassador Animal Policy should address the specific messages related to the use of Ambassador Animals, as well as the need to be cautious about hidden or conflicting messages (e.g., "petting" an animal while stating verbally that it makes a poor pet). This section may include or reference the AZA Conservation Messages.

Although education value and messages should be part of the general collection planning process, this aspect is so critical to the use of Ambassador Animals that it deserves additional attention. In addition, it is highly recommended to encourage the use of biofacts in addition to or in place of the live animals. Whenever possible, evaluation of the effectiveness of presenting Ambassador Animals should be built into education programs.

VI. Human Health and Safety

The safety of our staff and the public is one of the greatest concerns in working with Ambassador Animals. Although extremely valuable as educational and affective experiences, contact with animals poses certain risks to the handler and the public. Therefore, the human health and safety section of the policy should address:

1. Minimization of the possibility of disease transfer from non-human animals to humans, and vice-versa (e.g., hand washing stations, no touch policies, use of hand sanitizer).
2. Safety issues related to handlers' personal attire and behavior (e.g., discourage or prohibit use of long earrings, perfume and cologne, not eating or drinking around animals, smoking, etc.).

AZA's Animal Contact Policy provides guidelines in this area; these guidelines were incorporated into accreditation standards in 1998.

VII. Animal Health and Welfare

Animal health and welfare are the highest priority of AZA accredited institutions. As a result, the Institutional Ambassador Animal Policy should make a strong statement on the importance of animal welfare. The policy should address:

1. General housing, husbandry, and animal health concerns (e.g. that the housing and husbandry for Ambassador Animals meets or exceeds general AZA standards and that the physical, social and psychological needs of the individual animal, such as adequate rest periods, provision of enrichment, visual cover, contact with conspecifics as appropriate, etc., are accommodated).

2. Where ever possible provide a choice for animal program participation, e.g., retreat areas for touch tanks or contact yards, evaluation of willingness/readiness to participate by handler, etc.)
3. The empowerment of handlers to make decisions related to animal health and welfare; such as withdrawing animals from a situation if safety or health is in danger of being compromised.
4. Requirements for supervision of contact areas and touch tanks by trained staff and volunteers.
5. Frequent evaluation of human / animal interactions to assess safety, health, welfare, etc.
6. Ensure that the level of health care for the Ambassador Animals is consistent with that of other animals in the collection.
7. Whenever possible have a “cradle to grave” plan for each Ambassador Animal to ensure that the animal can be taken care of properly when not used as a Ambassador Animal anymore.
8. If lengthy “down” times in Ambassador Animal use occur, staff should ensure that animals accustomed to regular human interactions can still maintain such contact and receive the same level of care when not used in programs.

VIII. Taxon Specific Protocols

We encourage institutions to provide taxonomically specific protocols, either at the genus or species level, or the specimen, or individual, level. Some taxon-specific guidelines may affect the use of Ambassador Animals. To develop these, institutions refer to the Conservation Programs Database.

Taxon and species -specific protocols should address:

1. How to remove the individual animal from and return it to its permanent enclosure, including suggestions for operant conditioning training.
2. How to crate and transport animals.
3. Signs of stress, stress factors, distress and discomfort behaviors.

Situation specific handling protocols (e.g., whether or not animal is allowed to be touched by the public, and how to handle in such situations):

1. Guidelines for disinfecting surfaces, transport carriers, enclosures, etc. using environmentally safe chemicals and cleaners where possible.
2. Animal facts and conservation information.
3. Limitations and restrictions regarding ambient temperatures and or weather conditions.
4. Time limitations (including animal rotation and rest periods, as appropriate, duration of time each animal can participate, and restrictions on travel distances).
5. The number of trained personnel required to ensure the health and welfare of the animals, handlers and public.
6. The level of training and experience required for handling this species
7. Taxon/species-specific guidelines on animal health.
8. The use of hand lotions by program participants that might touch the animals

IX. Logistics: Managing the Program

The Institutional Policy should address a number of logistical issues related to Ambassador Animals, including:

1. Where and how the Ambassador Animal collection will be housed, including any quarantine and separation for animals used off-site.
2. Procedures for requesting animals, including the approval process and decision-making process.
3. Accurate documentation and availability of records, including procedures for documenting animal usage, animal behavior, and any other concerns that arise.

X. Staff Training

Thorough training for all handling staff (keepers, educators, and volunteers, and docents) is clearly critical. Staff training is such a large issue that many institutions may have separate training protocols and procedures. Specific training protocols can be included in the Institutional Ambassador Animal Policy or reference can be made that a separate training protocol exists.

It is recommended that the training section of the policy address:

1. Personnel authorized to handle and present animals.

2. Handling protocol during quarantine.
3. The process for training, qualifying and assessing handlers including who is authorized to train handlers.
4. The frequency of required re-training sessions for handlers.
5. Personnel authorized to train animals and training protocols.
6. The process for addressing substandard performance and noncompliance with established procedures.
7. Medical testing and vaccinations required for handlers (e.g., TB testing, tetanus shots, rabies vaccinations, routine fecal cultures, physical exams, etc.).
8. Training content (e.g., taxonomically specific protocols, natural history, relevant conservation education messages, presentation techniques, interpretive techniques, etc.).
9. Protocols to reduce disease transmission (e.g., zoonotic disease transmission, proper hygiene and hand washing requirements, as noted in AZA's Animal Contact Policy).
10. Procedures for reporting injuries to the animals, handling personnel or public.
11. Visitor management (e.g., ensuring visitors interact appropriately with animals, do not eat or drink around the animal, etc.).

XI. Review of Institutional Policies

All policies should be reviewed regularly. Accountability and ramifications of policy violations should be addressed as well (e.g., retraining, revocation of handling privileges, etc.). Institutional policies should address how frequently the Ambassador Animal Policy will be reviewed and revised, and how accountability will be maintained.

XII. TAG and SSP Recommendations

Following development of taxon-specific recommendations from each TAG and SSP, the institution policy should include a statement regarding compliance with these recommendations. If the institution chooses not to follow these specific recommendations, a brief statement providing rationale is recommended.

Appendix G: Browse and Forage Species Used in Zoological Institutions in North America

Any browse or forage fed to capybara must be clean, free of contaminants (dust, stones, plant detritus, animal matter, fecal matter, etc.) and pesticides. Avoid using plants grown along roads and highways because the plants can contain petroleum residues and chemicals used for road maintenance. Commercial suppliers should be asked about pesticide use for their products and/or purchase edible plant materials from organic producers. When possible, plants and produce should be washed.

Ingestion of poisonous plants or chemicals: General symptoms caused by the ingestion of poisonous plants or chemicals are often sudden onset and can include the mouth (frothing, blisters, burned tissue, panting); eyes (dilated pupils, glassy eyes); pulse (arrhythmia (erratic heartbeat) or a weak, rapid pulse); lack of muscle coordination (staggering, incoordination, inability to stand); excitability; and, convulsions. Specific reactions to toxic plants can be found on some of the below reference sites. ***If plant or chemical toxicity is suspected, veterinary professionals should be consulted immediately.***

Plant toxicity websites: The following websites can be consulted for information on the toxicity of various plants.

http://aggie-horticulture.tamu.edu/plantanswers/publications/poison/poison.html
http://www.ansci.cornell.edu/plants/index.html
http://poisonousplants.ansci.cornell.edu/
http://www.accessdata.fda.gov/scripts/plantox/
http://www.cbif.gc.ca/eng/species-bank/canadian-poisonous-plants-information-system/?id=1370403265036
http://www.library.illinois.edu/vex/toxic/comlist.htm
http://www.ars-grin.gov/duke/highchem.html
http://research.vet.upenn.edu/Default.aspx?alias=research.vet.upenn.edu/poisonousplants

Forage and Browse Used in North American Zoological Institutions (Organized alphabetically by genus)

Genus	Species	Common Name
Abies	*A. balsamea*	Balsam Fir
Abies	*A. concolor*	White Fir
Abies	*A. fraseri*	Frasier Fir
Abies	*P. abies*	Norway Spruce
Acacia	*A. auriculiformis*	Ear Leaf Acacia
Acacia	*A. cultriformis*	Knife Acacia
Acacia	*A. cyclopsis*	Coastal Wattle
Acacia	*A. dealbata*	Fern Leaf Acacia
Acacia	*A. decurrens*	Green Wattle
Acacia	*A. farnessiana*	Sweet Acacia
Acacia	*A. karroo*	Karroo Thorn
Acacia	*A. longifolia*	Long Leaf Acacia
Acacia	*A. longifolia*	Sydney Golden Wattle
Acacia	*A. melanoxylan*	Black Acacia
Acacia	*A. redolens*	Desert Carpet
Acacia	*A. saligna*	Blue Leaf Wattle
Acacia	*A. saligna*	Weeping Wattle
Acacia	*A. woodii*	Paperbark Acacia
Acacia	*A. xanthophloea*	African Fever Tree

Acer	*A. accharinum*	Silver Maple
Acer	*A. egundo*	Box Elder
Acer	*A. ginnala*	Amur Maple
Acer	*A. negundo*	Baron Manitoba Maple
Acer	*A. palmatum*	Japanese Maple
Acer	*A. platanoides*	Crimson King
Acer	*A. platanoides*	Norway Maple
Acer	*A. psuedoplatanus*	Sycamore
Acer	*A. saccharum*	Sugar Maple
Actinidia	*A. chinensis*	Kiwi Fruit
Agropyron	*A. smithii*	Wheatgrass
Albizia	*A. julibrissin*	Mimosa
Albizia	*A. julibrissin*	Silk Tree
Alcea	*A. rosea*	Hollyhock
Allium	*A. schoenoprasum*	Chives
Allium	*A. tuberosum*	Garlic Chives
Alnus	*A. crispa*	Green Alder
Alnus	*A. glutinosa*	Balck Alder
Alnus	*A. rhombiflora*	White Alder
Alnus	*A. rubra*	Red Alder
Alnus	*A. tenuifolia*	Mountain Alder
Alnus	*A. viridis*	European Alder
Aloysia	*A. triphylla*	Lemon Verbena
Alpinia	*A. purpurata*	
Alpinia	*A. zerumbet*	Ginger
Amelanchier	*A. alnifolia*	Saskatoon Serviceberry
Amelanchier	*A. canadensis*	Serviceberry
Amelanchier	*Amelanchier spp.*	Amelanchier or Shadbush
Andopogon	*Andopogon spp.*	Blue Stem
Anethum	*A. graveolus*	Dill
Antirrhinum	*Antirrhinum spp.*	Snapdragons
Apium	*A. graveolus*	Celery
Aronia	*A. melanocarpa*	Black Chokeberry
Artemesia	*A. dracunculus*	Tarragon
Arundinaria	*Arundinaria spp.*	Bamboo
Arundo	*A. donax*	Elephant Grass
Arundo	*Arundo sp.*	
Asparagus	*A. sprengeri*	Asparagus Fern
Bambusa	*B. beechyan*	Beechy Bamboo
Bambusa	*B. glaucescens*	Hedge Bamboo
Bambusa	*B. multiplex*	Golden Goddess Bamboo
Bambusa	*B. nuda*	

Bambusa	*B. oldhamii*	Oldham Bamboo
Bambusa	*B. textillis*	Weaver's Bamboo
Bambusa	*B. Tuldoide*	Punting Pole Bamboo
Bambusa	*B. ventricosa*	Budda's Belly Bamboo
Bambusa	*B. vulgaris cv. Vittata*	Painted Bamboo
Bashania	*B. fargesii*	
Bauhinia	*B. blakeana*	HongKong Orchid Tree
Bauhinia	*B. variegate*	Variegatedorchid Tree
Berberis	*B. julianae*	Evergreen Barberry
Berberis	*B. koreana*	Korean Barberry
Berberis	*B. vulgaris*	Red Barberry
Beta	*B. vulgaris*	Beet
Betula	*B. nigra*	River Birch
Betula	*B. papyrifera*	Paper Birch
Betula	*B. pendula laciniata*	Cutleaf Weeping Birch
Bischofia	*B. javonica*	Bischofia
Bischofia	*B. trifoliate*	Toob
Borago	*B. officionalis*	Borage
Boutelaua	*Boutelaua sp.*	Grass
Brachychiton	*B. discolor*	Queensland Lace Bark
Brassaia	*B. actynophylla*	UmbrellaTree
Brassica	*B. juncea*	Mustard Greens
Brassica	*B. oleracea*	Collard Greens
Bryanthus	*Bryanthus spp.*	Pampas Grass
Bucida	*B. buceras*	Black Olive
Buddleia	*B. cloudii*	Butterfly Bush
Calamagrostis	*Calamagrostis spp.*	Reed Grass
Calendula	*C. officinalis*	Calendula Marigold
Calistemon	*C. virrinalis*	Bottle Brush
Calliandra	*C. haematocephala*	Powder Puff
Callicarpa	*Callicarpa sp*	Beautyberry
Camellia	*C. sasanqua*	Camelia
Canna	*C. generalis*	Canna
Caragana	*C. arborescens*	Siberian Pea Shrub
Carica	*C. papaya*	Papaya
Carissa	*C. grandiflora*	Natal Plum
Carnegia	*C. gigantica*	Saguaro
Carpenus	*Carpinus spp.*	Hornbean
Carya	*C. glabra*	Pignut Hickory
Carya	*C. illinoinensis*	Pecan
Carya	*Carya spp.*	Hickory
Caryota	*Caryota spp.*	Fishtail Palm

Catalpa	*C. speciosa*	Catalpa
Celtis	*C. australis*	Mediterranean Hackberry
Celtis	*C. laevigata*	Sugar Hackberry
Celtis	*C. occidentalis*	Hackberry
Centraderia	*Centraderia spp.*	Centraderia
Ceratonia	*C. siliqua*	Carob Tree
Cercidium	*C. floridum*	Palo Verde
Cercis	*C. canadensis*	Redbud
Chaenomeles	*C. japonica*	Floweringquince
Chimnobambusa	*C. quadrangularis*	Square Bamboo
Chlorophytum	*Chlorophytum spp.*	Spider Plant
Chrysalidocarpus	*C. lutescens*	Areca Palm
Chrysobalanus	*C. icaco*	Cocoplum
Cinnamamum	*C. camphora*	Camphor Laurel
Citharexylum	*C. fruticosum*	Fiddlewood
Citrellus	*C. lanatus*	Watermelon
Citrus	*C. sinensis*	Orange
Citrus	*Citrus spp.*	Lime, Lemon
Cladrastis	*C. tinctoria*	Yellowwood
Conocarpus	*C. erectus*	Buttonwood
Coprosma	*Coprosma spp.*	Coprosma
Coriandrum	*C. sativum*	Cilantro
Cornus	*C. florida*	Dogwood
Cornus	*C. kausa*	Korean Dogwood
Cornus	*C. sericea*	American Dogwood
Cornus	*C. stolonifera*	Yellow Twig Dogwood
Cortaderia	*C. selloana*	Hardy Pampas Grass
Corylus	*C. avellana*	Hazelnut
Costus	*C. speciosus*	Crepe Ginger
Costus	*C. spiralis*	Spiral Ginger
Cotoneaster	*C. lacteus*	Cotoneaster
Crataegus	*Crataegus spp.*	Hawthorn
Cucumis	*C. sativus*	Cucumber
Cucurbita	*C. pepo*	Squash Pumpkin
Cupaniopsis	*C. anacardioides*	Carrotwood
Cymbopogon	*C. citratus*	Lemongrass
Cynara	*C. scolymus*	Artichoke
Cynodon	*C. dactylon*	Coastal Bermudagrass
Cyperus	*C. alterrifolius*	Umbrella Plant
Cyrrbopogon	*C. citratus*	Lemon Grass
Dendrocalamus	*D. brandisii*	Kurz
Dendrocalanus	*Dendrocalanus spp.*	Bamboo

Dendrocalarrus	*D. giganteus*	Giant Bamboo
Dianthus	*D. caryophyllus*	Carnations
Digitaria	*D. californica*	Cottontop Grass
Diospyros	*D. kaki*	Persimmon
Dracaena	*D. fragrans*	Corn Plant
Duranta	*D. repens*	Skyflower
Ebbingei	*E. elaeganus*	Silverberry
Eichhornia	*E. crassipes*	Water Hyacinth
Elaeagnus	*E. pungens*	Thorny Eleagnus
Elaeagnus	*E. commutate*	Silverberry
Elaeocarpus	*E. decipiens*	Japanese Blueberry
Eleagnus	*E. angustifolia*	Russian Olive
Eleagnus	*E. umbellata*	Autumn Olive
Epipremnum	*Epipremnum spp.*	Pothos
Eragrostis	*E. curvula*	Love Grass
Erianthus	*E. ravennae*	Hardy Pampas Grass
Eriobotrya	*E. japonica*	Loquat
Escallonia	*E. rubra*	Redclaws
Escallonia	*Escallonia spp.*	Escalonia
Eucalyptus	*E. bigalerita*	Woolybutt
Eucalyptus	*E. camaldulensis*	Murray red gum
Eucalyptus	*E. cinerea*	Eucalyptus
Eucalyptus	*E. cinerea*	Silver-Dollar Tree
Eucalyptus	*E. cornuta*	Yate Tree
Eucalyptus	*E. dalrympleana*	Mountain Gum
Eucalyptus	*E. globulus*	Blue Gum
Eucalyptus	*E. goniocalyx*	Bundy
Eucalyptus	*E. grandis*	Rose Gum
Eucalyptus	*E. gunnii*	Blue Gum
Eucalyptus	*E. gunnii*	Cider Tree
Eucalyptus	*E. lehmannii*	Lehmann's Gum
Eucalyptus	*E. leucoxylon*	White Ironbark
Eucalyptus	*E. macarthurii*	Camden Woollybutt
Eucalyptus	*E. melliodora*	Yellow Box
Eucalyptus	*E. microcorys*	Tallowwood
Eucalyptus	*E. microtheca*	Blue Ghost
Eucalyptus	*E. muellerana*	Yellow Stringybark
Eucalyptus	*E. nicholii*	Narrow-leaved Black Peppermint
Eucalyptus	*E. occidentali*	Flat-topped Yate
Eucalyptus	*E. ovata*	Swamp Gum
Eucalyptus	*E. pauciflor*	Cabbage Gum
Eucalyptus	*E. perriniana*	Round-leaved Snow Gum

Eucalyptus	*E. polyanthemos*	Silver-dollar Gum
Eucalyptus	*E. polycarpa*	Bloodwood
Eucalyptus	*E. pulverulenta*	Silver-leaved Mountain Gum
Eucalyptus	*E. punctata*	Gray Gum
Eucalyptus	*E. regnans*	Giant Gum
Eucalyptus	*E. resinifera*	Red Mahogany
Eucalyptus	*E. robusta*	Swamp Mahogany
Eucalyptus	*E. rudis*	Desert Gum
Eucalyptus	*E. saligna*	Sydney Blue Gum
Eucalyptus	*E. sideroxylon*	Red Ironbark
Eucalyptus	*E. tereticornis*	Forest Red Gum
Eucalyptus	*E. viminali*	Manna Gum
Eucalyptus	*Eucalyptus spp.*	Broad-leafed Peppermint
Eucalyptus	*Eucalyptus spp.*	Fuzzy Leaf Eucalyptus
Eugenia	*E. cumini*	
Eugenia	*E. jambos*	
Eugenia	*E. peniculata*	
Euonymous	*E. fortunei*	Winter Creeper
Fagus	*F. grandifolia*	American Beech
Fagus	*F. sylvatica*	European Beech
Fargesia	*F. fungosa*	
Fargesia	*F. nitida*	
Feijoa	*F. sellowiana*	Pineapple Guava
Ficus	*F. aurea*	Stangler Fig
Ficus	*F. benghalensis*	Banyon Tree
Ficus	*F. citrifolia*	Shortleaf Fig
Ficus	*F. elastica*	Indian Rubber Tree
Ficus	*F. berjinina*	Weeping Fig
Ficus	*F. afzelii*	Afzelius Fig
Ficus	*F. Benjamin*	Benjamin Tree
Ficus	*F. elastica decorum*	Indian Rubber Plant
Ficus	*F. florida*	
Ficus	*F. lyrata*	Fiddle Leaf Fig
Ficus	*F. macrophylla*	Moreton Bay Fig
Ficus	*F. microcarpa*	Chinese Banyan Tree
Ficus	*F. microcarpa nitida*	
Ficus	*F. microcarpa retusa*	
Ficus	*F. mysorensis*	
Ficus	*F. natalensis*	
Ficus	*F. nekbud*	Zulu Fig Tree
Ficus	*F. pumila*	Creeping Fig
Ficus	*F. religiosa*	Bo Tree

Ficus	*F. rubiginosa*	Rusty Leaf Fig
Ficus	*F. rumphii*	
Ficus	*F. sycomorus*	Sycamore Fig
Ficus	*F. thoningii*	
Ficus	*F. trichopoda*	
Ficus	*F. watkinsiana*	
Ficus	*F.rubiginos*	Rusty Fig Mulberry
Firmiana	*F. simplex*	Chinese Parasol Tree
Foericulum	*F. vulgare*	Fennel
Forestiera	*F. neo-mexicana*	New Mexican Privet
Forsythia	*Forsythia spp.*	Forsythia
Fragaria	*Fragaria spp.*	Strawberry
Fraxinus	*F. americana*	American Ash
Fraxinus	*F. pennsylvanica*	Green Ash
Fraxinus	*F. velutina*	Velvet Ash
Galium	*G. odoratum*	Sweet Woodruff
Gingko	*G. biloba*	Ginkgo Tree
Gleditisia	*G. triacanthos*	Honeylocust
Gleditsia	*Gleditsia triacanthos*	Thorny Honey Locust
Grewia	*G. microsus L.*	
Grewia	*G. occidentalis*	
Hamamelis	*H. intermedia*	Witchhazel
Hamelia	*H. patens*	Firebush
Harpephyllum	*H. caffrum*	Kaffir Plum
Hedera	*H. helix*	English Ivy
Hedychium	*H. coronarium*	Shell Ginger
Hedychium	*H. gardenarium*	White Ginger Lilly
Helianthus	*H. anuus*	Sunflower
Heliconia	*H. caribea*	Lobster Claw
Hemerocallis	*Hemerocallis spp.*	Daylilly
Hibiscus	*H. mascheutos*	Swamp Mallow
Hibiscus	*H. mutabilis*	Confederate Rose
Hibiscus	*H. rosa-sinensis*	Rose of China
Hibiscus	*H. syriacus*	Rose of Sharon
Hibiscus	*H. tiliaceus*	Beach Hibiscus
Hippophae	*H. rhamnoides*	Common Seabuckthorn
Hydrocotyle	*H. umbellate*	Dollar Weed
Ilex	*Ilex spp.*	Ilex
Inga	*I. vera*	
Jacaranda	*J. mimosifoli*	Jacaranda Tree
Jacaranda	*J. obtusifolia*	Jacaranda Tree
Jasminum	*Jasminum spp.*	Jasmine

Juniperus	*J. communis*	Effusa Common Juniper
Juniperus	*J. sabrina*	Savin Juniper
Kerria	*Kerria spp.*	Kerria
Koelreuteria	*K. paniculata*	Golden Raintree
Lactuca	*L. sativa*	Lettuce
Larix	*L. laricina*	Larch
Larix	*L. siberica*	Siberian Larch
Lavendula	*L. angustiflolia*	English Lavender
Levisticum	*L. officinale*	Lovage
Leycesteria	*L. formosa*	Himalayan Honeysuckle
Ligustrum	*L. igustrum japonicum*	Wax Leaf Privet
Ligustrum	*L. sinense*	Chinese Privet
Liquidambar	*L. styraciflua*	Sweetgum
Liriodendron	*L. tulipifera*	TulipTree
Liriope	*L. platyphylla*	Monkey Grass
Lonicera	*L. fragrantissima*	Honeysuckle
Lonicera	*L. tatarica*	Honeysuckle
Lonicera	*Lonicera spp.*	Bush Honeysuckle
Lonicera	*Lonicera spp.*	Honey Locust
Lonicera	*L. japonica*	Japanese Honeysuckle
Lonicera	*L. maachii*	Amur Honeysuckle
Magnolia	*Magnolia sp*	Magnolia
Mahonia	*Mahonia spp.*	Mahonia
Malus	*Malus spp.*	Crab Apple
Malvaviscus	*Malvaviscus spp.*	Turk's Cap
Mangifera	*M. indica*	Mango
Marrubium	*M. vulgare*	Common Horehound
Medicago	*M. sativa*	Alfalfa
Melissa	*M. officionalis*	Lemon Balm
Mentha	*M. piperita*	Peppermint
Miscanthus	*M. sinensis*	Chinese Silvergrass
Miscanthus	*Miscanthus spp.*	Japanese Silver Grass
Miscanthus	*Miscanthus spp.*	Maiden Grass
Monard	*M. fistulosa*	Bee Balm
Morus	*M. rubra*	Red Mulberry
Morus	*Morus alba*	White Mulberry
Morus	*Morus nigra*	Black Mulberry
Morus	*Morus rubra*	Mulberry
Musa	*M. acuminate*	Banana
Musa	*M. cavendishii*	Banana
Musa	*M. paradisiaca*	Banana
Myrica	*M. cerifera*	Wax Myrtle

Nandina	*N. domestica*	Nandina
Nasturtium	*N. trapaeolummquis*	
Nasturtium	*T. majus*	Indian Cress Nasturtium
Nasturtium	*T. minus*	Dwarf Nasturtium
Nasturtium	*T. tuberosum*	Anu, Ysanu, Mashua
Nepeta	*N. cataria*	Catnip
Nyssa	*N. sylvatica*	Sour Gum
Ocimum	*Ocimum spp.*	Basil
Oncidium	*Oncidium spp.*	Butterfly Orchid
Opuntia	*O. englemanni*	Prickly Pear
Opuntia	*Opuntia spp.*	Spineless Cactus
Origanum	*O. majorana*	Marjoram
Origanum	*O. vulgare*	Oregano
Paricum	*P. vergatum*	Switchgrass/
Parkinsonia	*P. microphylla*	Palo Verde
Parthenocissus	*P. quinquefolia*	Kahili Ginger
Passiflora	*P. incarnate*	Passion Flower Vine
Pelargonium	*Pelargoniumspp.*	Scented Geranium
Pennisetum	*P. rubrum*	red fountain grass
Pennisetum	*P. typhoideum*	Kikuyu
Pennisetum	*Pennisetum spp.*	Fountain grass/
Pennisetum	*Pennisetum spp.*	Ornamental grass/
Perovskia	*P. atriplicafolia*	Russian Sage
Petroselinum	*P. crispum*	Parsley
Phaseolus	*P. vulgaris*	Common Green Bean
Phaseolus	*Phaseolus spp.*	Bean Vine
Philadelphus	*Philadelphus spp.*	Mock Orange
Phoenix	*P. reclinata*	Senegal Date Palm
Photinia	*P. frazeri*	Photinia
Photinia	*P. glabra*	Red Tip Photinia
Photinia	*P. serrulata*	Photinia
Phyllostachys	*P. dulcis*	Sweet Shoot Bamboo
Phyllostachys	*P. rubr omarginata*	Red-margined Bamboo
Phyllostachys	*Phyllostachys spp.*	Bamboo
Phyllostachys	*P. aurea*	Golden Bamboo
Phyllostachys	*P. aureosulcata*	Yellow Groove Bamboo
Phyllostachys	*P. bambusoides cv. Castillion*	
Phyllostachys	*P. meyeri*	Meyer's Bamboo
Phyllostachys	*P. nigr*	Black Bamboo
Phyllostachys	*P. nuda*	Nude Sheath Bamboo
Phyllostachys	*P. vivax*	

Picea	*P. glauca*	Alberta Spruce
Picea	*P. glauca*	White Spruce
Picea	*P. pungens*	Blue Spruce & CO Spruce
Pinus	*P. contorta latifolia*	Lodgepole Pine
Pinus	*P. mugo mugo*	Mugho Pine
Pinus	*P. nigra*	Austrian Pine
Pinus	*P. ponderosa*	Ponderosa Pine
Pinus	*P. strobus*	White Pine
Pinus	*P. thumbergiana*	Japanese Black Pine
Pistacia	*P. clinensis*	Chinese Pistachio
Pisum	*P. sativum*	Pea Vine
Pittosporum	*Pittosporum spp.*	Pittosporum
Platanus	*P. occidentalis*	American Planetree
Platanus	*P.acerifolia*	London Plane Tree
Plumbago	*P. auriculata*	Plumbago
Polygonum	*Polygonum spp.*	Knotweed
Polystichum	*P. munitum*	Sword Fern
Populus	*P. acuminata*	Lanceleaf Poplar
Populus	*P. angustifolia*	Narrowleaf Cottonwood
Populus	*P. balsamifera*	Balsam Poplar
Populus	*P. deltoides*	Eastern Cottonwood
Populus	*P. deltoids*	Cottonwood
Populus	*P. fremontii*	Fremont Cottonwood
Populus	*P. sargentii*	Plains Cottonwood
Populus	*Populus spp.*	Aspen
Populus	*Populus spp.*	Poplar Hybrids
Portulaca	*P. oleracea*	Summer Purselane
Prosopis	*Prosopis spp.*	Mesquite
Prospis	*P. veluntina*	Velvet Mesquite
Prunus	*P. americana*	Apricot
Prunus	*P. americana*	Wild Plum
Prunus	*P. persica*	Peach
Prunus	*P. tomentosa*	Nanking Cherry
Prunus	*P. dulcis*	Almond
Prunus	*Prunus spp.*	Plum
Pseudosasa	*P. japonica*	Arrow Bamboo
Psuedotsuga	*P. menziesii*	Douglas Fir
Pueraria	*Pueraria spp.*	Kudzu
Purica	*P. granatum*	Pomegranate
Pyracantha	*P.coccinea*	Scarlet Firethorn
Pyrus	*Pyrus spp.*	Fruiting Pear
Quercus	*Q. alba*	White Oak

Quercus	*Q. ilex*	Holm Oak
Quercus	*Q. rubra*	Red Oak
Quercus	*Q. velutina*	Black Oak
Quercus	*Q. virginiana*	Live Oak
Rhamnus	*R. cathartica*	Buckthorn
Rhododendron	*Rhododendron spp.*	Rhododendron
Rhus	*R. glabra*	Smooth Sumac
Rhus	*R. typhina*	Staghorn Sumac
Rosa	*R. banksiae*	Lady Banks Rose
Rosa	*Rosa spp.*	Rose
Rosemarius	*R. officinalis*	Rosemary
Rubus	*Rubus spp.*	Blackberry
Rubus	*Rubus spp.*	Bramble
Rubus	*Rubus spp.*	Raspberry, Blackberry
Sabal	*S. minor*	Blue Palmetto
Sabal	*S. palmetto*	Palmetto
Saccharum	*S. officinarum*	Sugar Cane
Salix	*S. alba*	White Willow
Salix	*S. Austree*	Willow
Salix	*S. babylonia*	Weeping Willow
Salix	*S. caesia*	Blue Willow
Salix	*S. caprea*	Goat Willow
Salix	*S. caroliniana*	Carolina Willow
Salix	*S. discolor*	American Pussy Willow
Salix	*S. exigua*	Coyote Willow
Salix	*S. goodingii*	Gooding's Willow
Salix	*S. matsudana*	Corkscrew Willow
Salix	*S. matsudana*	Dragon's Claw Willow
Salix	*S. matsudana alba*	Golden Weeping Willow
Salix	*S. nigra*	Black Willow
Salix	*S. pentadra*	Laurel Leaf Willow
Salvia	*S. elegans*	Pineapple Sage
Salvia	*S. officianalis*	Sage
Sanguisorba	*S. minor*	Salad Burnet
Santolina	*S. chamaecyparissus*	Cotton Lavender
Sasa	*Sasa spp.*	Bamboo
Sassafras	*S. albidum*	Sassafras
Scheffelara	*S. actinophylla*	Australia Umbrella Tree
Schinus	*S. molle*	California Pepper Tree
Schinus	*S. terebinthifolius*	Brazilian Pepper
Scirpus	*Scirpus spp.*	Bulrush
Sechium	*S. edule*	Choyote

Semiarundinaria	*Semiarundinaria spp.*	Bamboo
Sheffelera	*S. arboricola*	Dwarf Scheffelera
Shibataea	*Shibataea spp.*	Bamboo
Simmondsia	*S. chinensis*	Jojoba
Sinarundinaria	*Sinarundinaria spp.*	Bamboo
Smilax	*Smilax spp.*	Greenbrier
Solanum	*S. lycopersicum cerasiforme*	Cherry Tomato
Sophora	*S. japonica*	Japanese Pagoda Tree
Spathodea	*S. campanulata*	African Tulip Tree
Spinacia	*S. oleracea*	Spinach
Stenotaphrum	*S. secundatum*	Saint Augustine Grass
Strelitzia	*S. nicolai*	Giant Bird of Paradise
Strelitzia	*S. reginae*	Bird of Paradise
Swieteria	*S. mahagoni*	WestIndies Mahogany
Symphoricarpos	*Symphoricarpos spp.*	Snowberry
Symphytum	*Symphytum spp.*	Comfrey
Syringa	*S. vulgaris*	Lilac
Syzygium	*S. cumini*	Java Plum
Syzygium	*S. jambos*	Rose Apple
Syzygium	*S. cumini*	Java Plum
Syzygium	*S. paniculatum*	Eugeria
Tagetes	*T. patula*	Marigold
Tamarindus	*T. indica*	Tamarind
Tanacetum	*Tanacetum spp.*	Tansy
Taraxacum	*T. officinale*	Dandelion
Tecomaria	*T. capensis*	Cape Honeysuckle
Terminalia	*Terminalia spp.*	Tropical Almond
Tetrastigma	*T. voirieranum*	Virginia Creeper
Thamnocalamus	*Thamnocalamus spp.*	Bamboo
Thuja	*Thuja spp.*	Arborvitae
Thymus	*T. vulgaris*	Thyme
Tilia	*T. americana*	Basswood
Tilia	*Tilia spp.*	Linden Basswood
Tipuana	*T. tipu*	Tipu Tree
Tradescantia	*Tradescantia spp.*	Spiderwort
Trifolium	*Trifolium spp.*	Clover
Triticum	*T. aestivum*	Wheat
Tsuga	*T. Canadensis*	Canadian Hemlock
Tulbaghia	*T. violacea*	Society Garlic
Typha	*Typha spp.*	Cat Tails
Ulmus	*U. americana*	American Elm
Ulmus	*U. crassifolia*	Cedar Elm

Ulmus	*U. parviflora*	Chinese Elm
Ulmus	*U. procera*	English Elm
Ulmus	*U. purrila*	Siberian Elm
Ulmus	*U. rubra*	Slippery Elm
Vaccinium	*Vaccinium spp.*	Blueberry
Viburnum	*V. rhytidophyllum*	Leatherleaf Viburnum
Viburnum	*Viburnum spp.*	
Viola	*V. wittrockiana*	Pansy
Virurnum	*V. lentago*	Nannyberry Viburnum
Vitis	*V. arizonica*	Arizona Grape
Vitis	*V. riparia*	River Bank Grape
Vitis	*V. vivifora*	Grapevine
Washingtonia	*W. robusta*	Mexican Fan Palm
Xylosma	*X. congestum*	Waxy Bush
Zea	*Z. Mays*	Corn
Zelkova	*Z. serrate*	Japanese Zelkova

Appendix H: Assessing Hay

When it is possible, managed browsing species should be fed browse from plant species they would normally eat in the wild. However, in North America, it is difficult to provide the variety of browse and to provide it year-round. In addition, the amount of wild browse needed to support healthy GIT function in managed animals is often beyond the ability of most organizations to obtain. For example, a moose calf would need 4 to 10 kg of browse daily.

Therefore, when browse is not available or it is not available in the amount needed, manage browsing species should be fed appropriate quality hay.

This information will assist you in rating your hay, grass and alfalfa:

1. **Stage of cut:** Your hay should have a leafiness of **65–75%.** If the hay has a lot of buds at the stem tips but no purple flowers, it was cut at the "bud stage" and the stems will be very leafy, small in diameter and pliable. If the hay has some flowers but is still leafy, it was likely cut "early bloom" and the stems will be larger but still pliable. If the hay was cut "post blossom", there is less leaf and more stem, and the stems are larger in diameter, woody and brittle. "Post blossom" hay is a lower quality of hay. Every cut of hay (first cut, second cut, third cut) can have bud stage, early bloom stage and post bloom stage. A first cut hay alfalfa hay of the season often has grass hay and may be called "mixed hay". Therefore, for browsing species, you should purchase only hay (usually second or third cut) that is pure alfalfa. It is recommended that you speak with your hay provider and get a detailed description of the hay you purchase.
2. **Growing conditions:** Your hay provider can also tell you about the growing conditions that will affect the quality of your hay. For example, if the growing conditions are hot and the plant grows fast, the bloom stage of hay can be lower quality (less leaf, more woody) than hay grown slower in cooler conditions.
3. **Hay analysis:** Most hay providers can give you an analysis of their current hay crops. It is recommended that you purchase a nutrient analysis of your hay if your hay provider does not give you the nutrient analysis. Another option is to access hay analysis information on the internet. For example, most provincial ministries of agriculture will provide seasonal information on hay crops with warnings about nutrient excesses or deficiencies in the current year's hay crops for the province. As well, some laboratories will provide links to this information. For example, the websites for Agri-Food Laboratories, Dairy One, Covance, Cumberland Valley Analytical, Midwest Labs, etc. provide the current analysis. Ideally, you should have a nutrient analysis done for each hay delivery. Typically, the minimum nutrient information you need is crude protein, calcium and phosphorus. Some analyses will typically provide the percent dry matter (DM) and acid detergent fibre (ADF). However, most laboratories also have mineral packages that give the percent calcium, phosphorus, potassium, magnesium and sodium as well as the parts per million (ppm) of zinc, manganese, copper and selenium. This data can be important for you to know if the hay in your agricultural region has deficiencies or excesses of some nutrients. Some of these tests would be standard in an analysis, but some tests may have to be added to a "package" and these tests would have individual pricing additional to the package price. There are two methods that your laboratory can use: wet chemistry or near infrared (NIR) analysis. Your laboratory can help you decide what analytic package is best for you. In addition, please contact me at any time if you have questions or would like some assistance in this process.
4. **Hay sampling:** If your hay provider cannot give you an analysis of your hay and you decide to have your hay analyzed (recommended), there is a sampling procedure you should follow. The correct method to sample hay is to use a hay corer. "Core samples" are taken from 10 to 20 bales selected randomly. For smaller bales (rectangles), the sample is taken from the small end about center and for larger bales (typically round), the sample is taken midpoint on the curve. The corer must be pushed as far into the bale as possible. You will need about 1 lb (1/2 kg) of hay total. All the samples should be put into one plastic bag that can be sealed and then sent immediately for analysis. If you do not have a corer or cannot borrow one, then the next best method is to collect your samples throughout one to two weeks of feeding. You need to sample 10 to 20 bales. For each bale that you use, select a sample from inside the bale as close to center as you can get. Each sample should be a mix of leaves and stem although – if your hay is extra-leafy – you will

have a lot of leaf. The sample must be representative of your hay. Place the sample in a plastic bag that can be sealed and keep this bag in an area without temperature or light extremes. For example, the bag of samples should not be exposed to cold, heat or sunlight. Add your bale samples to the bag until you reach your goal of 10 to 20 samples. Mix together well and submit to the laboratory.

5. **Hay weight:** You will need to have an average weight of your hay bales. Your hay provider should be able to give you this information.
6. **Colour:** Quality alfalfa hay will be green and usually a darker green than quality grass hay. Alfalfa hay can sometimes be a very bright green colour and this bright green colour means the alfalfa hay has probably been treated with propionic acid (or a similar biological) as a preservative and fungicide.
7. **Appearance:** In addition to 65%-75% leaf, good hay will have short stems that are pliable (easy to bend without breaking); no obvious mold (usually visible in flakes); and, no foreign matter (sticks, weeds, dirt, dust, stones, etc).

Weighing hay: It is not necessary to weigh your hay at every feeding. However, you must get an estimation of the weight of a hay bale. Most hay bales range in weight from 14 kg to 25 kg. Your supplier should be able to get an average weight for you before delivery since most feed supply companies have larger weight scales. If your supplier cannot tell you the average weight of your hay bales, then there is an efficient method to do this yourself. To get an average weight: weigh 3 hay bales, add the weights of the 3 bales, divide by 3 and the amount you get will be what you will use for the average weight of your hay. I suggest getting an average bale on each new delivery of hay.

Example: Three hay bales are weighed and they weigh 20 kg, 22 kg and 25 kg.
20 kg + 22 kg + 25 kg = 67 kg
67 kg /3 = 22.3 kg average weight of a bale

To calculate the daily feed amounts:
Example: If the amount to Feed out = 6 kg
22.3 kg bale/6 kg amount to feed = 3.7 (round up to 4)
Therefore, ¼ of a bale is approximately 6 kg

Appendix I: Example Animal Data Transfer Form

AMERICAN ASSOCIATION of ZOO KEEPERS

ANIMAL DATA TRANSFER FORM

1. Curator's copy of information on new arrival Date:
2. Keeper's copy of information on new arrival
3. Copy for zoo files and/or veterinarian

*****Please send a copy of this form to shipping institution and state condition of animal(s)*****

Previous Institution (s):
Current Institution

Contact Person:
Title:
Email: Phone:
Receiving Institution:
Common name: Scientific name:

Zoo ID#	House Name	Sex	Hatch / Birth Date*	Tattoo/Band/ Tag #	Weight*	Transponder	Studbook # (Regional/International)

*** Note if it is actual or estimated.**
DIET: Present diet and supplements, favored items, problem foods, feeding procedures.

BEHAVIORAL HISTORY & SPECIFICATIONS: Please list any unique behavioral traits, problems with aggression, safety concerns, or other behavioral problems that may affect management.

General Disposition (skittish, prefers males over females, imprinted, aggressive, etc.):
-
Stereotypic behavior (frequency, duration, triggers):
-None
Methods used for managing stereotypic behavior:
-
Does the animal have a history of aggression towards keepers and/or other animals? ☒ No ☐ Yes
If yes, please explain:
-

What are the conditions and behavioral precursors to the aggression?

What successful strategies are used for dealing with the aggression?

General comments or describe other behaviors that require further explanation:
MEDICAL HISTORY OR PHYSICAL CONDITION: Medication techniques, immobilization techniques, chronic medical problems, Vet Contact.
-
ENCLOSURE DATA: Exhibit dimensions and description, disinfection/cleaning needs, temperature and climate control needs.

Exhibit Features: (When offered or provided, please list or check where applicable. Add comments where necessary)
Exhibit dimensions
-
Substrates: sand ☐ gunite ☒ mulch ☒ leaf litter ☐ soil ☐ other ☐ pine shavings
Exhibit Furniture: deadfall ☒ live trees ☐ rockwork ☒ perching ☐ termite mounds ☐ other ☐
Water features :
Holding Area: indoor ☒ outdoor ☒ None (see above) ☐
Substrates: sand ☐ mulch ☒ leaf litter ☐ soil ☐ other ☒ cement floors
Holding Furniture:
How frequently rotated:
SOCIAL HISTORY *(check all that apply)*
Rearing type: ☒ dam, parent or family reared ☐ hand reared (☐ with conspecifics ☐ without conspecifics) ☐ puppet
☐ supplemental ☐ foster reared (☐ by same species ☐ by different species) ☐none ☐ autonomous
☐ colony / peer
Comments:

Animal housed: ☒ individually ☒ with conspecifics ☒ with mixed species (List species & # of each -
☐ other, please describe:
Housed on exhibit ☒ off-exhibit ☒ access to both ☐
Comments:

REPRODUCTIVE HISTORY: Relevant information, introduction techniques, behavior toward young, specific concerns.

ENRICHMENT HISTORY (Please attach any relevant schedules, approved item lists, sample calendars, etc.)
Goals for the enrichment: stimulation

Enrichment activities offered in exhibit:

Enrichment offered: daily ☒ weekly ☒ monthly ☐ scheduled ☐ other☐
How frequently rotated: several times a week
Enrichment activities offered in holding (if different from those offered on exhibit):

Food Enrichment
Diet Presentation: see above
# of feedings per day: Varied times: ☐ When: Food scattered ☐ Hidden ☐
Novel Foods (please list or attach approved list of food items, frequency and amounts offered and presentation):
Enrichment Devices / Items
PVC feeders ☐ Tires ☐ Burlap/towels ☐ Plastic containers ☐ Puzzle feeders ☐
☐ Cardboard boxes/tubes/bags ☐ Ropes/vines/fire hose ☐ Balls/kegs/barrels ☐Toys (Kong®, dog chews, etc.)
Attachment methods used (*chain, rope, bungee*):

Preferred enrichment for this animal (list):

Safety Concerns (ingests cloth, has become impacted, displays at cage mates with large items, etc.):

General Comments (including expanding on any of the data entered, above):

TRAINING OR BEHAVIORAL CONDITIONING:
Training goals for this animal (list general behavioral goals and indicate which goals have been achieved and/or which goals were partially shaped but not complete at time of shipment):

How long has animal participated in a behavioral conditioning program?
Frequency and Duration of Training Sessions:
☐ once daily ☐ twice daily ☐ once weekly ☐ twice weekly
☒ other, please specify:
Average length of training session (minutes):
Animal attitude/demeanor towards/during training:
Level of contact between the keeper and animal: ☒ free contact ☐ protected contact.

Social arrangement during training sessions:
☒ housed individually and trained individually ☐ separated from conspecifics for training
☐ training with conspecifics present ☐ trained with mixed species present
☐ other, please describe:

Animal conditioned to enter crate/chute/cage for transport? (circle appropriate device*) ☒ No ☐ Yes
Length: Width: Height:
*Attach pictures if necessary to describe training area or device (crates, chutes, etc.).
Reinforcers: ☐verbal ☒food List type and amount used:
☐tactile ☐combination of all of the above

Bridging stimulus: ☐clicker ☒verbal, describe:
☐whistle ☐other, describe:
How are undesirable behaviors addressed? ☐Time-out ☒Ignore ☐re-direct ☐incompatible behavior ☐other (describe):

Which methods have been most successful?

BEHAVIORS TRAINED (Please provide a brief summary – more detail can be added in subsequent section)

Behavior	Verbal cue/command	Visual cue	Criteria for reinforcement	Devices used

Please attach list of behaviors if more room is needed.

General Training Comments: